Pulse train solutions in a time-delayed opto-electronic oscillator

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Abstract

In this master's thesis, we study an optoelectronic time-delay oscillator. In its optical part, the light of a semiconductor laser passes a Mach-Zehnder modulator and shines on a photodiode to produce a radio frequency signal that is filtered and amplified in the electronic part. This voltage in return controls the optical part through the Mach-Zehnder modulator. The major part of the time-delay is due to an optical delay line that produces time-delayed feedback, implying rich dynamics, the topic of this thesis.

For this optoelectronic oscillator, we find novel pulse train solutions that appear when the oscillator is first seeded with an initial pulse at certain parameter values. We are able to control the width of the pulses over a range of three orders-of-magnitude with a minimum pulse width of 150 ps by adjusting the bandpass filter characteristics in the experiment.

Furthermore, the dynamic equations in the pulsing regime show strong similarities to the delayed feedback FitzHugh-Nagumo system in neuroscience, which is a generic model for excitability. The analogy goes so far that both systems show similar nullclines and the same bifurcation mechanism between oscillatory and excitable regimes.

These nullclines help to describe the shape of the waveforms in the pulsing and oscillatory regime as well as the shape of the trajectory on its way to chaos for high feedback gains. Furthermore, they help to derive analytic expressions for the pulse width and the oscillation period as a function of the low-frequency cutoff of the bandpass filter, a finding that is in agreement with experiments and numerical simulations. Using the nullclines, we can derive the Hopf bifurcation point between the excitable and oscillatory regimes exactly, a feature which was only approximately possible in previous works.

Zusammenfassung

Die vorliegende Masterarbeit behandelt einen optoelektronischen Oszillator mit Zeitverzögerung. Im optischen Teil des Oszillators trifft das Licht eines Halbleiterlasers nach dem Passieren eines Mach-Zehnder-Modulators auf eine Photodiode. Im elektronischen Teil des Oszillators wird das resultierende Radio-Frequenz-Signal frequenzgefiltert, verstärkt und steuert den Mach-Zehnder-Modulator, sodass es zu einer Rückkopplung kommt. Diese Rückkopplung ist durch eine optische Verzögerungschleife zeitverzögert: Die resultierenden reichhaltigen Dynamiken werden in dieser Arbeit untersucht.

In dem hier untersuchten optoelektronischen Oszillator haben wir neuartige Pulslösungen gefunden, bei denen sich Pulse mit der Zeitverzögerung der Rückkopplung wiederholen. Diese Pulslösungen können für geeignete Parameter mit einem Anfangspuls angeregt werden. Wir können die Breite der Pulse über drei Größenordnungen kontrollieren und finden eine kleinsten Pulsbreite von 150 ps im Experiment.

Wir zeigen, dass sich der optoelektronische Oszillator ganz ähnlich wie das FitzHugh-Nagumo-Modell aus der Neurophysik verhält, welches ein generisches Modell für Anregbarkeit ist. Diese Analogie geht so weit, dass beide Systeme ähnliche Nullklinen und den gleichen Bifurkationsmechanismus zwischen oszillatorischem und anregbarem Regime besitzen.

Mithilfe dieser Nullklinen beschreiben wir die Zeitserien der Pulslösungen, der Oszillationen und der Transienten zum Chaos. Außerdem leiten wir mit den Nullklinen den analytischen Zusammenhang zwischen Pulsbreite bzw. Oszillationsperiode und der oberen Grenzfrequenz des Bandpassfilters her, welcher mit den Experimenten und den Simulationen übereinstimmt. Mit den Nullklinen können wir auch den Hopfbifurkationspunkt zwischen anregbarem und oszillatorischem Regime exakt bestimmen, welches bisher nur mit Näherungen möglich war.

Contents

Glossary

Δ	Bandwidth
γ	Feedback gain
$\hat{H}(\omega)$	Transfer function of a two pole bandpass filter
$\mathcal{F}\{\cdot\}(\omega)$	Fourier transform
ω_+	High frequency cutoff
ω_-	Low frequency cutoff
ω_0	Center frequency
τ	Time delay
ε	Filter parameter
d	Amplifier parameter
g	Constant gain
g_{MD}	Linear gain of the MD
$H(\omega)$	Transfer function
m	Phase shift parameter
P_{in}	Input power of the MZM
P_{out}	Output power of the MZM
T	Period of oscillations
U	Filter variable
V	Measured voltage
V_{in}	Input voltage of the open loop setup
$V_{\mathrm{in}}^{\mathrm{BP}}$	Input voltage of the bandpass
$V_{\mathrm{in}}^{\mathrm{MD}}$	Input voltage of the MD
V_{out}	Output voltage of the open loop setup
$V_{\mathrm{out}}^{\mathrm{BP}}$	Output voltage of the bandpass
$V_{\mathrm{out}}^{\mathrm{MD}}$	Output voltage of the MD
V_{sat}	Saturation voltage of MD
V_{DC}	voltage applied to the DC port of the MZM
V_{rf}	voltage applied to the rf port of the MZM
$V_{\pi,\mathrm{DC}}$	π-voltage of the DC port of the MZM
$V_{\pi,\mathrm{rf}}$	π-voltage of the rf port of the MZM
w	Pulse width at half max
DDE	delay differential equations
FHN	FitzHugh-Nagumo
MD	modulator driver
MZM	Mach-Zehnder modulator
OEO	opto-electronic oscillator
rf	radio frequency

1
Introduction

1.1 Excitability

Excitability is an important characteristic of many biological systems, such as neural networks [1] and the heart [2] and is also studied in optics [3, 4, 5, 6]. Excitability can be found in the Hodgkin-Huxley model [7, 8], which describes the membrane potential of a giant squid axon. It is a realistic model in computational neuroscience and can be studied with the methods of nonlinear dynamics [9]. The Hodgkin-Huxley model was soon simplified to the FitzHugh-Nagumo (FHN) model [10, 11] as a generic model of excitability and is still extensively studied. When using the FHN model to describe single neurons, coupling between them can be modeled with the Pyragas delayed feedback [12] which induces complex dynamics in general networks [13, 14, 15, 16, 17] and in networks of FHN models [18, 19]. The effect of time-delayed feedback on noise-induced oscillations in the FHN model is also studied extensively [13, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29]. The suppression of excitation waves by time-delayed feedback has been shown within the FHN model under the influence of spatial diffusion [30, 31, 32]. While the FHN model describes type-II excitability associated with a Hopf bifurcation, time-delayed feedback has also been studied in type-II excitability associated with a global SNIPER bifurcation [33, 34]. Furthermore, two coupled systems as a network motif of neurons or neuron populations are studied within the FHN model, focusing on deterministic pulse trains that are periodic with the time delay [18, 19, 26, 35, 36, 37]. A super-threshold input pulse excites the system out of the asymptotically stable fixed point into a pulse train. In this thesis the main question is whether pulse trains can also be excited in a time-delayed optoelectronic oscillator.

1.2 Optoelectronic oscillators

A dynamic system with optical delay was first studied by Ikeda [38] in the form of a ring cavity in 1979 as sketched in Fig. 1.1. The dynamics of the absorption in a sample cell are dependent on a delayed feedback due the optical path around the cavity. This setup was experimentally realized by Gibbs *et al.* [39] two years later. A time-delayed optoelectronic oscillator (OEO) similar to the one used in this thesis was firstly proposed by Neyer and Voges [40] in 1982. The key difference to the experiment in this thesis is that their components are DC- rather than AC-coupled. In AC-coupled devices the constant part of the voltage is blocked and low frequency signals are attenuated. We use AC-coupled electronic devices, which is a property of most electronic equipment that operates at broadband frequency ranges. It was shown that the low frequency cutoff has a significant influence on the dynamics [41, 42]. We study the OEO for different frequency cutoffs and find how these cutoffs influence the dynamics quantitatively. OEOs similar to our system are studied by Larger and collaborators. They are interested in secure chaos communication [43] and the dynamic solutions in general. They study dynamics like breathers [44, 45], scenarios which we also deal with in this thesis, and low-phase-noise oscillations [46]. Recently, with the same setup used in this thesis, Callan *et al.* studied the transition from the unique fixed point to broadband chaos and the switching between both states [47] which is an intriguing feature of optoelectronic chaos [48]. We also study the transition to chaos. Optoelectronic oscillators can be used to build pure frequency sources [49, 50, 51]. A network of two OEO [52] was studied by Illing *et al.* The results of our work are highly applicable, because in our setup we use off-the-shelf telecommunication equipment.

Figure 1.1: Ikeda ring cavity.

1.3 Delay Differential Equations

Problems that involve a time delay occur in many fields, such as mechanical, electronic, optical and biological systems [53] and, therefore, need to be modeled with delay differential equations (DDE). In applied problems where engineers need their models to behave close to the real process, which might include time delays, DDE become more important [54]. Delayed feedback can be used to control chaos [13, 55] and to stabilize fixed points [56, 57, 58, 59, 60], which introduces a time delay to ordinary differential equations.

In DDE, the time evolution $\dot{x}$ of the system variable $x(t)$ depends on the system variable not only at times t but also at previous times, which is in our case the time $t - \tau$. DDE belong to the class of functional differential equations [54], where the right hand side is a functional, dependent on the whole function $x(\cdot)$. Dealing with one constant delay, we are investigating a relatively simple representative out of the array of possible DDE that could include time varying, state dependent or distributed delays [61].

As an initial condition, a whole initial history function on the initial time interval $[-\tau, 0]$ needs to be specified. Since a function is continuous, there are infinitely many ways to define the history function, and therefore DDE are infinitely dimensional [53].

In the framework of nonlinear dynamics, analytical investigations of DDE concentrate on simple solutions because general solutions of DDE are difficult to find [61, 62]. The simplest conceivable solution is a fixed point, a scenario for which DDE simplify to algebraic equations. Another straightforward solution is a limit cycle. When such solutions are found scientists in nonlinear dynamics consider the solutions' stability and their bifurcations. An analytical tool to find such solutions and to study bifurcations, the nullcline analysis, will be explained and used in this thesis.

To obtain approximative solutions for DDE, it is possible to simulate them numerically. Even numerical simulations are especially complicated for DDE [63]. A variable step size, which is useful for problems with time scale separations, is hard to realize for DDE. In this thesis we deal with the numerical difficulties of DDE when simulating the dynamic equations.

1.4 Overview

In chapter 2, we introduce our experimental system and the underlying dynamic equations, then, in chapter 3, we summarize mostly analytic work already done on the dynamic equations. In chapter 4, we find novel pulse train solutions for the OEO. We show how to calculate the second phase space variable from the measured voltage, leading to a more complete picture of pulse trains in phase space. A similarity between the OEO and the FitzHugh-Nagumo system encourages us to call our system excitable. We calculate the Hopf bifurcation points using τ-periodic nullclines. These nullclines characterize the phase trajectory and the time series of the pulse trains and, in chapter 5, help to find a relation between pulse width and the bandpass filter parameters that agrees with simulations and experiments. In chapter 6 we apply the nullcline method to four additional dynamic scenarios which are two different oscillatory states, breather solutions, and transitions to chaos. Furthermore, we relate the period of the oscillations to the bandpass filter parameters, describe how the duty cycle of the oscillations changes with system parameters, and calculate the waveform of one oscillatory scenario. In chapter 7, we summarize our findings, consider applications, and address future research. The analytical, numerical and experimental methods are described finally in the Appendix (chapter 8). There we describe the numerical integrators, the nullcline approach for delay differential equations, the automation of the experiment, and the experimental devices.

2
System description

We extend the optoelectronic oscillator (OEO) that was used by Kristine Callan and collaborators [47, 64]. We add devices to send in initial pulses, use additional bandpass filters, add different delay lines, and automate parts of the setup. Moreover, we measure the fixed and variable parameters with higher precision.

In this chapter, we give a short overview over the whole setup followed by a detailed description and characterization of the parts that are most important in this thesis. In the second part of this chapter, we derive the dynamic equations which are a system of two delay differential equations (DDE). The model numbers of the devices and the automation to control the parameters are shown in the Appendix.

2.1 Experimental setup

The OEO is schematically sketched in Fig. 2.1. In the optical part of the OEO, light of a $1.55\,\mu$m semiconductor laser is coupled into a single mode optical fiber, passes sequentially through a stepwise variable optical attenuator, a polarization controller, a Mach-Zehnder modulator (MZM), a delay line and is incident on an inverting AC-coupled photodetector. In the electronic part, the voltage emitted by the photodetector passes through a bandpass filter and a power splitter. One half of the split voltage, defined as the dynamic variable and denoted by V, is amplified by an inverting modulator driver (MD). The other half of the signal is split again and measured with a high-speed oscilloscope, which has an 8 GHz

Figure 2.1: Schematic of the setup.

analog bandwidth and a 40 GS/s sampling rate. Therefore, the oscilloscope detects half of the dynamic variable $V/2$. In order to influence the dynamics and especially to provide pulses as initial functions, a defined electrical pulse from a pattern generator can be combined with V at the input of the MD. The pattern generator is isolated from the system with an electrical switch. The voltage exiting the MD is fed back to the MZM via its radio frequency (rf) input port to close the feedback loop. The optical power transmission of the MZM is a function of the applied voltages as described later on. Besides the time dependent voltage exiting the MD, we apply a bias to the DC port of the MZM.

The OEO is a time delayed feedback system, because the rf signal that controls the transmission of the MZM is a function of the intensity of light leaving the MZM a time delay earlier. The time delay is large compared to the time scale of the dynamics, plays a crucial and cannot be neglected. Another important feature of the OEO is that it is bandpass filtered which is the starting point to derive the dynamic equation of the OEO.

In the next sections, we explain the following devices in detail: the MZM, the MD, and the bandpass filter.

2.1.1 Mach-Zehnder modulator

Because it is a polarization sensitive device, we use a polarization controller prior to the MZM. It is sufficient to tune the angle of polarization with the polarization controller, because the light that the laser emits is already linearly polarized.

The optical power transmission of the MZM ($P_{\mathrm{out}}/P_{\mathrm{in}}$), where P_{in} is the input and P_{out} the ouput power of the MZM, is a nonlinear function of the applied voltage V_{rf},

$$P_{\mathrm{out}}/P_{\mathrm{in}} \propto \cos^2\left[\pi V_{\mathrm{DC}}/(2V_{\pi,\mathrm{DC}}) + \pi V_{\mathrm{rf}}/(2V_{\pi,\mathrm{rf}}) + \phi\right], \tag{2.1}$$

where V_{DC} is the applied bias and V_{rf} is the applied radio frequency signal, $V_{\pi,\mathrm{DC}}$, $V_{\pi,\mathrm{rf}}$ are the π-voltages of the respective ports, and ϕ is a small, but not neglectable, phase difference that is due to temperature fluctuations slowly varying. The voltage difference between two minimums of the $\cos^2$-transmission curve is $2V_\pi$.[1]

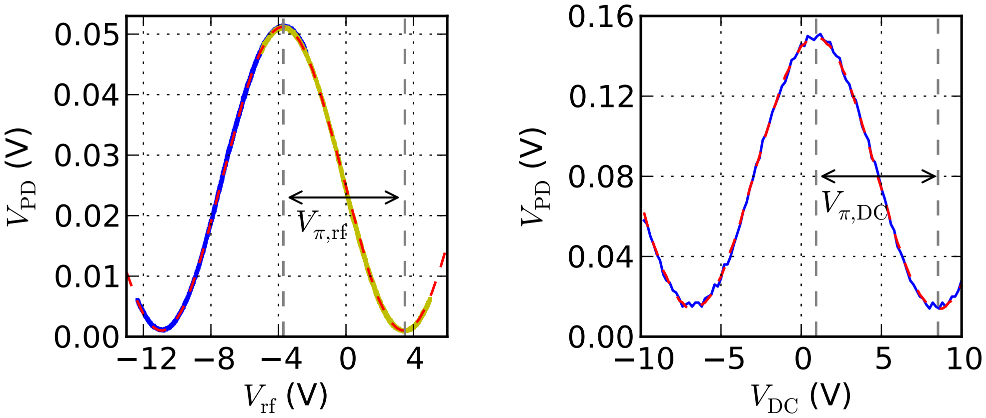
2.1.1.1 Working principle

The working principle of a MZM is Pockels electrooptic effect [65], where an applied voltage changes the index of refraction linearly for a certain polarization. A polarized electromagnetic wave, traveling through a special birefringent medium with an applied Pockels voltage, acquires a phase difference ϕ_0 in respect to a beam traveling through the same medium without the voltage. A simple MZM splits the incident light into two beams and applies a Pockels voltage to one beam which therefore slows down. When recombining both beams, a phase difference ϕ_0 occurs. Adding two oscillating electro-magnetic waves that have a phase difference ϕ_0 results in an amplitude modulation with a $\cos(\phi_0)$-function. Therefore, the intensity is modulated with a $\cos^2(\phi_0)$-function. The phase difference is proportional to the independently applied voltages V_{dc} and V_{rf}, leading to Eq. (2.1).

2.1.1.2 Characterization

The π-voltage of the DC port, $V_{\pi,\mathrm{DC}}$, is measured by applying a saw tooth voltage to the DC port in the range from -10 to $10\,\mathrm{V}$ and fitting the resulting transmission of the MZM to Eq. (2.1), as shown in Fig. 2.2b. This measurement is automated with LabVIEW as described in Sec. 8.3.1. The best fit value is

[1] Applying the π-voltage to one port shifts the phase between two optical beams in the MZM from 0 to π, hence from constructive to destructive interference.



(a) rf-port (b) DC-port

Figure 2.2: (a) Measurement of the MZM transmission curve of the rf port around a phase shift of $\pi/4$ (blue solid line) and around $-\pi/4$ (yellow solid line). Least square fit (red dashed line) with best fit parameter $V_{\pi,\mathrm{rf}} = 7.2$. (b) Measurement of the MZM transmission for the DC port with the LabVIEW program (blue solid line). Least square fit (red dashed line) with best fit parameter $V_{\pi,\mathrm{DC}} = 7.6$. The photodetector voltage, V_{PD}, is proportional to the transmission, $P_{\mathrm{out}}/P_{\mathrm{in}}$.

$V_{\pi,\mathrm{DC}} = 7.6$. To measure the π-voltage of the rf port, $V_{\pi,\mathrm{rf}}$, we similarly apply a saw tooth voltage from -5 to 5 V. Since the range from -5 to 5 V does not cover the whole transmission curve, we measure it twice for two different biases V_{DC}, corresponding to phase shifts of $\pi/4$ and $-\pi/4$. With both measurements the whole transmission curve can be reconstructed and fitted to the model Eq. (2.1). This is shown in Fig. 2.2a with a least square fit with best fit value $V_{\pi,\mathrm{rf}} = 7.2$. The saw tooth voltage has a frequency of 1 kHz and the transmission is measured with a DC-coupled photodetector (Thorlabs DETOKFC).

In Sec. 8.3.1, it is explained how we use the automation of the setup to measure the phase shift ϕ and apply the voltage V_{DC} corresponding to the desired phase of the transmission curve (Eq. (2.1)). We define

$$m = \frac{\pi V_{\mathrm{DC}}}{2V_{\pi,\mathrm{DC}}} + \phi, \tag{2.2}$$

as the phase shift parameter that we set through the automation software. This leads to a transmission curve of

$$P_{\mathrm{out}}/P_{\mathrm{in}} \propto \cos^2\left[m + \pi V_{\mathrm{rf}}/(2V_{\pi,\mathrm{rf}})\right]. \tag{2.3}$$

2.1.2 Modulator driver

The MD amplifies the voltage emitted by the photodetector to drive the MZM with voltages similar to the π-voltage and therefore the transmission of the MZM can be reasonably changed. The MD that we use saturates for high voltages which we model with a sigmoid, namely a hyperbolic tangent function,

$$V_{\mathrm{out}}^{\mathrm{MD}} = V_{\mathrm{sat}} \tanh\left(\frac{g_{\mathrm{MD}} V_{\mathrm{in}}^{\mathrm{MD}}}{V_{\mathrm{sat}}}\right), \tag{2.4}$$

where g_{MD} and V_{sat} are linear gain and saturation voltage, respectively.

We measure these two constants, g_{MD} and V_{sat}, by propagating a sine wave of amplitude $V_{\mathrm{in}}^{\mathrm{MD}}$ into the MD and measuring the amplitude $V_{\mathrm{out}}^{\mathrm{MD}}$ of the output sine wave. The experimental measurement for an input frequency of 15 MHz with a least square fit is shown in Fig. 2.3. The measurement is taken with an increased point density between $V_{\mathrm{in}}^{\mathrm{MD}} = 0.2\,\mathrm{V}$ and $V_{\mathrm{in}}^{\mathrm{MD}} = 0.42\,\mathrm{V}$, the voltage range where important dynamics occur which is why we want to have best agreement in this range.[1] The fit does not account for data points with $V_{\mathrm{in}}^{\mathrm{MD}} > 0.7\,\mathrm{V}$, voltages that do not occur in the experiments. The best fit values for this measurement are $g_{\mathrm{MD}} = -17$ and $V_{\mathrm{sat}} = 4.8\,\mathrm{V}$. It can be seen from Fig. 2.3 that the saturation voltage is not the actual saturation voltage but the best fit parameter to describe the dynamics in the important voltage range.

The amplifying characteristic of the MD shows weak frequency dependence that could be due to an impedance of the MD that is not precisely equal to $50\,\Omega$ [66]. In the model, we assume that the MD has a frequency independent gain characteristic which could lead to discrepancy between experiments and theory.

[1] The high level of pulses and low and high levels of the oscillatory dynamics lie in this voltage range.

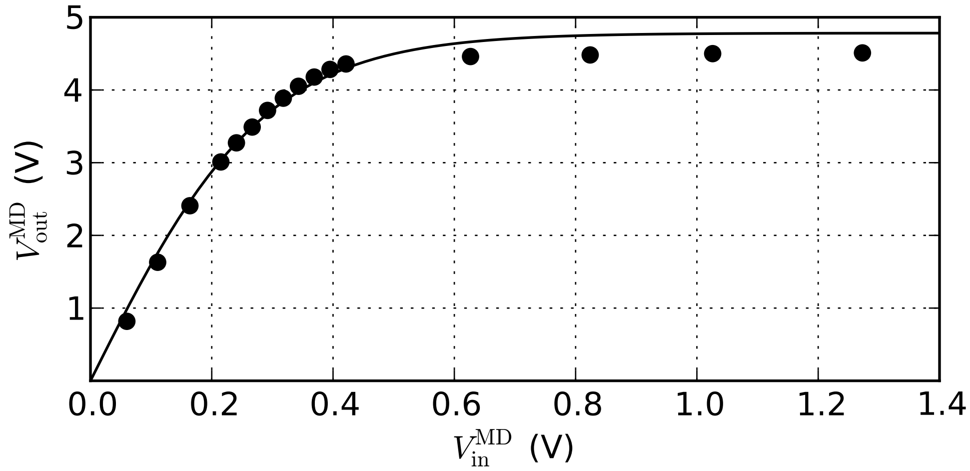


Figure 2.3: Measurement of the MD amplifying characteristic with 15 MHz sine waves of different amplitudes, $V_{\rm in}^{\rm MD}$, and output sine waves of amplitude $V_{\rm out}^{\rm MD}$ (circles). Least square fit with hyperbolic tangent function of the data from 0 to 0.7 V (solid line). The point density is chosen to be higher in regions where the dynamics mainly occur.

2.1.3 Bandpass filter

The MD and the photodetector are considered AC-coupled devices and therefore do not amplify low frequency modulations of the incident light. They possess high and low frequency cutoffs, hence are acting as a bandpass filter. We use another bandpass filter after the photodetector to control the bandwidth. The bandpass characteristics of all system components are modeled as one bandpass filter after the photodetector. In the next sections we derive the mathematical description of bandpass filters which builds the foundation of the dynamic equations of the OEO.

2.1.3.1 Transfer function

A frequency dependent response of a device is described by the transfer function $H(\omega)$, which relates the Fourier mode of input, $V_{\rm in}^{\rm BP}$, and output signal, $V_{\rm out}^{\rm BP}$, for all frequencies ω

$$\mathcal{F}\left\{V_{\text{out}}^{\text{BP}}(t)\right\}(\omega) = \gamma H(\omega)\mathcal{F}\left\{V_{\text{in}}^{\text{BP}}(t)\right\}(\omega). \tag{2.5}$$

where γ is the gain of the bandpass filter and

$$\mathcal{F}\left\{f(t)\right\}(\omega) = \int_{-\infty}^{\infty} f(t)e^{i\omega t}\,\mathrm{d}t \tag{2.6}$$

denotes the Fourier transform.

The transfer function $H(\cdot)$ is normalized for the frequency of maximum transmission. The feedback gain, γ, models the gain or attenuation of the system and is independent of the frequency. γ can be greater than one, because we describe the whole system including the amplifiers as the bandpass filter.

2.1.3.2 Two-pole bandpass filter

We model the bandpass characteristic of the system as a bandpass filter with two frequency cutoffs of first order: a low frequency cutoff, ω_-, and a high frequency cutoff, ω_+.

Multiplying the functions of a first order lowpass and a first order highpass filter and normalizing, results in the transfer function of a two-pole bandpass filter [42]

$$\hat{H}(\omega) = \frac{\Delta i\omega}{-\omega^2 + i\Delta\omega + \omega_0^2}. \tag{2.7}$$

Here, $\Delta = \omega_+ - \omega_-$ is the bandwidth and $\omega_0 = \sqrt{\omega_+\omega_-}$ is the center frequency. At the center frequency, ω_0, the transfer function in normalized $\hat{H}(\omega_0) = 1$. The absolute value of the transfer function

$$\left|\hat{H}(\omega)\right| = \frac{\Delta\omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + \omega^2\Delta^2}} \tag{2.8}$$

at each cutoff frequency is $1/\sqrt{2}$ (on a logarithmic scale approximately $-3\,\mathrm{dBm}$) and, therefore, the power transmission, $|\hat{H}(\cdot)|^2$, is half at the cutoff frequencies. The imaginary part of the transfer function accounts for a phase shift.

Figure 2.4: Setup for the measurement of the transfer function, $|H(\omega)|$, and the gain, γ, of the system. The transfer function of the OEO is measured as in (b), but we assume that it was measured as in (a) for the derivation of the DDE.

2.1.3.3 Transfer function of the OEO

The experimental frequency response, $|H(\omega)|$, of the system is more complex and is fitted for each filter to the simple description of a two-pole bandpass filter, $|\hat{H}(\omega)|$ (Eq. (2.8)), with two fit parameters, ω_- and ω_+. We have six different bandpass filters at our disposal with model numbers listed in table 8.1 on page 80.

To measure the frequency response, $|H(\omega)|$, of each filter, we open the circuit as shown in Fig. 2.4b, send a low-amplitude sine-wave (amplitude V_{in}) with power $-30\,\text{dBm}$ of a certain frequency, ω, into the open loop and record the amplitude of the output signal, V_{out}. We repeat this measurement for different frequencies, ω, leading to the transfer function $|\gamma H(\omega)| = (V_{\text{out}}/V_{\text{in}})|_{\omega}$.

The theory of transfer functions assumes linear responses where V_{out} has the same frequency as V_{in}. Therefore, we need to operate the system in the linear regime by tuning the transmission of the MZM to the most linear part at a phase shift of $m = -\pi/4$ (see Eq. (2.3)). The low amplitude of the input signal ensures that we stay in the linear regime.

To acquire a fit that is sensitive for both cutoff frequencies, we measure the frequencies with a linear spacing of 2 points per MHz and fit with an exponential

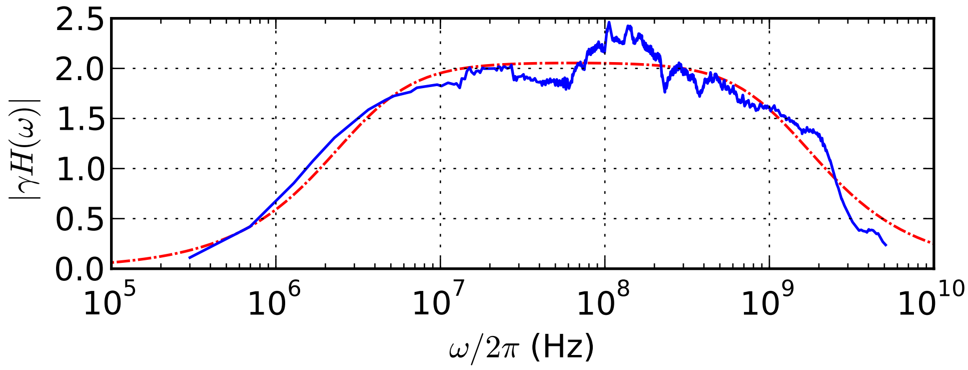


Figure 2.5: Transfer function $|\gamma H(\omega)|$ (blue solid line) of the OEO with filter #4 (see Table 2.1). For the measurement we used the setup shown in Fig. 2.4b. Fit with a two-pole bandpass characteristic $|\gamma \hat{H}(\omega)|$ (red dashed line). The fit parameters are $\gamma = 2.05$, $\omega_-/2\pi = 3.32\,\text{MHz}$, and $\omega_+/2\pi = 1.22\,\text{GHz}$.

weight $\propto \exp(-x)$. This method is expected to give the same result as a logarithmic sampling. Both cutoff frequencies differ by several orders of magnitude, so that logarithmic plotting and sampling is essential to resolve both cutoffs with similar precision. For values of the low frequency cutoff $\omega_- < 1\,\text{MHz}$ we fitted ω_- separately with a denser sampling.

For a frequency range from 130 kHz to 20 GHz, the data acquisition is automated by a spectrum analyzer coupled to a frequency generator. For lower frequencies, the measurement needs to be taken by hand for each frequency.

Figure 2.5 shows the transfer function, $|H(\omega)|$, of the OEO with a specific bandpass filter and the fit to the two-pole bandpass filter, $|\hat{H}(\omega)|$. The transfer function, $|H(\omega)|$, and the fit, $|\hat{H}(\omega)|$, differ from each other. For a frequency $\omega/2\pi = 105\,\text{MHz}$, where the frequency response has large ripples, the transfer function and fit differ by 19%.

The difference between $|H(\omega)|$ and $|\hat{H}(\omega)|$ is a source of error because it leads to an uncertainty in γ but also to differences between the model and the experiments. In the model we approximate the transfer function, $|H(\omega)|$, by a two-pole bandpass filter, $|\hat{H}(\omega)|$. Therefore, we do not expect experiments and theory to match perfectly. Nevertheless, the mathematical description that we derive from a two-pole bandpass characteristic is in good agreement with the

Filter	$\omega_-/(2\pi)$ (Hz)	$\omega_+/(2\pi)$ (Hz)	ε	Δ (1/ns)
#1	52.9×10^3	1.74×10^9	3.04×10^{-5}	10.9
#2	460×10^3	3.90×10^9	1.18×10^{-4}	24.5
#3	1.48×10^6	463×10^6	3.22×10^{-3}	2.90
#4	3.32×10^6	1.22×10^9	2.74×10^{-3}	7.64
#5	28.2×10^6	1.11×10^9	2.67×10^{-2}	6.80
#6	119×10^6	1.53×10^9	9.15×10^{-2}	8.87

Table 2.1: Characteristics of bandpass filters used in experiment and simulation. The filter numbers are related to the model numbers in table 8.1 on page 80. The transfer functions are shown in Sec. 8.4.2 on page 81.

experiment [41, 42, 47].

The transfer functions of the other filters are shown in Sec. 8.4.2 on page 81. The cutoff frequencies and the filter parameters, Δ and ε, of the filters used in this thesis, are shown in table 2.1. The filter parameter ε is introduced later on and plays the role of a time scale separation in the dynamic equations of the OEO. All filters with exception of filter #2 and #6 are amplifiers combined with an attenuator and are inverting components (see also table 8.1 on page 80).

2.1.4 Feedback gain γ

The feedback gain, γ, is the response of the system to small sine waves with a frequency corresponding to the center frequency, ω_0, when we assume a transmission of a two-pole bandpass filter. In other words, γ is one of the fit parameters, when fitting the frequency response to a two-pole bandpass filter. Changing the attenuation of the laser with the stepwise variable optical attenuator in Fig. 2.1 and exchanging delay lines does not change the transfer function, $|H(\omega)|$, but only γ. To measure γ without remeasuring the whole spectrum, we measure the gain for an input sine wave of a certain frequency, $\tilde{\omega}$, and amplitude, V_{in}, and output amplitude, V_{out}, of the output sine wave according to Fig. 2.4b, and build

Filter	#1	#2	#3	#4	#5	#6
$\tilde{\omega}/(2\pi)$ (MHz)	20	20	30.5	20	202.5	195
$\frac{1}{\vert H \vert}$	0.998	1.103	0.908	1.033	0.997	0.980

Table 2.2: Measurement of γ: frequencies $\tilde{\omega}$ where we measure $\tilde{\gamma}$ and conversion factors $\frac{1}{|H|}$ to calculate γ according to Eq. (2.10).

the ratio

$$\tilde{\gamma} = V_{\text{out}}/V_{\text{in}}. \tag{2.9}$$

We relate this ratio to the feedback gain, γ, as

$$\tilde{\gamma}|_{\tilde{\omega}} = \gamma|H(\tilde{\omega})| \Rightarrow \gamma = \frac{\tilde{\gamma}}{|H(\tilde{\omega})|}. \tag{2.10}$$

The frequency, $\tilde{\omega}$, is taken in a linear region out of the spectrum, $|H(\omega)|$, as shown in Table 2.2.

2.2 System equations

We use the transfer function of a two-pole bandpass filter to derive an integro differential equation that leads to an integro delay differential equation, when considering the influence of the whole circuit. The integral can be substituted through another variable, leading to the DDE that is used throughout this thesis and by many researchers in the field [44, 45, 47, 67].

We model the bandpass filter as if it were located after the photodetector, i.e. as if we measured it as in Fig. 2.4a, even though it is measured as in Fig. 2.4b. This is an approximation that simplifies the model, because we do not need to model the filter characteristics of each device separately. This approximation is valid because the bandpass filter has a narrow pass band compared to the other devices. Many researchers dealing with OEOs use the same approximation even when an additional bandpass filter after the photodetector is not included [44, 45, 47, 67].

2.2.1 Dynamic equation of a two-pole bandpass filter

Inserting (2.7) into (2.5) leads to

$$\left(1+\frac{i\omega}{\Delta}+\frac{\omega_0^2}{\Delta i\omega}\right)\mathcal{F}\left\{V_{\mathrm{out}}^{\mathrm{BP}}(t)\right\}(\omega)=\gamma\mathcal{F}\left\{V_{\mathrm{in}}^{\mathrm{BP}}(t)\right\}(\omega). \tag{2.11}$$

Applying the properties

$$i\omega\mathcal{F}\left\{f(t)\right\}(\omega)=\mathcal{F}\left\{\frac{\mathrm{d}}{\mathrm{d}t}f(t)\right\}(\omega) \tag{2.12}$$

$$1/(i\omega)\mathcal{F}\left\{f(t)\right\}(\omega)=\mathcal{F}\left\{\int^t f(s)\,\mathrm{d}s.\right\}(\omega), \tag{2.13}$$

and the linearity of the Fourier transform, we obtain

$$\mathcal{F}\left\{\left(1+\frac{1}{\Delta}\frac{\mathrm{d}}{\mathrm{d}t}+\frac{\omega_0^2}{\Delta}\int^t\right)V_{\mathrm{out}}^{\mathrm{BP}}\right\}(\omega)=\mathcal{F}\left\{\gamma V_{\mathrm{in}}^{\mathrm{BP}}\right\}(\omega), \tag{2.14}$$

which becomes

$$\left\{\left(1+\frac{1}{\Delta}\frac{\mathrm{d}}{\mathrm{d}t}+\frac{\omega_0^2}{\Delta}\int^t\right)V_{\mathrm{out}}^{\mathrm{BP}}\right\}=\gamma V_{\mathrm{in}}^{\mathrm{BP}}(t), \tag{2.15}$$

when comparing the arguments of the Fourier transform. By the left hand side we mean

$$V_{\mathrm{out}}^{\mathrm{BP}}(t)+\frac{1}{\Delta}\left.\frac{\mathrm{d}}{\mathrm{d}s}V_{\mathrm{out}}^{\mathrm{BP}}(s)\right|_t+\frac{\omega_0^2}{\Delta}\int^t V_{\mathrm{out}}^{\mathrm{BP}}(s)\mathrm{d}s. \tag{2.16}$$

2.2.2 Dynamic equations of the OEO

The bandpass filter is located after the photodetector (Fig. 2.1), so that the input voltage of the bandpass, $V_{\mathrm{in}}^{\mathrm{BP}}$, is the output voltage of the photodetector and the output voltage of the bandpass, $V_{\mathrm{out}}^{\mathrm{BP}}$, is the input voltage of the first power splitter. We substitute these voltages in terms of V by taking the effects of the optical and electrical components into account. V is the voltage entering the MD. Considering the system components, it holds that

$$V_{\mathrm{out}}^{\mathrm{BP}}=2V, \tag{2.17a}$$

because the electric power splitter divides the signal in two, and

$$V_{\text{in}}^{\text{BP}} = \frac{4V_{\pi,\text{rf}}}{\pi g_{\text{MD}}} \cos^2\left\{m + \frac{\pi V_{\text{sat}}}{2V_{\pi,\text{rf}}} \tanh\left[\frac{g_{\text{MD}}}{V_{\text{sat}}} V(t-\tau)\right]\right\}, \tag{2.17b}$$

where the $\cos^2$ argument is due to the MZM (Eq. (2.3)) and the tanh argument with parameters is due to the MD (Eq. (2.4)) which is connected to the rf port of the MZM: $V_{\text{rf}} = V_{\text{out}}^{\text{MD}}$. The time delay, τ, accounts for the time the signal $V(t)$ needs to travel through the feedback loop. The feedback gain, γ, and the scaling factor in front of the $\cos^2$ argument was chosen so that γ is the linear gain of the system. The linear gain, γ, can be measured for $m = -\pi/4$ and low amplitude sine waves with a frequency corresponding to the center frequency of the two-pole bandpass filter, as described in Sec. 2.1.4.

Inserting Eq. (2.17) into Eq. (2.15) and using the following abbreviations

$$\varepsilon := \frac{\omega_0^2}{\Delta^2}, \tag{2.18a}$$

$$d := \frac{\pi V_{\text{sat}}}{2V_{\pi,\text{rf}}}, \tag{2.18b}$$

$$g := \frac{V_{\text{sat}}}{g_{\text{MD}}}, \tag{2.18c}$$

leads to

$$\left(1 + \frac{1}{\Delta}\frac{\mathrm{d}}{\mathrm{d}t} + \Delta\varepsilon\int^t\right) V = \gamma\frac{g}{d}\cos^2\left\{m + d\tanh\left[\frac{V(t-\tau)}{g}\right]\right\}. \tag{2.19}$$

Equation (2.19) is an integro delay differential equation, because it contains an integral, a time delay, and a derivative. It is advantageous to substitute for the integral by introducing another variable

$$U = \Delta\varepsilon\int^t V - \gamma\frac{g}{d}\cos^2(m). \tag{2.20}$$

The arbitrary offset $-\gamma(g/d)\cos^2(m)$ shifts the variable U, leading to a fixed point value of $U^* = 0$ which we derive later. This is legitimate because the physical quantity V is not changed. Using Eq. (2.20) to substitute for the integral in

Table 2.3: Fixed parameters in the system.

Description	Symbol	Value
Linear gain of the MD	g_{MD}	-17
Saturation voltage of the MD	V_{sat}	$4.8\,\mathrm{V}$
π-voltage of the rf port of the MZM	$V_{\pi,\mathrm{rf}}$	$7.2\,\mathrm{V}$
π-voltage of the DC port of the MZM	$V_{\pi,\mathrm{DC}}$	$7.6\,\mathrm{V}$
Saturation parameter	d	1.0
Constant gain	g	$-0.28\,\mathrm{V}$

Eq. (2.19), leads to the DDE

$$\frac{\dot{V}(t)}{\Delta} = -V(t) - U(t) + \gamma\frac{g}{d}\left\{\cos^2\left[m + d\tanh\left(\frac{V(t-\tau)}{g}\right)\right] - \cos^2\left[m\right]\right\}, \tag{2.21a}$$

$$\frac{\dot{U}(t)}{\Delta} = \varepsilon V(t). \tag{2.21b}$$

We describe the dynamics of the system, namely the dynamics of V, with these DDE of Eq. (2.21). The filter variable U cannot be measured because it is only an auxiliary quantity.

2.2.3 Parameters

The values of the filter parameters are shown in table 2.1 and are measured as described in Sec. 2.1.3. The constant parameters d and g are determined by the characteristics of the modulator driver, g_{MD} and V_{sat} (Section 2.1.2), and also the π-voltage of the radio frequency port of the MZM, $V_{\pi,\mathrm{rf}}$ (Section 2.1.1), leading to a saturation parameter $d = \pi V_{\mathrm{sat}}/2V_{\pi,\mathrm{rf}} = 1.0$ and constant gain $g = V_{\mathrm{sat}}/g_{\mathrm{MD}} = -0.28\,\mathrm{V}$. These parameters are shown in table 2.3.

Time delay, τ The time delay, τ, can be changed through the delay line. Without an optical delay line added, the delay due to the connectors and the optical and electrical parts is 24 ns. We add fiber spools (delay lines) to adjust the time delay to $\tau = 65$ ns, 1.5 μs, and 93 μs.

Feedback gain, γ The dimensionless gain, γ, accounts for all gains and losses in the system: the light power exiting the MZM at maximum transmission, the attenuation of the optical delay line, the linear conversion of the photodetector, the loss in the bandpass filter, and the loss through the power splitter. The measurement of the gain is described in Sec. 2.1.4. The gain can be adjusted with the optical attenuator (see Sec. 8.3.2 on page 79) from 0 to 10, but, depending on the attenuation of the fiber spool and the frequency filter, the range of gain can be smaller. When the bandpass filter is inverting the gain has negative sign.

Phase shift, m The parameter m determines the phase shift of the MZM transmission curve (Eq. (2.3)) and can be freely adjusted from $-\pi/2$ to $\pi/2$. In Sec. 8.3 it is shown how we use the automation of the setup to apply a voltage corresponding to the desired phase.

Time scale separation, ε The filter parameter, ε separates the time scales between changes in V and U (see Eq. (2.21)). The time scale separation can be used for analytical studies because, under certain conditions, it allows changes in U to be neglected because they are slow compared to changes in V. The time scale separation in the system ranges over four orders of magnitude, from 10^{-5} to 10^{-1} (numerical values in table 2.1).

2.2.4 Dimensionless form

In the scientific literature of OEOs [45, 47], it is common to render time, t, and the variables V and U dimensionless. This can be realized by rescaling

$$s = \Delta t, \tag{2.22a}$$

$$\tilde{\tau} = \Delta \tau, \tag{2.22b}$$

$$x = V/g, \tag{2.22c}$$

$$y = U/g, \tag{2.22d}$$

leading to system equations of

$$\dot{x} = -x - y + \frac{\gamma}{d}\left\{\cos^2\left[m + d\tanh\left(x(s-\tilde{\tau})\right)\right] - \cos^2\left[m\right]\right\}, \tag{2.23a}$$

$$\dot{x} = \varepsilon y. \tag{2.23b}$$

The system equations in the dimensionless form (Eq. (2.23)) are more convenient for analytic studies but we use the DDE in Eq. (2.21) because, then, the results can be directly compared to the experiment.

2.2.5 Initial conditions

The setup (Fig. 2.1) allows us to insert pulses that are added to the dynamic variable $V(t)$. When the dynamics are at rest, $V(t) = 0$ when inserting pulses, the signal from the pattern generator can be considered an initial function. The pattern generator emits square pulses of adjustable pulse widths and amplitudes with rise and fall time $< 50\,\mathrm{ps}$. The pulse pattern is repeated in periods of microseconds. To send in single pulses, we use an electrical switch that opens for the period of the pulses (Sec. 8.3.3 on page 80).

3 Previous Work

3.1 Fixed point and stability

The fixed point of the system equations (Eq. (2.21)) and the Hopf bifurcation were studied previously [41, 44, 45, 47, 68]. Recent interest lies in the transition from the fixed point to chaos [44, 47, 64]. Since the fixed point and its stability are fundamental to this thesis and linear stability analysis is an important tool in nonlinear dynamics [69], we present the fixed point analysis, as carried out by Callan [47, 64]. In Sec. 4.6, we find another simpler and purely analytic way to come to the same result. Even though Callan used the dimensionless model equation (Eq. (2.23)) we do the analysis with dimensions (Eq. (2.21)) to be consistent in this thesis.

A fixed point, denoted by V^* and U^*, is a constant solution of a differential equation. Therefore, $\dot{V} = 0$, $\dot{U} = 0$ and $V(t-\tau) = V(t) = V^*$, which, after inserting into the system equation (Eq. (2.21)) and solving for $V(t) = V^*$ and $U(t) = U^*$, gives

$$(V^*, U^*) = (0, 0). \tag{3.1}$$

The linear stability analysis examines the stability of the fixed point by assuming small perturbations so that the system equations can be linearized. An initial small perturbation is assumed to evolve according to the linearized differential equations. Linearizing the system equations (Eq. (2.21)) around the fixed point, leads to

$$\frac{\delta \dot{V}(t)}{\Delta} = -\delta V(t) - \delta U(t) - \gamma \sin(2m)\delta V(t-\tau), \tag{3.2a}$$

$$\frac{\delta \dot{U}(t)}{\Delta} = \varepsilon \delta V(t), \tag{3.2b}$$

where δ denotes small perturbations: $(V(t), U(t)) = (V^* + \delta V, U^* + \delta U)$. The time evolution of these small perturbations, because they are described by linear differential equations, can be found with the exponential ansatz

$$\delta U = e^{\lambda t}, \tag{3.3}$$

where λ is called the local Lyapunov exponent. The Lyapunov exponent determines whether the fixed point is asymptotically stable ($\mathrm{Re}(\lambda) < 0$), unstable ($\mathrm{Re}(\lambda) > 0$) or invariant ($\mathrm{Re}(\lambda) = 0$).[1] After inserting Eq. (3.3) into Eq. (3.2), we derive the characteristic equation

$$\lambda^2 + \Delta\lambda + \Delta^2\varepsilon + \gamma \sin(2m)\lambda e^{-\lambda\tau} = 0. \tag{3.4}$$

At the bifurcation point, small perturbations neither grow nor decay hence $\mathrm{Re}(\lambda) = 0$. Nevertheless, λ can have an imaginary part, $\mathrm{Im}(\lambda) = \Omega$. Inserting these conditions into the characteristic equation, Eq. (3.4), and separating for real and imaginary parts, results in the condition for the bifurcation point

$$0 = -\Omega^2 + \Delta^2\varepsilon + \Delta\gamma \sin(2m)\Omega \sin(\Omega\tau) \tag{3.5a}$$

$$0 = 1 + \gamma \sin(2m) \cos(\Omega\tau). \tag{3.5b}$$

These are implicit equations for the bifurcation point. Equations (3.5) have the symmetry $\Omega \to -\Omega$ and require $\Omega \neq 0$ because $\varepsilon > 0$ in Eq. (3.5a). Therefore, a pair of complex conjugate eigenvalues crosses the imaginary axis at the bifurcation point, otherwise known as the Hopf bifurcation.

Inserting Eq. (3.5) into the trigonometric identity $\sin^2(\Omega\tau)+\cos^2(\Omega\tau) = 1$, we can find a solution for Ω. This solution can be inserted back into Eq. (3.5b), to derive a transcendental equation for $\gamma \sin(2m)$. The smallest γ which solves the transcendental equation can be considered the bifurcation point, γ_{H}. When the

[1]When $\mathrm{Re}(\lambda) = 0$, higher order terms determine the stability.

system is in the fixed point and we increase γ to the Hopf feedback gain, $\gamma = \gamma_{\mathrm{H}}$, the Hopf bifurcation occurs and the system leaves the fixed point.

The transcendental equation for γ_{H} was numerically solved for the smallest γ, leading to

$$\gamma_{\mathrm{H}} \approx \pm 1/\sin(2m), \tag{3.6}$$

for our system with filter parameters, ε and Δ, of filter #1 in table 2.1 on page 14 [64]. For $m \to 0$, $\gamma_{\mathrm{H}} \to \infty$ and, therefore, at $m = 0$ the fixed point can be assumed to be asymptotically stable for any γ.

The bifurcation point for varying m is given by

$$m_{\mathrm{H}} \approx \pm \arcsin(1/\gamma)/2 + n\pi, \tag{3.7}$$

with $n \in \mathbb{Z}$. Callan *et al.* [47] showed that, instead of a Hopf bifurcation, a global bifurcation occurs for $m = 0$ and $\gamma > 4$, as demonstrated in the next section.

3.2 One-dimensional map

In this section we concur with Callan *et al.* [47] that a global bifurcation from the fixed point to chaos occurs in the vicinity of $m = 0$. A special interest in this study lies in the transition to chaos. The method to find this global bifurcation is a one-dimensional map that can be derived by a nullcline analysis.

For $m = 0$, the transition from the fixed point to chaos consists of a train of short pulses that start as a noise peak and grow in a train of short pulses to a maximum amplitude [47]. Several pulse trains start to excite a single long pulse,[1] after which the time series becomes chaotic. Callan *et al.* [47] argued that an initial noise peak excites a pulse after the time delay, τ, through the delayed feedback. Another pulse is excited after another time delay, τ, and so on. The amplitude of these pulses is described as a map with the pulse number as an index.

Depending on γ and m, pulses either grow or decay, implying an unstable or stable the fixed point, respectively. It is our task to find $\gamma(m)$, the bifurcation

[1] In Sec. 6.4 we study the long pulse in the transition to chaos.

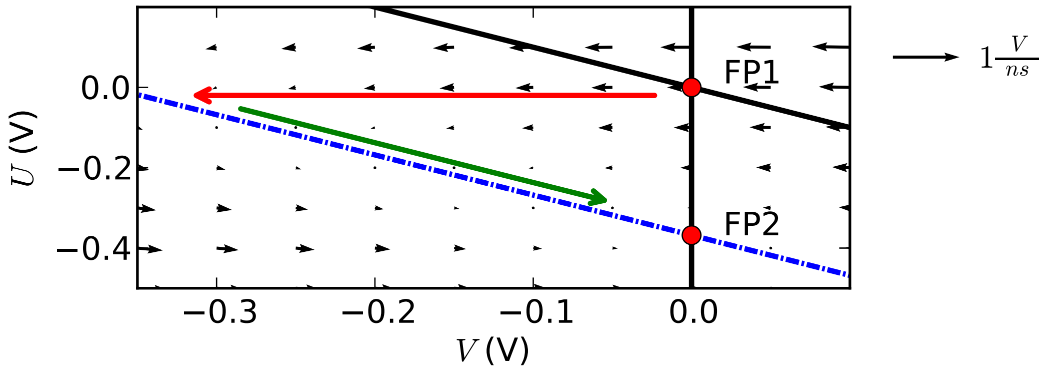


Figure 3.1: Nullclines for $V_0 = 0$ (black solid line) and $V_0 = 0.5\,\mathrm{V}$ (dashed blue line) and velocity field for $V_0 = 0.5\,\mathrm{V}$ (black arrows). The other parameters are $m = 0$, $\gamma = -2$, $\Delta = 1$, $\varepsilon = 0.1$. The fixed point is labeled as FP1 for $V_0 = 0$ and FP2 for $V_0 = 0.5$. The transition from FP1 to FP2 happens along the red arrow and then along the green arrow.

point where the pulse amplitude stays constant. Callan *et al.* [47] considered the phase space with the nullclines of the system equations (Eq. (2.21)) as an analytical tool. Here we assume $V(t-\tau) = V_0$, with V_0 the pulse amplitude of the preceding pulse, leading to

$$\dot{V} = 0 \Rightarrow U = -V + F(V_0), \tag{3.8a}$$

$$\dot{U} = 0 \Rightarrow V = 0, \tag{3.8b}$$

$$F(V_0) = \gamma \frac{g}{d} \left\{ \cos^2 \left[m + d \tanh\left(\frac{V_0}{g}\right)\right] - \cos^2[m] \right\} \tag{3.8c}$$

The idea behind a nullcline analysis is explained in the materials and methods section, Sec. 8.2 on page 77. The phase portrait with the nullclines for $V_0 = 0$ and $V_0 = 0.5\,\mathrm{V}$ is shown in Fig. 3.1 with the velocity field for $V_0 = 0.5\,\mathrm{V}$ superimposed. Most of the time $V_0 = 0$, meaning that the fixed point (V^*, U^*) is asymptotically stable, but when a pulse appears in the delayed argument the delayed voltage is $V_0 = 0.5\,\mathrm{V}$. Then, the nullcline is shifted down, implying a shifted fixed point at $(V, U) = (F(V_0), 0)$. Because of the time scale separation (as shown in the velocity field, *i.e.*, horizontal arrows), the phase trajectory from the previous fixed point to the new one moves first towards the shifted nullcline and subsequently toward the new fixed point along the shifted nullcline. Callan *et al.* [47] argued that the pulses are short in time so that the pulse will approach the shifted nullcline but will no longer travel along the shifted nullcline. When the trajectory arrives at the shifted nullcline, the pulse $V_0 = 0.5$ ends and $V_0 = 0$ again, resulting in the previous fixed point at the origin. This argument is reasonable because the time the trajectory travels along the shifted nullcline is a factor $1/\varepsilon \gg 1$ larger than the time it takes for the trajectory to move from $(0, 0)$ to the shifted nullcline. Since the shifted nullcline has slope -1 the new pulse has the amplitude

$$V_1 = F(V_0). \tag{3.9}$$

This approach makes no statements about the time evolution of V and U but describes instead a map of amplitudes V_n at discrete times $n\tau$ of pulses in a train. Using Eq. (3.9), the map is described by

$$V_{n+1} = F(V_n). \tag{3.10}$$

This map can be analyzed for fixed points, corresponding to limit cycles, with the conditional equation

$$V^* = F(V^*). \tag{3.11}$$

Callan *et al.* [47] found three fixed points for $m = 0$: the quiescent state, $V^*_{\mathrm{s1}} = 0$, the separatrix, V^*_{u}, and the pulsing state, V^*_{s2}. Where 's' and 'u' denote the stability as stable and unstable, respectively. For $m = 0$, the separatrix was found to be [47]

$$V^*_{\mathrm{u}} \approx -g/\gamma d^2. \tag{3.12}$$

If the mean noise amplitude is larger than the separatrix, the system bifurcates into the pulsing state, thereby leading to a noise induced bifurcation. Experimentally, the stable state V^*_{s2} was not observed. Equation (3.12) could be tested with pulses that were injected into the OEO and showed good agreement with experiments and simulations [47].

3.3 FitzHugh-Nagumo model with time delayed feedback

In this thesis we find an analogy between the OEO and the delayed feedback FitzHugh-Nagumo (FHN) model. Here we describe pulse trains that are generated in the delayed feedback FHN model. The FHN model [10, 11] is a generic model for excitability in neuroscience and receives frequent attention in the scientific literature (see Sec. 1.1). The delayed feedback, which was not in the original work of FitzHugh, is modeled by a Pyragas [12, 13] control term that is proportional to the difference between delayed and current state ($\propto [x(t-\tau) - x(t)]$). The Pyragas control is non-invasive, implying a vanishing control force when the dynamics rest at the fixed points (*i.e.* $x(t) = x(t-\tau) = x^*$) or, alternatively, show τ-periodic limit cycles. It was shown that the Pyragas control can stabilize τ-periodic orbits [56]. The introduction of a delayed feedback term means that we have to deal with DDE. The system equations in most publications about the FHN with time delayed feedback are

$$\varepsilon \dot{x} = x - \frac{x^3}{3} - y + K\left[x(t-\tau) - x(t)\right], \tag{3.13a}$$

$$\dot{y} = x + a, \tag{3.13b}$$

where activator, $x(t)$, and inhibitor, $y(t)$, are the system variables, a is the threshold parameter, $0 < \varepsilon \ll 1$ is the time scale separation, and C is the coupling strength. In neuroscience the activator x corresponds to a membrane voltage of neurons.

3.3.1 Pulse train solutions

A typical pulse train in the delayed feedback FHN model is shown in Fig. 3.2a as a time series resulting from numerical simulation [37]. Similar pulse trains are discussed in Chapter 4 in greater detail. The activator, x, gets excited out of the fixed point, A, and increases rapidly to point B where the trajectory decreases slowly to point C. Here another rapid decrease to point D happens followed by a slow increase to the fixed point, A. The phase from B to C can be considered as

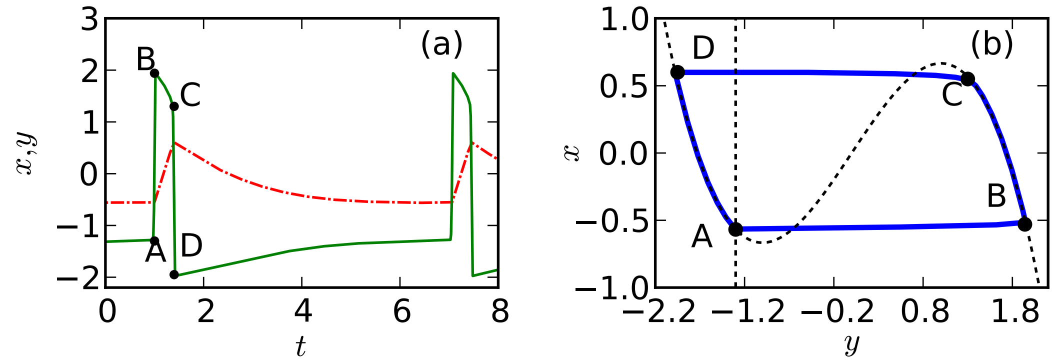


Figure 3.2: Trajectory of the simulated FHN system Eq. (3.13) as (a) a time series of $x(t)$ (green solid line) and $y(t)$ (red dashed line), and (b) a phase portrait of $(x(t), y(t))$ (blue solid line) with the τ-periodic nullclines (black dashed line) superimposed. A, B, C, D mark corresponding points in the time series and phase portrait. The parameters are $\varepsilon = 0.01$, $a = 1.3$, $K = 0.5$, $\tau = 6$.

the pulse high level and the phase from D to A as a refractory phase, where the activator is negative. After the delay, τ, this pulse at time $t - \tau$ excites another pulse at time t. Therefore the trajectory is τ-periodic.[1]

We gain more insight into the dynamics of the pulse train by considering the phase space. The phase space is structured by the nullclines as described in Sec. 8.2 on page 77. Assuming τ-periodic solutions ($V(t-\tau) = V(t)$), the τ-periodic nullclines of Eq. (3.13) are

$$\dot{x} = 0,\ x(t-\tau) = x(t) \Rightarrow y = x - \frac{x^3}{3}, \tag{3.14a}$$

$$\dot{y} = 0 \Rightarrow x = -a. \tag{3.14b}$$

The phase space is shown in Fig. 3.2b with nullclines superimposed. The trajectory gets excited from point A and travels horizontally toward point B. Then, it travels along the right branch of the cubic nullcline to point C where it moves horizontally toward point D after which the trajectory moves along the left branch of the cubic nullcline toward point A.

The phases from A to B and from C to D are fast compared to the phases from B to C and from D to A because of the time scale separation, ε. The trajectory

[1] The period is actually $\tau + \delta$ with the negligibly small excitation time δ [18].

describes a delay-induced limit cycle in phase space. We describe similar pulse trains in Sec. 4.2 in greater detail.

3.3.2 Excitation mechanism

To understand the excitation from point A toward the right branch of the nonlinear nullcline, we need to consider the dynamic nullclines. For a short excitation time, δ, it is $V(t) \neq V(t-\tau)$ because the dynamics need a finite time to respond to the excitation [18]. During the time δ, the τ-periodic nullclines (Eq. (3.14)) are not the nullclines of the system. Instead, the actual nullclines are shifted upward so that point A is located below the cubic nullclines, a position where the velocity field is directed toward the right branch of the nonlinear nullcline. This mechanism was used to calculate the excitation border of pulse train solutions [37].

3.3.3 Bifurcations

The FHN system shows excitability for $|a| > 1$, and, when coupled with a feedback control as in Eq. (3.13), a whole pulse train can be induced [18]. For $a = \pm 1$, a Hopf bifurcation happens. A linear stability analysis to calculate the Hopf bifurcation point analogous to section 3.1 can be found in [18, 35, 37]. For $|a| < 1$ the fixed point lies on the middle branch of the cubic nullcline and is no longer stable. Instead, τ-periodic limit cycle oscillations occur around the cubic nullcline.

The FHN model also shows canard explosions [70, 71] after the Hopf bifurcations for $|a| \lesssim 1$, so that the Hopf oscillations 'explode' in amplitude immediately subsequent to the Hopf bifurcation [72].

4

Pulse train solutions

We experimentally find novel pulse train solutions in the OEO that was introduced in Chapter 2. These dynamics have not been previously reported in the scientific literature, and the key to finding them was to control the parameters m, but especially γ, with high precision. Such high precision was achieved through an implementation of a measurement automation program as described in Sec. 8.3 on page 77.

We saw these pulse trains first by decreasing γ in small steps starting from a chaotic state of the OEO. Later, we realized that these pulses can be excited in our system by sending in pulses as an initial condition when the system is in the resting state for parameters m in the vicinity of $m = 0$ and $m = \pm\pi/2$ and γ in the range $2 \lesssim \gamma \lesssim 3$.

Various dynamics for the OEO have been reported in the literature, such as different oscillatory behaviors [44, 45], breather modes [44, 45] and chaos [43, 47, 67]. Because the OEO is a time delayed system the dynamics are rich and highly dependent on the initial conditions and, therefore, it is not astonishing that novel dynamics can be found by varying the initial conditions.

In this chapter we introduce the pulse train solutions with an experimental waveform, explain how the filter variable, $U(t)$, can be calculated from $V(t)$, show the phase trajectory, compare the experiments to numerical simulations, compare the OEO in this regime to the FHN model, and finally calculate the Hopf bifurcation points of the OEO using a novel method.

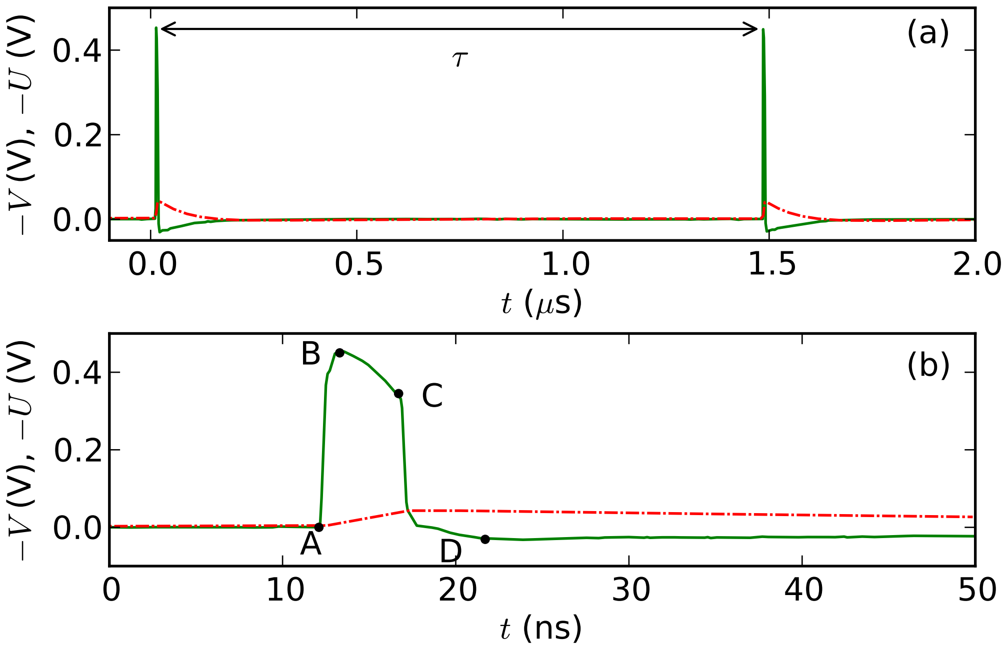


Figure 4.1: Waveform of $V(t)$ (solid green) and calculated $U(t)$ (dashed red) for $m = 0$, $\gamma = -2.54$, $\tau = 1.47\,\mu$s, $g = -0.28\,$V, $d = 1.0$, and filter parameters of filter #4 in table 2.1. (a) Large time scale: the pulse pattern repeats with time delay τ. (b) Smaller time scale showing one pulse in detail; labeled are four points A to D, which are referred to in the text. The variable $U(t)$ was calculated according to Eq. (4.1). The waveform is averaged over 20 measurements.

4.1 Waveform of pulse train solutions

An experimental waveform of the pulse train dynamics is shown in Fig. 4.1. The pulse train was excited with an initial pulse, while the dynamics were in the steady state (fixed point). This initial pulse was sent in with the pattern generator as a square pulse (rise and fall time $< 50\,$ps) with a width of $1\,$ns and amplitude of $0.5\,$V.

The waveform, $V(t)$, shown in Fig. 4.1a over a time interval of $2\,\mu$s, is a stable pulse train. The pulses in the train have a negative sign because, in this case, the filter inverts the measured voltage. We plot $-V(t)$ to aid interpretation. It can be seen that the period of the pulse train is τ and that right after each pulse the voltage has the opposite sign of the pulse a phase which we call refractory period.

In Fig. 4.1b the waveform of a single pulse out of the train is shown in greater detail, with four points A to D marked on the pulse. By reference to these four points, we describe the characteristic shape of the pulse. The voltage increases rapidly from zero (point A) to the maximum amplitude (point B), then it decreases slowly to point C, before it rapidly drops past zero to point D. After the pulse, the voltage slowly recovers back to zero, the resting state of the system (the fixed point). Thus, the segment from A to B corresponds to the rise, B to C to the high level, and C to D to the fall of the pulse. The segment from D back to the fixed point is the refractory phase analogous to the time series of the FHN model in Sec. 3.3. Another pulse with the same shape occurs after the time delay, τ.

4.2 Experimental phase portrait and nullclines

To analyze the trajectory in phase space, we need to know both system variables, $V(t)$ and $U(t)$, but we can measure only the voltage, $V(t)$. In the next section, we show how the filter variable, $U(t)$, can be calculated from the voltage, $V(t)$.

4.2.1 Calculating the filter variable U

Using Eq. (2.21), we can calculate $U(t)$ from $V(t)$ except for an offset $U(0)$, resulting in

$$U(t) = \Delta\varepsilon \int_0^t V(s)\,\mathrm{d}s - U(0). \tag{4.1}$$

For pulsing solutions, we have already seen that the system relaxes back to the fixed point $(U^*, V^*) = (0, 0)$ after a pulse. This behavior is used to determine the offset $U(0)$ for pulse train solutions. For other dynamic states in this thesis, we determine $U(0)$ by lining up the phase trajectory with the nullclines that we derive later on.

When calculating the integral in Eq. (4.1), a constant voltage must be subtracted from V, so that the mean of V becomes zero: $V \to V - \langle V \rangle$. This constant voltage is typically small (for example, $\langle V \rangle = 1.3\,\mathrm{mV}$ for the waveform in Fig. 4.1) and is erroneously produced either by the oscilloscope or the photodetector. The photodetector should not produce a DC voltage because it is AC-coupled. A DC voltage in V does not change the dynamics because the MD is AC-coupled and does not amplify constant voltages. We use Simpson's rule for the numerical integration.

Figure 4.1 also shows the resulting calculated filter variable $U(t)$. When V is excited out of the resting state $(V^*, U^*) = (0, 0)$ (points A, B) U slowly rises until the pulse in V ends (points C, D), and when V is the refractory phase (after point D) U decreases slowly. Finally, both variables V and U approach zero, the fixed point, before the trajectory gets excited again.

4.2.2 τ-periodic nullclines

As seen earlier in Sec. 3.2, nullclines are a useful tool to study phase space dynamics. The idea behind a nullcline analysis is explained in the materials and methods section, Sec. 8.2 on page 77. We utilize τ-periodic nullclines to analyze the OEO for τ-periodic pulse trains. Therefore, we assume $V(t)$ to be τ-periodic (*i.e.* $V(t-\tau) = V(t)$) so that the nullclines are not dependent on the delayed variable. The τ-periodic nullclines of Eq. (2.21) are

$$\dot{V} = 0 \Rightarrow \; U(V) = -V + \frac{\gamma g}{d}\left\{\cos^2\left[m + d\tanh\left(\frac{V}{g}\right)\right] - \cos^2 m\right\} \tag{4.2a}$$

$$\dot{U} = 0 \Rightarrow \; V = 0, \tag{4.2b}$$

and are shown in Fig. 4.2a as dashed lines. The nonlinear nullcline Eq. (4.2a) possesses both a minimum and a maximum. The intersection point of both nullclines, point A in Fig. 4.2a, is the fixed point $(U^*, V^*) = (0, 0)$.

For τ-periodic solutions, both nullclines structure the phase space as follows.

- For $V(t)$ left (right) to the $(\dot{U} = 0)$-nullcline (Eq. (4.2b)) (*i.e.* left (right) half space), the velocity field is directed downward (upward), and for $V(t)$ on this nullcline there is no component in the U-direction.
- For $U(t)$ above (below) the $(\dot{V} = 0)$-nullcline (Eq. (4.2a)), the velocity field is directed left (right), and for states on this nullcline there is no component in the V-direction.

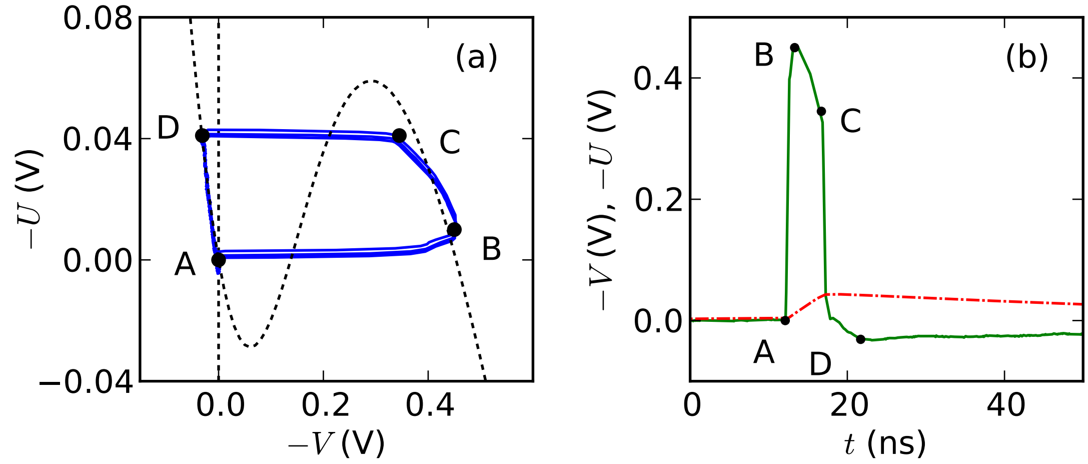


Figure 4.2: (a) Phase portrait $(V(t), U(t))$ (solid blue line) of the experimental time series shown in (b) with the τ-periodic nullclines superimposed (black dashed). (b) Fig. 4.1b repeated. Parameters, also those for the nullclines, as in Fig. 4.1. The variable $U(t)$ was calculated according to Eq. (4.1). The waveform is averaged over 20 measurements. Labeled are four corresponding points A to D on both plots.

We are able to make these statements because the velocity field is continuous and zero in the respective direction on the nullcline (see Sec. 8.2).

In the dynamics, the left and right branch of the nonlinear nullcline (Eq. (4.2a)) are attracting in V and the middle branch is repulsive. Combined with the time scale separation (Sec. 2.2.3), these statements are very useful in the description of the τ-periodic dynamics as follows.

4.2.3 Experimental phase portrait

Figure 4.2a shows the phase portrait (V, U) of the experimental time series in Fig. 4.2b and Fig. 4.1. Four points, A-D, are indicated on the phase plane trajectory and correspond to the points A-D in the time series Fig. 4.2b. A pulse at time $t - \tau$ excites the trajectory starting at the fixed point (point A) to the right branch of the nullcline (point B). The excitation mechanism is explained in Sec. 4.2.4. The trajectory travels along the right branch of the nullcline toward point C where, due to the delayed feedback, the trajectory moves back to the left branch of the nullcline (point D). Finally, the trajectory travels along the left branch of the nullcline back toward the fixed point. The delay induced limit cycle repeats after the delay τ following the same scheme.

The direction of the velocity field with respect to the nullcline (see Sec. 4.2.2) explains the path. It is directed upward along the right branch of the nullcline (path from B to C) and downward along the left branch of the nullcline (path from D to A). Between the middle branch and the right branch of the nullcline the velocity field is directed right (path from A to B), and between middle and left branch the velocity field points left (path from C to D). The trajectory moves along the outer branches of the nullcline because they are attractive in V and leaves these branches due to time delayed feedback as described in Sec. 4.2.4 The filter parameter, $0 < \varepsilon \ll 1$, acts as a time scale separation between changes in V and U as explained in Sec. 2.2.3. For transitions from A to B and from C to D, mostly V changes and for the transitions from B to C and from D to A, mostly U changes. Thus, the former transitions happen on the time scale of V (fast) and the latter on the time scale of U (slow). V is on the order of $1/\varepsilon$ faster than U. This explains why the transitions from A to B and from C to D proceed horizontally in phase space and quickly in the time series, leading to a pulse shape.

4.2.4 Excitation mechanism with dynamic nullcline

The excitation mechanism for pulse trains in an OEO can be explained with dynamic nullclines analogously to the excitation mechanism in the FHN system in Sec. 3.3.2. When one pulse excites the next pulse after the time, τ, the second pulse needs a short excitation time, δ, to become excited. For the time δ, $V(t) \neq V(t-\tau)$, with the effect that the nullcline (Eq. (4.2a)) shifts up, so that the nullcline is located above point A. Below the nullcline, the velocity field is directed right, with the result that the trajectory gets excited toward the right branch.

The excitation time, δ, changes the period of the pulse train to $\tau + \delta$, but δ is small compared to τ, so that we neglect it. The excitation time, δ, was studied by Dahlem *et al.* [18] for the FHN model.

4.2.5 Connection to the one-dimensional map

The τ-periodic nullclines can be seen as an extension to the three fixed points of the one-dimensional map in Sec. 3.2 [73]. When inserting $U = 0$ into the nullcline Eq. (4.2a), we arrive at

$$V = \frac{\gamma g}{d} \left\{ \cos^2 \left[m + d \tanh \left(\frac{V}{g} \right) \right] - \cos^2 m \right\} , \tag{4.3}$$

the conditional equation for the fixed points of the one-dimensional map (Eq. (3.11)). Therefore the fixed points of the one-dimensional map correspond to the nullcline for $U = 0$.

When γ is smaller than a threshold, γ_c, two fixed points of the map vanish (likely in a saddle node bifurcation) [47], which is expressed in the nullclines as the maximum becoming negative. In this case, the right branch of the nullcline is lower than 0, and, therefore, pulses can not be excited to it. Hence, γ_c describes a threshold for the minimal γ necessary for the pulse trains to occur. The unstable fixed point of the one-dimensional map, $V_\mathrm{u}^* \approx -g/\gamma d^2$ (Eq. (3.12)), describes the excitation border. The unstable fixed point, V_u^*, and the critical gain, γ_c, are approximated in [47]. Callan *et al.* [47] also calculated a gain γ (called γ_th) as a function of the noise intensity D for which pulses in the transition to chaos are stochastically excited. It is interesting to study whether the pulse trains in this thesis can also be stochastically excited and whether the threshold, γ_th, stays the same.

4.3 Numerical studies of the pulse train solutions

By numerical simulation of the model equations, Eq. (2.21), we can arrive at approximate solutions. The numerical simulation method is described in Sec. 8.1.2 on page 76. To acquire a pulse train corresponding to the experimental waveform in Fig. 4.1, we simulate the system equations with the same parameter values (see caption in Fig.4.1). As an initial function, we use a Gaussian function with width 2 ns and amplitude 1 V. The resulting waveform and phase portrait, together with the nullclines (Eq. (4.2)), is shown in Fig. 4.3. Here, the system equations were simulated for a preceding time of 300 μs, implying the assumption

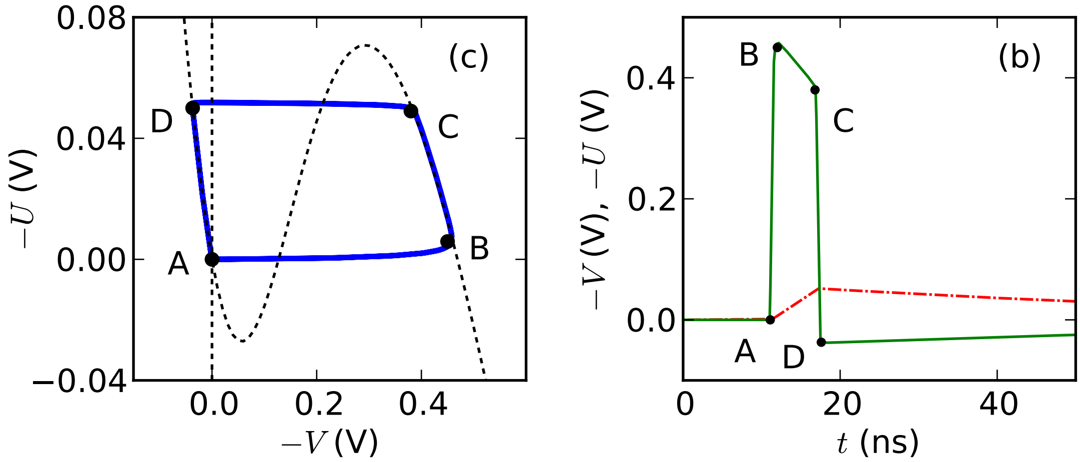


Figure 4.3: Numerical simulation of the system equations with parameters as in Fig. 4.1. (a) Phase portrait of $(U(t), V(t))$ (solid blue line) with nullclines (Eq. 4.2) (dashed black line). (b) Waveform of $V(t)$ (solid green line) and $U(t)$ (dashed red line). Four respective points A-D are labeled on both plots.

that a stable limit cycle was reached. That the simulation time was chosen long enough can also be deduced from the phase portrait, Fig. 4.3a, which shows a trajectory of 100 pulses that overlap each other (*i.e.*, the pulse shape is no longer changing).

4.3.1 Comparison between experiments and simulations

The data from the numerical simulations (Fig. 4.3) is very similar to the experimental data (Fig. 4.2). The time series of the numerical simulation looks cleaner around point D, and the phase trajectory of the simulation looks cleaner around point C compared to the experimental data. The difference between experiments and simulations could be due to the nonideal bandpass characteristics of the experimental components (Sec. 2.1.3.3) or the nonideal and frequency dependent amplifying characteristics of the MD (Sec. 2.1.2).

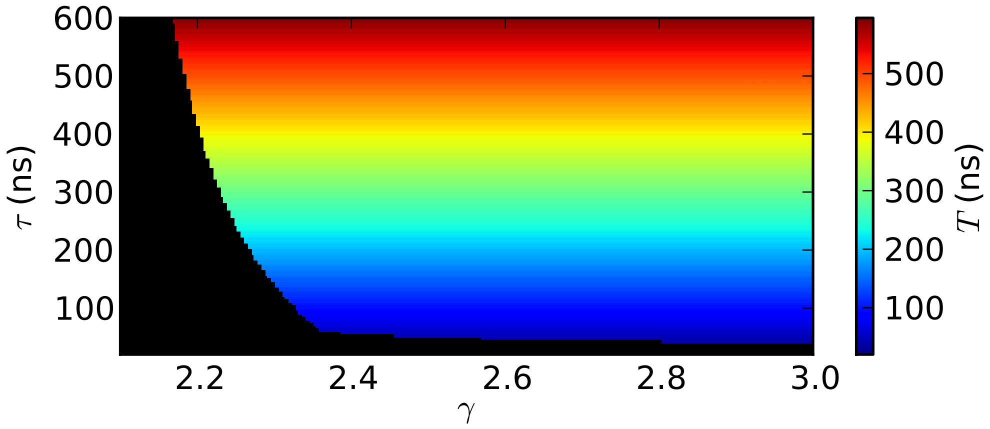


Figure 4.4: Numerical simulation of the system equations for different parameters γ and τ and fixed parameters $m = 0$, $g = -0.28\,\mathrm{V}$, $d = 1.0$. The period of pulse trains is color coded. Regions where the initial pulse decays to zero and the dynamics rest in the fixed point are color black.

4.3.2 Excitation border

In Fig. 4.4, we simulate the model (Eq. (2.21)) for different parameters γ and τ and recorded the period, T, of the pulse train for a bias point $m = 0$. The region in parameter space, where the initial pulse decays back to the fixed point and does not excite a stable pulse trains, is colored black. It can be seen from the color code that the period, T, of pulse trains equals the time delay, τ. It is remarkable that a similar parameter plane exists for the time delayed FHN model [35, Fig. 10]. For the time delayed FHN model, the excitation border, the border between pulse trains and resting in the fixed point, could be calculated analytically [37].

For the acquisition of the parameter plane, we simulate the model equations with the Adams-Bashforth-Moulton algorithm, described in Sec. 8.1.1. We use a constant time step of $\mathrm{d}t = 50\,\mathrm{ps}$, simulate for a time of $30\,\mu\mathrm{s}$, and initialize the pulse trains with an initial Gaussian pulse of amplitude $0.5\,\mathrm{V}$ and pulse width of $10\,\mathrm{ns}$.

4.4 Pulse trains for different bias points m

The nullclines change with parameters m and γ. In this section, we study how the nullclines and the resulting pulse trains change when m is altered. We find that a change in m can change the dynamics from positive to negative pulses and from excitable to oscillatory behavior. We assume a moderate gain, $2 \leq |\gamma| \leq 3$, to exclude chaotic dynamics.

4.4.1 Nullcline analysis

The nullclines and the velocity field for different m is shown in Fig. 4.5 on page 43. The nullclines in Fig. 4.5a with $m = 0$ correspond to the pulse trains shown in Fig. 4.2. For $m = 0.1$ (Fig. 4.5b), the maximum of the nullcline has shifted downward below zero, with the result that the right branch was too low for pulses to get excited to this branch, because pulses start from the fixed point, $V^* = U^* = 0$, and move only horizontally (due to the time scale separation). The system is no longer excitable, and staying at the fixed point is the only possible dynamic scenario. When increasing m from 0 to 0.1, the pulses narrow until the system is no longer excitable. Changing m from 0 to negative values makes the pulses wider (Fig. 4.5c) until m is so negative that both nullclines intersect at the minimum of the nullcline (Fig. 4.5d). At this point the system becomes oscillatory as described below.

In Fig. 4.5e ($m = 1.3$), the extrema of the nullcline moved from positive V to negative V. The system is still not excitable because no branch of the nullcline exist on the left or right of the fixed point for $U = 0$.

Both extrema shift down for $m = \pi/2$ in Fig. 4.5f. For this value of m, the system is excitable again this time with negative pulses that can be described by the nullclines in manner analogous to the positive pulses in Sec. 4.2.3. We also see the same nullclines for $m = -\pi/2$ because of the symmetry in the nullclines[1].

For $m \approx -0.2$ (Fig. 4.5d), the minimum of the nullcline coincides with the fixed point, describing the Hopf bifurcation point. Shifting the minimum of the

[1] For $m = \pm\pi/2$ the nonlinear nullcline can be written as $U(V) = -V + \gamma G/(d) \sin^2 \left[d \tanh\left(G^{-1}V\right)\right]$.

nullcline past the fixed point, changes the dynamics to be oscillatory. The Hopf bifurcation point separates excitatory and oscillatory regimes and is quantitatively described by Eq. (3.6). The nullcline description is used in Sec. 4.6 to calculate the Hopf bifurcation point. In Chapter 6, we add further detail to the nullcline description in the oscillatory regime.

4.4.2 Intuitive interpretation of positive and negative pulses

For m in the vicinity of $m = 0$ and $m = \pi/2$, we see positive and negative pulses, respectively (assuming positive γ). We analyze the transmission of the MZM for pulse train solutions and these two values of m. The transmission of the MZM, using V_{in} as the input to the MD, is proportional to (Eq. (2.3))

$$P_{\text{out}}/P_{\text{in}} \propto \gamma \cos^2 \left[m + d \tanh \left(V_{\text{in}}/g\right)\right]. \tag{4.4}$$

For $m = \pi/2$, let us first consider the time when the spike is not in the MZM, but elsewhere in the oscillator. Then, the floor of the spike is coming into the MZM and the transmission $P_{\text{out}}/P_{\text{in}} = 0$ (for $m = \pi/2$ and $V_{\text{in}} = 0$), hence the MZM is shut and allows no laser light to pass through. When the pulse (high-level) arrives at the rf port of the MZM, the transmission of the MZM changes to a high value and shifts back to zero after the electrical pulse ends. Therefore, the MZM is open for the pulse width, producing a bright pulse in the optical part. Because both, the photodetector and MD, are inverting devices we measure negative pulses with the oscilloscope and send positive pulses into the MZM. Additionally, most filters that we use are inverting so that we measure positive electric pulses with the oscilloscope for $m = \pi/2$.

For $m = 0$, the MZM is open when no pulse is coming in. Because the photodetector is AC-coupled, no voltage can be measured after the photodetector when constant laser power is incident on it. A pulse that enters the MZM closes the MZM for approximately the pulse width, after which the MZM opens. Therefore, we observe a short negative dip in the optical power, which is converted to a positive pulse by the inverting AC-coupled photodetector. Most of the filters invert the signal which is then measured as a negative pulse with the oscilloscope. Such optical power dips are called dark pulses. Because most experimental work on fiber optics deals with bright pulses rather than dark pulses [74] bright pulses are considered to be technologically more attractive.

4.5 Analogy to the FitzHugh-Nagumo model

The delayed feedback FHN model (*i.e.*, FHN model with Pyragas feedback control) was introduced in Sec. 3.3.

Even though the mathematical expression for the τ-periodic nullclines of the OEO (Eq. (4.2)) and the nullclines of the FHN model (Eq. (3.14)) are different, both have similar shapes (*e.g.* the τ-periodic nullclines of the OEO oscillator in Fig. 4.2a and the τ-periodic nullclines of the delayed feedback FHN model in Fig. 3.2b look similar). Furthermore, both systems possess a time scale separation.

In both systems, the phase space trajectory can be described using nullcline analysis and the time scale separation in a similar way. Due to the similar shape of the nullclines, the waveforms of the pulse trains also look very similar (*e.g.*, compare the waveforms of pulse trains in the OEO in Fig. 4.1 and pulse trains in the FHN model in Fig. 3.2a). Another analogy between the FHN model and OEO is that trains of positive and negative pulses exist in both systems. The FHN system is excitable for $|a| > 1$ when both nullclines intersect on the left or right branch of the nonlinear nullcline, similar to the OEO, as discussed in Sec. 4.4.1. The strong similarity between the OEO and the FHN model, a generic model for excitability, implies that calling the pulse train regime excitable in the OEO is warranted. Analogous to the FHN system the excitability of the OEO is of type II.

Some findings for the commonly studied delayed feedback FHN model (see work cited in Sec. 1.1), might also extend to the OEO. In that manner, it is interesting to study whether coherence resonance occurs in OEOs [36].

The FHN system exhibits Hopf bifurcations at $a = \pm 1$. Specifically, for $|a| > 1$ the FHN system is excitable and for $|a| < 1$ it is oscillatory. The Hopf bifurcation happens when the $(\dot{y} = 0)$-nullcline passes the minimum of the $(\dot{x} = 0)$-nullcline. Using the nullclines, the Hopf bifurcation points in the OEO can be derived analytically as shown in the next section. The FHN system exhibits another bifurcation, the canard explosion, which occurs immediately subsequent to the Hopf bifurcation [70, 71]. Whether a canard explosion happens in the OEO is unclear at this point and warrants further research.

4.6 Hopf bifurcation

The Hopf bifurcation separates the excitable and oscillatory regimes in the OEO and is already well explored for the OEO [41, 44, 45, 47, 68]. We summarize the mathematical derivation of the Hopf bifurcation point in Sec. 3.1. Using rather complicated methods including approximate numerical methods to solve implicit equations (Sec. 3.1), it was found that the Hopf bifurcation point is given by

$$\gamma_{\mathrm{H}} \approx \pm 1/\sin(2m). \tag{4.5}$$

When m and γ are opposite in sign, nullcline analysis provides a much easier, completely analytic, and exact method. For m and γ of the same sign, we use approximate nullclines.

The Hopf bifurcation is manifested in the nullclines when the minimum of the nonlinear nullcline shifts over the $(\dot{U}=0)$-nullcline. We also know that the intersection point of both nullclines is always the fixed point $(U^*, V^*) = (0,0)$. First, let us consider the extrema of the nonlinear nullcline $U(V)$ (Eq. (4.2))

$$U'(V) \stackrel{!}{=} 0 \Rightarrow \tag{4.6}$$

$$0 = -1 - \frac{\gamma g}{d} 2\cos\left[m + d\tanh\left(V/g\right)\right]\sin\left[m + d\tanh\left(V/g\right)\right]\frac{d}{g}\operatorname{sech}^2\left(V/g\right). \tag{4.7}$$

The Hopf bifurcation happens when one extrema coincides with the fixed point $V^* = 0$. Therefore, we can set $V = 0$ in Eq. (4.7), which gives

$$1 = -2\gamma_{\mathrm{H}}\cos\left(m\right)\sin\left(m\right), \tag{4.8}$$

leading to[1]

$$\gamma_{\mathrm{H}} = \frac{-1}{\sin\left(2m\right)}. \tag{4.9}$$

[1] $2\cos(m)\sin(m) = \sin(2m)$.

The other Hopf bifurcation point separates pulsing solutions from 2τ-periodic solutions that we describe later in Sec. 6.1.5. There, we find that the nullclines of these solutions are approximately given by

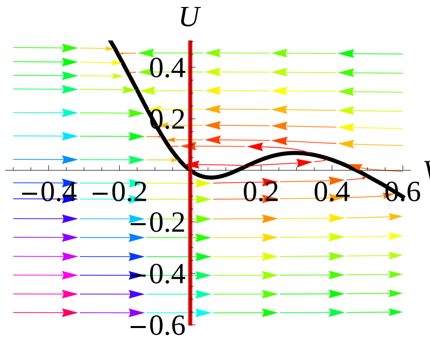
$$U(V) = -V + \frac{\gamma g}{d}\left\{\cos^2\left[m + d\tanh\left(-V/g\right)\right] - \cos^2 m\right\} \tag{4.10a}$$

$$V = 0, \tag{4.10b}$$

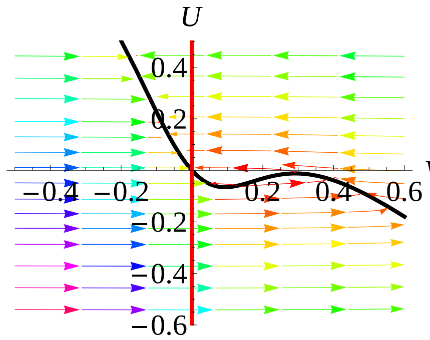
which leads, by analogous derivation to the one above, to

$$\gamma_{\mathrm{H}} = \frac{+1}{\sin\left(2m\right)}. \tag{4.11}$$

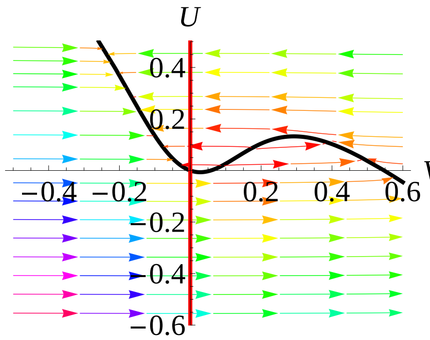
For both cases we arrive at the same Hopf condition as described in the scientific literature but with a shorter derivation and without the burden of approximate numerical calculations. For the first Hopf point we do not need to use approximations. The common derivation of the Hopf point, described in Sec. 3.1, uses a rather complicated numerical search for the smallest solution of a transcendental equation.



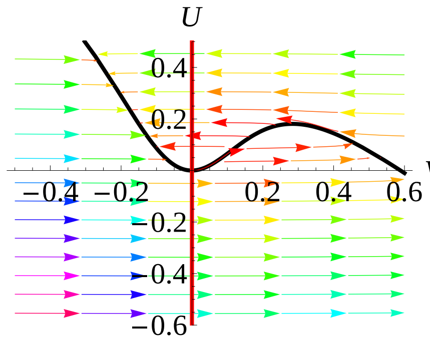
(a) $m = 0$



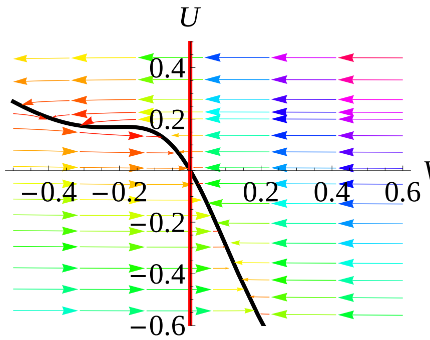
(b) $m = 0.1$



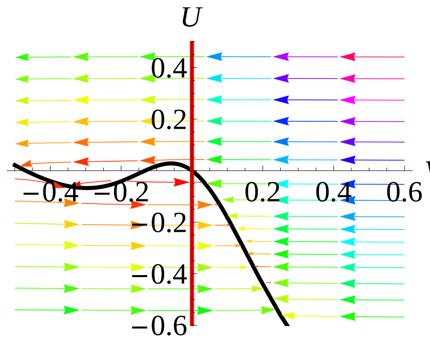
(c) $m = -0.1$



(d) $m = -0.2$



(e) $m = 1.3$



(f) $m = \pi/2$

Figure 4.5: The nullclines and the velocity for different m. The ($\dot{V} = 0$)-nullcline (nonlinear nullcline) in black and the ($\dot{U} = 0$)-nullcline in red. The velocity field is indicated with arrows for $\varepsilon = 0.01$ and $\Delta = 1$, where blue, green and red arrows denote high, medium and low velocities, respectively. $\gamma = 2.6$, $g = -0.28\,\mathrm{V}$, $d = 1.0$ and m as shown below the figures.

5
Controlling the pulse width

Using the nullclines from the previous chapter, we find a quantitative dependency between pulse widths of the pulse train solutions and the low frequency cutoff, ω_-, of the bandpass filter in the form of a power law with exponent -1, a relation which allows the pulse width to be tuned by adjusting ω_-. Testing this analytical dependency experimentally, we find that we can tune the pulse width over a range of three orders of magnitude. We find good agreement among experiment, simulation, and theory.

Periodic trains of optical pulses become more useful at low pulse width because their frequency spectrum has a broader bandwidth with decreasing pulse width. Therefore, the ability to tune pulse widths narrower is very desirable [75].

Illing and Gauthier [41] discussed how the presence of AC coupled elements alters the dynamics of a delayed system. Here, we show that the pulse width is inversely proportional to the low frequency cutoff, a finding that extends their work.

5.1 Power law of the pulse width

The analytic considerations in this section are guided by Fig. 4.2. The experimental pulse width is measured as width of a pulse at a voltage corresponding to half the amplitude of the pulse (pulse width at half max). In the theoretical calculation, we approximate this pulse width, w, as the time the trajectory needs to move from point B to point C ($T_{\mathrm{B}\to\mathrm{C}}$) in Fig. 4.2, the duration of the excited phase,

$$w = T_{\mathrm{B}\to\mathrm{C}} = \int_{\mathrm{B}}^{\mathrm{C}} \mathrm{d}t. \tag{5.1}$$

Fast transitions from A to B and from C to D compared to the rather slow passage from B to C (due to the time scale separation) allow for this approximation. We note that this approximation might worsen for filters with weaker time scale separation $1/\varepsilon$.

We substitute $\mathrm{d}t = \mathrm{d}U/\dot{U}$ in Eq. (5.1) and adjust the integral boundaries of the points $\mathrm{B} = (U_{\mathrm{B}}, V_{\mathrm{B}})$ and $\mathrm{C} = (U_{\mathrm{C}}, V_{\mathrm{C}})$

$$w = \int_{U_{\mathrm{B}}}^{U_{\mathrm{C}}} \frac{1}{\dot{U}}\,\mathrm{d}U. \tag{5.2}$$

From Eq. (2.21b), we also substitute $\dot{U} = \Delta\varepsilon V$ into Eq. (5.2). Furthermore, the path from B to C in Fig. 4.2a is approximately given by the nullcline Eq. (4.2a). Therefore we express V through the inverted nullcline $V(U)$

$$w = \frac{1}{\Delta\varepsilon}\int_{U_{\mathrm{B}}}^{U_{\mathrm{C}}} \frac{1}{V(U)}\,\mathrm{d}U = f(\omega_-,\omega_+)c(m,\gamma). \tag{5.3}$$

The pulse width, w, separates into a frequency dependent and a parameter (m, γ) dependent function[1]

$$f(\omega_-,\omega) := \frac{1}{\Delta\varepsilon} = \frac{\omega_+ - \omega_-}{\omega_+\omega_-} \approx \frac{1}{\omega_-}, \tag{5.4}$$

$$c(m,\gamma) := \int_{U_{\mathrm{B}}}^{U_{\mathrm{C}}} \frac{1}{V(U)}.\,\mathrm{d}U \tag{5.5}$$

The approximation of Eq. (5.4) holds for $\omega_- \ll \omega_+$. Later we show numerically that the integral borders, U_{B} and U_{C}, do not vary with the cutoff frequencies ω_- and ω_+.

After fixing the parameters m and γ fixed in Eq. (5.3) and using the approximation of Eq. (5.4), we find that the pulse width, w, varies with the low frequency cutoff, ω_-, by way of a power law of exponent -1

$$w = c\cdot(\omega_-)^{-1}. \tag{5.6}$$

The pulse width is, therefore, inversely proportional to ω_- with constant of proportionality, $c(m,\gamma)$, as defined by Eq. (5.5).

[1] We consider d and g to be constants.

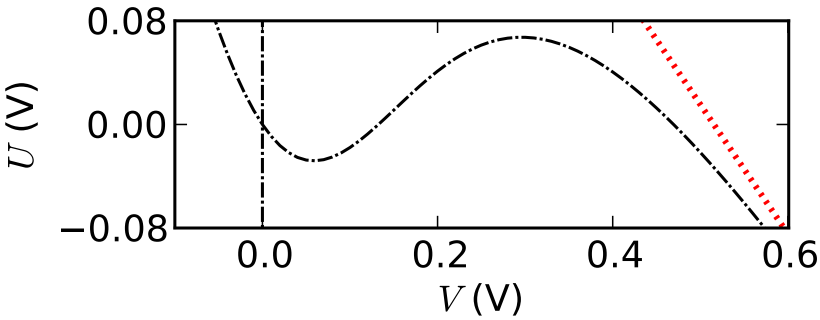


Figure 5.1: Nullclines of Eq. (4.2) (black dashed line) with approximation of the right branch of the nullcline in Eq. (5.7) (red dotted line). The parameters are $\gamma = 2.6$, $m = 0$, $g = -0.28$, $d = 1.0$.

5.1.1 Approximation of the scale factor

The idea that $c(m, \gamma)$ can be approximated in the following way originated from Eckehard Schöll.

In the excited phase (phase from B to C in Fig. 4.2a), we consider $|V| \gg |g|$ (here: $V/g \approx 1.4$), so that $\tanh(V/g) \approx 1$ (involving an error of 10%). Physically, this approximation assumes that the MD is saturated by the high level of a pulse. Inserting the tanh approximation into the nullcline (Eq. (4.2a)) leads to a rather simple approximation of the right branch of the nonlinear nullcline

$$U(V) = -V + F, \tag{5.7}$$

with a constant F, embodying the approximated nonlinearity

$$F = \frac{\gamma g}{d} \left\{ \cos^2 [m + d] - \cos^2 m \right\}. \tag{5.8}$$

Figure 5.1 shows the nullclines (Eq. (4.2)) and the approximated right branch of the nonlinear nullcline (Eq. (5.7)). One can see that the approximation involves an error in $V < 20\%$ in the pulse amplitude. The approximated nullcline (5.7) can be inverted

$$V(U) = -U + F. \tag{5.9}$$

We insert Eq. (5.9) into Eq. (5.5) and use $U_B \approx U_A = U^* = 0$ which accounts for the time scale separation (*i.e.*, U changes slowly across the path from A to B in

Fig. 4.2). This gives (with log as the natural logarithm)

$$c(m, \gamma) = \int_0^{U_\mathrm{C}} \frac{1}{-U + F} \mathrm{d}U = \log(F) - \log(F - U_\mathrm{C}). \tag{5.10}$$

The value of U_C still has to be determined from simulation or experiment. For $m = 0$, we use $\cos^2(d) + \sin^2(d) = 1$ to simplify Eq. (5.10) to

$$c(m, \gamma) = -\log\left(\frac{F - U_\mathrm{C}}{F}\right) = -\log\left(1 + \frac{U_\mathrm{C} d}{\gamma g \sin^2 d}\right). \tag{5.11}$$

From the simulation, we find $U(C) = 0.05$ for $\gamma = \pm 2.6$ and $m = 0$ independently of the filter parameters, ω_- and ω_+, leading to

$$c(m = 0, \gamma = \pm 2.6) = 0.1. \tag{5.12}$$

5.1.2 Parameter dependency of the scale factor

In Sec. 4.4, we have already discussed how the nullclines change with parameter m, leading to changes in the integral path that affects the integral defining $c(m, \gamma)$ (Eq. (5.5)). Increasing γ causes the minimum and maximum of the nonlinear nullcline to diverge, thereby leading to a longer path from A to B and hence a greater value of $c(m, \gamma)$. When m is close to the Hopf bifurcation, the fixed point is close to the minimum or maximum of the nullcline, leading to a longer path from B to C, hence a greater value of $c(m, \gamma)$. A greater value of $c(m, \gamma)$ means wider pulses so that the following qualitative statements hold for the parameter dependency of the pulse width, w.

- Increasing the gain, γ, leads to wider pulses.
- Parameter m close to the Hopf bifurcation leads to wider pulses.

In Fig. 5.2, we simulated the model equations (Eq. (2.21)) for different γ and m and recorded the pulse width. This parameter space was calculated similarly to that in Sec. 4.3.2 where the numerical parameters can be found. The nullclines for the three points (a), (b), (c) that are labeled on the parameter plane are shown in Fig. 4.5a,b,c, respectively. Comparing the nullclines to the respective

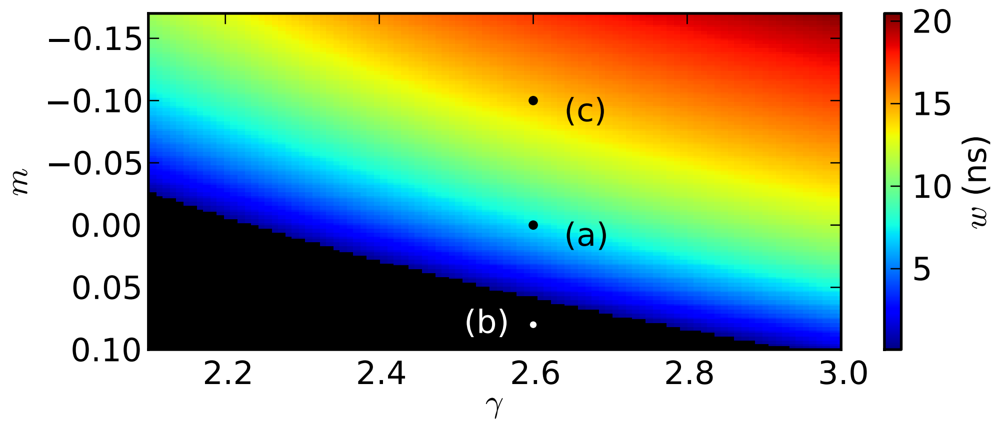


Figure 5.2: Numerical simulation of the system equations for different parameters m and γ for fixed parameters $\tau = 300\,\text{ns}$, $g = -0.28\,\text{V}$, $d = 1.0$. The pulse width of trains is color coded. Regions where the initial pulse decays to zero and we measure only the fixed point are colored black. The plots from Fig. 4.5a,b,c, showing the respective nullclines, are labeled.

points in the parameter plane, it can be seen that the pulse width is large when the path along the left branch of the nullcline is long.

The color coded pulse width is largest when γ is large and when m is close to the Hopf bifurcation, a scenario that occurs when $m_{\mathrm{H}} = -\arcsin(1/\gamma)/2 \approx -0.17$ for $\gamma = 3$ (Eq. (3.7)). Therefore, we can consider the statements above as valid.

5.1.3 Alternative approach

We can derive Eq. (5.3) with a different approach that was similarly undertaken for the FHN model [35, 36]. This leads to an explicit integral for the function $c(m, \gamma)$.

We abbreviate $F(\cdot)$ as the feedback function

$$F(V) = \gamma g/d \left\{\cos^2\left[m + d\tanh(V/g)\right] - \cos^2 m\right\}. \tag{5.13}$$

For this approach, we use the equation of the nullcline Eq. (4.2a) and take the derivative with respect to time t:

$$\dot{U} = \dot{V}\left(F'(V) - 1\right). \tag{5.14}$$

Performing calculations in a manner similar to those above and using Eq. (5.14), leads to

$$w = \int_{V_\mathrm{B}}^{V_\mathrm{C}} \frac{1}{\dot{V}}\,\mathrm{d}V = \int_{V_\mathrm{B}}^{V_\mathrm{C}} \frac{(F'(V)-1)}{\dot{U}}\,\mathrm{d}V = \frac{1}{\Delta\varepsilon}\int_{V_\mathrm{B}}^{V_\mathrm{C}} \frac{(F'(V)-1)}{V}\,\mathrm{d}V \tag{5.15}$$

$$= f(\omega_-,\omega_+)c(m,\gamma), \tag{5.16}$$

which is the same equation as Eq. (5.3) but with an explicit expression for the integrand

$$c(m,\gamma) = \int_{V_\mathrm{B}}^{V_\mathrm{C}} \frac{(F'(V)-1)}{V}\,\mathrm{d}V. \tag{5.17}$$

From Eq. (5.3), we calculated $c(m,\gamma)$, using approximate nullclines that could be inverted in contrast to the exact nullcline that cannot be inverted. Here, we do not need to invert the nullcline, an advantageous situation. We are still unable to calculate the integral in Eq. (5.17) without approximations because the nullcline involves complicated functions.

5.2 Power law in the simulations

The numerical simulations provide a way to test the power law given in Eq. (5.6). To do so, we keep the high frequency cutoff, $\omega_+/(2\pi) = 10\,\mathrm{GHz}$, fixed and simulate for different low frequency cutoffs, ω_-. We determine the pulse width, w, in the simulation as the pulse width at half of the pulse amplitude. We keep the other parameters $m = 0$ and $\gamma = 2.6$ fixed. The simulation method is explained in Sec. 8.1.2 on page 76.

We acquired wide pulses ($w = 161\,\mathrm{ns}$) for small ω_- as can be seen in Fig. 5.3a,b for $\omega_-/(2\pi) = 100\,\mathrm{kHz}$, and narrow pulses ($w = 1.68\,\mathrm{ns}$) for large ω_- as can be seen in Fig. 5.3d,e for $\omega_-/(2\pi) = 10\,\mathrm{MHz}$. The time scales of both pulses differ by a factor of 100 as the low frequency cutoffs, ω_-, do. This provides some initial evidence that condition Eq. (5.6) holds.

Dependence on the time delay, τ The time delay determines the period of the pulses and was also chosen differently in both time series. It was verified that the time delay, if large enough, had no influence on the pulse shape. After the

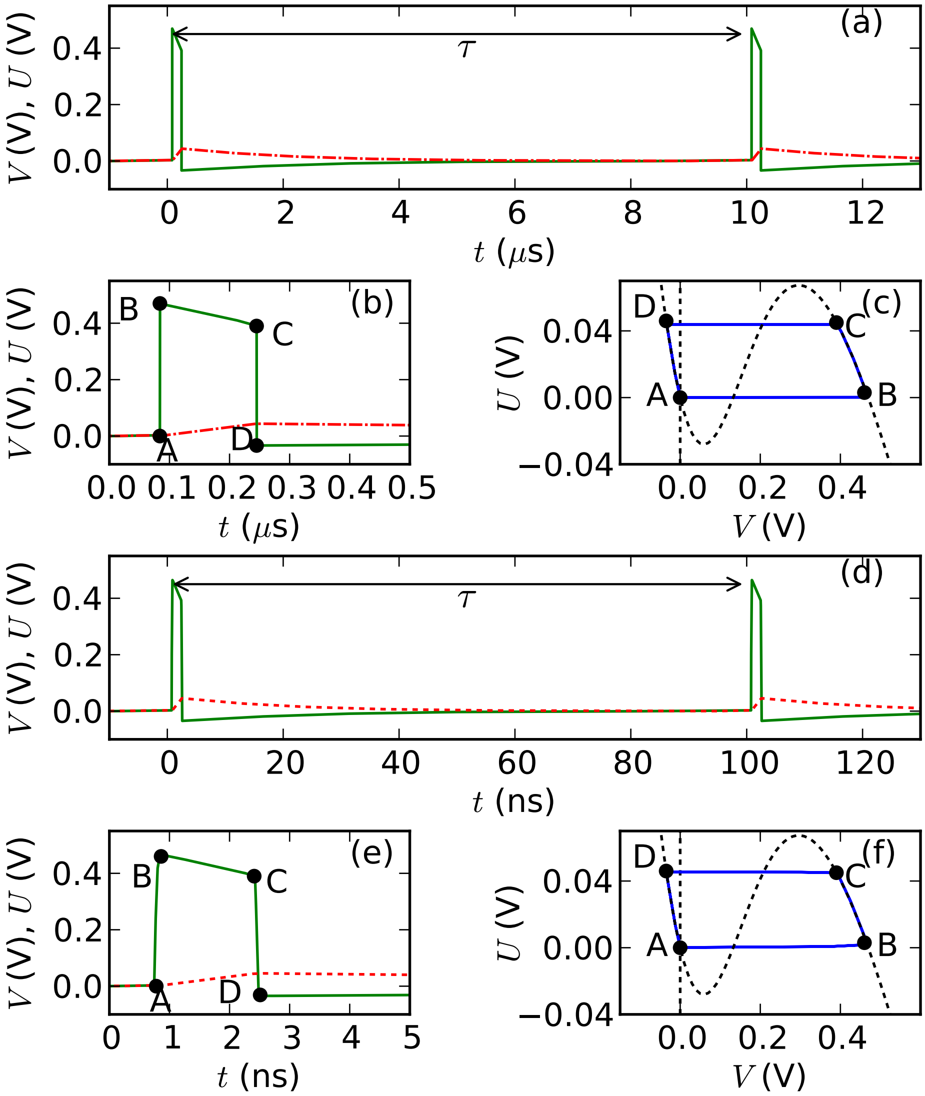


Figure 5.3: Waveforms of $V(t)$ (green solid line) and $U(t)$ (red dashed line) for $\omega_-/(2\pi) = 100\,\text{kHz}$ (a),(b) and $\omega_-/(2\pi) = 10\,\text{MHz}$ (d),(e) with respective phase protraits (c), (f), acquired by simulation of the model equations (2.21). The nullclines are shown black dashed. The other parameters are $m = 0$, $\gamma = 2.6$, $\omega_+/(2\pi) = 10\,\text{GHz}$, initial amplitude $1\,\text{V}$ and for (a)-(c) $\tau = 10\,\mu\text{s}$ and initial width $100\,\text{ns}$ for (d)-(f) $\tau = 100\,\text{ns}$ and initial width $1\,\text{ns}$. The pulse widths are $w = 161\,\text{ns}$ for (a)-(c) and $w = 1.68\,\text{ns}$ for (d)-(f).

pulse (after point D in Fig. 5.3b), the voltage V becomes negative and relaxes asymptotically back to zero. In the simulation, the voltage approaches effectively zero after finite time T_{ref} and stays in the fixed point until the excitation is triggered by the delayed signal $V(t-\tau)$. When choosing a time delay larger than T_{ref}, the trajectory waits in the fixed point for the time $\tau - T_{\mathrm{ref}}$. The time that the trajectory waits in the fixed point has no effect on the pulse shape but only on the period of the pulse train. Therefore, we can save computing time by choosing τ not to be unnecessarily large (not larger than T_{ref}).

Dependence on initial conditions It was also verified that the pulse shape does not depend on width and amplitude of the initial pulse in the history function when a sufficiently long simulation time is used. Only when the width is off by a factor greater than 10, pulse trains are not excited and the system instead relaxes back to and remains at the fixed point.

Dependence on the filter parameters The respective phase portraits of the waveforms can be found in Fig. 5.3c,f. They look similar, even though the low frequency cutoff ω_- was changed by a factor of 100. The phase trajectory has not changed, hence the integral path of $c(m,\gamma)$ is expected to be independent of ω_-. This observation stresses that the integral $c(m,\gamma)$ itself is independent of ω_- because the integrand is independent of the cutoff frequencies. In the simulation, we also confirmed that the integral borders U_{B} and U_{C} stay approximately constant across a range of ω_- and ω_+.

In Fig. 5.4, we show the pulse width as a function of the low frequency cutoff, ω_-, from simulations for a fixed high frequency cutoff, $\omega_+ = 10\,\mathrm{GHz}$. The inverse proportionality of Eq. (5.6), shown in Fig. 5.4 as a solid line with slope -1 in the Log-Log-plot, exhibits an offset that is determined by the scale factor from Eq. (5.12). For $\omega_-/(2\pi) \leq 100\,\mathrm{MHz}$, the simulated pulse width fall on this line, which suggests that the power law in Eq. (5.6) and the approximation of the scale factor in Eq. (5.12) are valid. For $\omega_-/(2\pi) > 100\,\mathrm{MHz}$ the points still fall close to the line.

Smaller frequencies than $\omega_-/(2\pi) = 10\,\mathrm{kHz}$ were not simulated because these solutions need a long delay $\tau > 2\,\mathrm{ms}$ that requires too much internal memory at the precision that we are using. For $\omega_-/(2\pi) > 350\,\mathrm{MHz}$, pulses are not possible

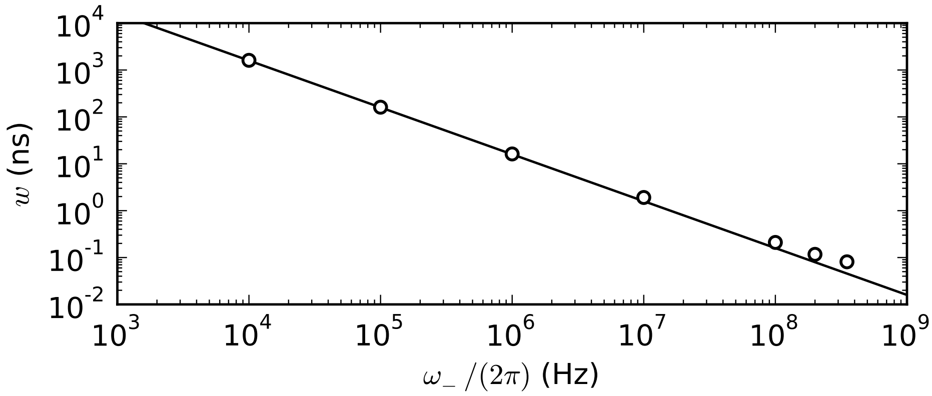


Figure 5.4: Pulse width w for different low frequency cutoffs ω_- and $m = 0$, $\gamma = 2.6$, $g = -0.28\,\mathrm{V}$, $d = 1.0$ and $\omega_+/(2\pi) = 10\,\mathrm{GHz}$ from simulations of Eq. (2.21) (Circles). the other parameters for data points 1-7 from left to right: time delay ($\tau = 100\,\mu\mathrm{s}$ for 1, $\tau = 10\,\mu\mathrm{s}$ for 2 and 3, and $\tau = 100\,\mathrm{ns}$ for the other data points), initial width ($1\,\mu\mathrm{s}$ for 1, $100\,\mathrm{ns}$ for 2, $10\,\mathrm{ns}$ for 3, and $100\,\mathrm{ps}$ for the other data points) and simulation time ($5\,\mathrm{ms}$ for 1 and $1\,\mathrm{ms}$ for the other data points) were chosen adequately to ω_-. Analytic condition Eq. (5.6) with constant of Eq. (5.12) is shown as a solid line.

for the bandwidth we use, $\Delta \approx \omega_+ = 10\,\mathrm{GHz}$. We hypothesize that pulses narrower than the achieved $w = 81\,\mathrm{ps}$ for $\omega_-/(2\pi) = 350\,\mathrm{MHz}$ would extend the bandwidth and are therefore not stable.

As a numerical test, we found that for $\omega_+/(2\pi) = 100\,\mathrm{GHz}$ and $\omega_-/(2\pi) = 2\,\mathrm{GHz}$, pulse widths $w < 8\,\mathrm{ps}$ are possible. A natural restriction is that electronic devices have finite response times and are therefore band limited with high frequency cutoffs $\omega_+/(2\pi)$ that do not extend to tens of GHz.

5.3 Power law in the experiments

We test Eq. (5.6) in the experiments by measuring pulse width of pulse trains for all of the six filters shown in table 2.1. Therefore, we tune the optical attenuator to a step that corresponds to the desired value of γ, set $m = 0$, and input a square pulse as an initial function using the pattern generator as described in Sec. 2.1. We set $m = 0$ and approximately the same $|\gamma| = 2.6$ for all filters so that the

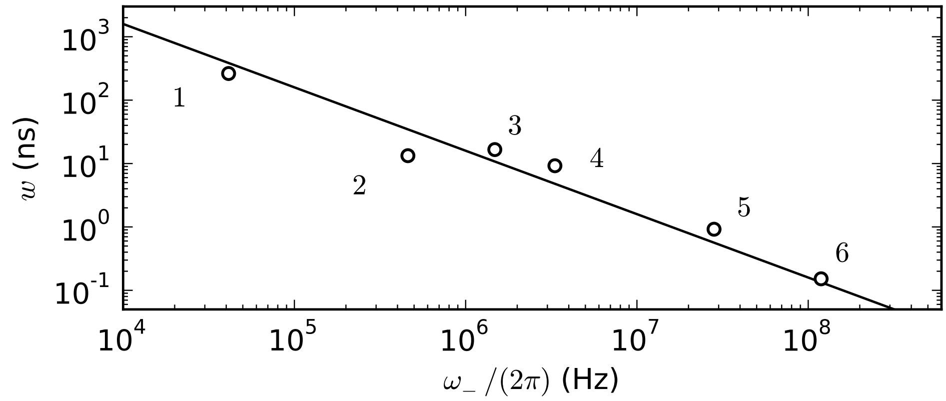


Figure 5.5: Pulse widths, w, of the pulse train solutions for different bandpass filters as shown in table 2.1 (labeled) as a function of ω_- from the experiment (circles). ω_+ is also different for each filter as shown in table 2.1. The parameters are $m = 0$, $g = -0.28\,\mathrm{V}$, $d = 1.0$, $\gamma \approx \pm 2.6$, except for filter #6 ($\gamma = 2.9$) and $\tau = 1.5\,\mu\mathrm{s}$, except for filter #1 and #2 ($\tau = 93\,\mu\mathrm{s}$). Analytic condition Eq. (5.6) with constant of Eq. (5.12) (solid line).

function $c(m, \gamma)$ (Eq. (5.5)) stays constant.

The measured pulse widths, w, are shown in Fig. 5.5 as a function of the low frequency cutoff $\omega_-/(2\pi)$. It can be seen that the trend of decreasing pulse width, w, for increasing low frequency cutoffs persists over three orders of magnitude. The solid line with slope -1 represents the inverse proportionality of Eq. (5.6) with the scale factor from Eq. (5.12). Since all measured widths fall close to the line, the power law Eq. (5.6) with scale factor Eq. (5.12) also approximately describes the experimental behavior.

The measured time series, corresponding to the data points in Fig. 5.5, are shown in Fig. 5.6. All time series with the exception of Fig. 5.6b show a clear refractory phase, indicating that the waveform in Fig. 5.6b (filter #2) might belong to a different dynamic state. This observation would also explain why the corresponding width is off in Fig. 5.5. The waveforms in Fig. 5.6a,e,f show distinct wiggles after a pulse indicating that γ was chosen to be too high. The feedback gain γ chosen for all filters was a compromise for which all filters except filter #6 showed pulse trains. The waveform Fig. 5.6f shows low frequency oscillations after the pulse, a finding that could be due to the non-ideal bandpass characteristics (see Fig. 8.6b on page 85). The experimental pulse widths were measured using

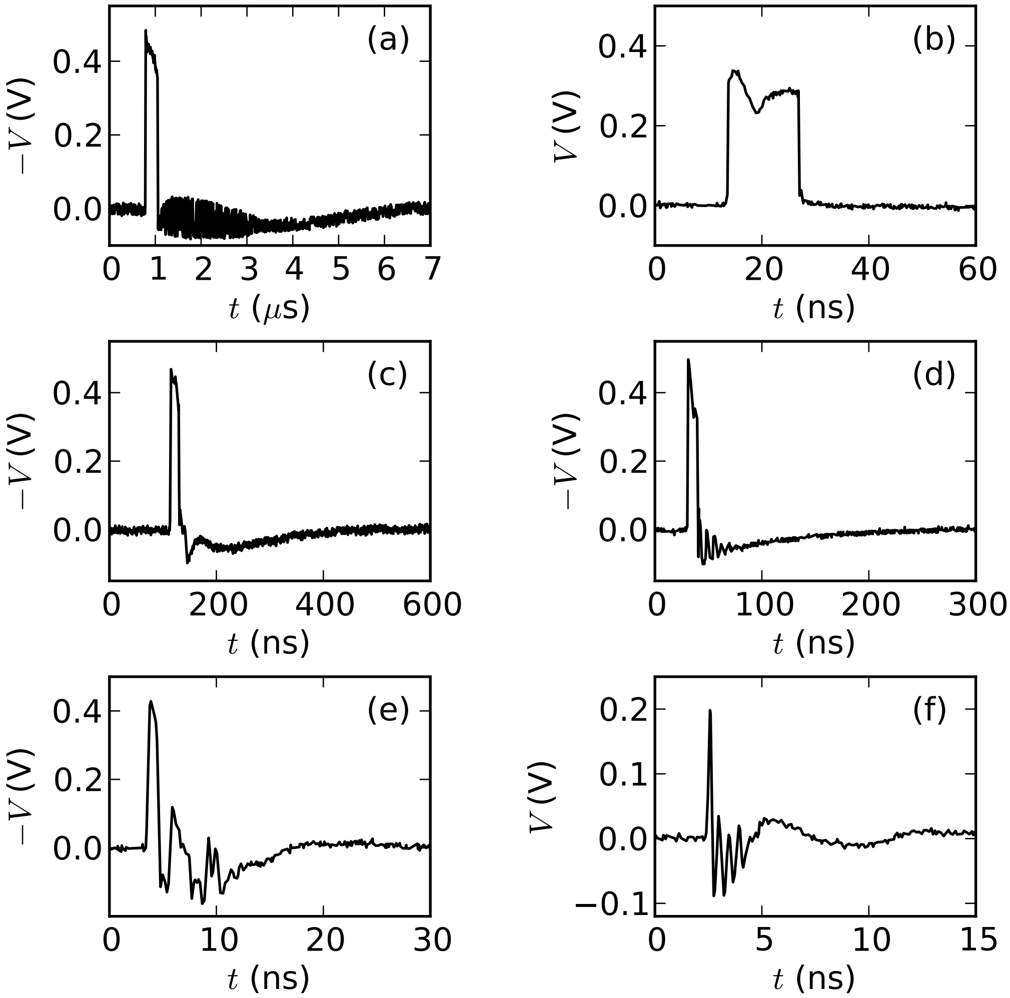


Figure 5.6: Experimental waveforms for the different filters shown in table 2.1, with parameters $m = 0$, $g = -0.28\,\mathrm{V}$, $d = 1.0$, $\gamma = \pm 2.6$ except for (f) ($\gamma = 2.9$) and $\tau = 1.5\,\mu\mathrm{s}$, except for (a),(b) ($\tau = 93\,\mu\mathrm{s}$). (a) Filter #1, (b) filter #2, (c) filter #3, (d) filter #4, (e) filter #5, (f) filter #6. These waveforms were used in Fig. 5.5 to calculate the pulse width.

an oscilloscope with an analog bandwidth of 8 GHz corresponding to a risetime of $\approx 50\,\mathrm{ps}$. Therefore, it can be expected that the smallest pulse width recorded (150 ps) is artificially high and could be as low as 50 ps [66].

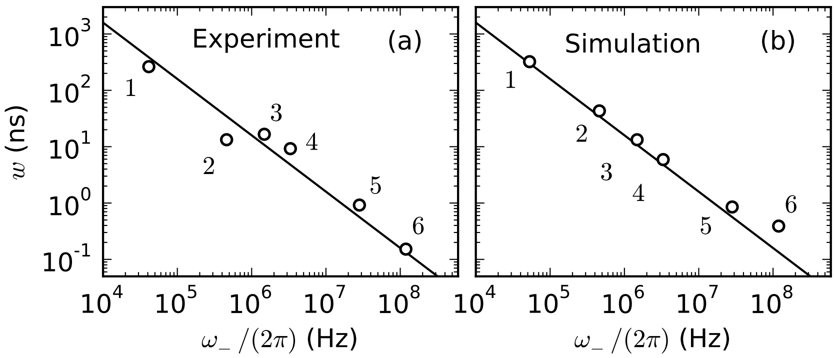


Figure 5.7: (a) Experimental pulse widths repeated from Fig. 5.5 and (b) associated numerical simulations for the same parameters (*i.e.* $m = 0$, $g = -0.28\,\mathrm{V}$, $d = 1.0$, $\gamma = 2.6$, except for filter #6 ($\gamma = 2.9$), $\tau = 1.5\,\mu\mathrm{s}$, except for filters #1 and #2 ($\tau = 93\,\mu\mathrm{s}$), and cutoffs ω_- and ω_+ shown in table 2.1). Solid lines: analytic condition Eq. (5.6) with constant of Eq. (5.12).

5.4 Comparing experimental and numerical results

Figure 5.7a shows again the experimental data from Fig. 5.5 which is compared to the simulation, using identical parameters especially the same cutoff frequencies, ω_- and ω_+, in Fig. 5.7b.

The solid line in both plots represents the analytic condition from Eq. (5.6) with constant defined by Eq. (5.12). It can be seen that the widths in experiment and simulation are similar and fall close to the solid line, as expected. We hypothesize that the difference between experiment and simulation is due to the non-ideal bandpass characteristics and an error in measuring γ in the experiment, as discussed in Sec. 2.1.3.3. By comparing the phase trajectories and the nullcline, we checked that the feedback gain $\gamma = 2.6$ except for filter #6 ($\gamma = 2.9$).

In the simulation of Fig. 5.7b, the data points of filters #5 and #6 differ from the analytic condition. For filter #5, this is due to the relatively low ω_+, which is different for each filter. When ω_+ approaches ω_-, the approximation of Eq. (5.4) worsens, leading to smaller pulse widths and other effects due to reduced bandwidth. Filter #6 has a larger pulse width than expected from the analytic condition. This finding is due to the higher feedback gain, γ, used for this data point ($\gamma = 2.9$ instead of $\gamma = 2.6$).

6 Oscillatory behavior

In the previous two chapters, we have described pulse trains as a novel dynamic scenario and found that τ-periodic nullclines are useful for describing them. In this chapter, we apply the τ-periodic nullclines to the oscillatory solutions for optoelectronic oscillators. Other research groups are especially interested in periodic solutions [44, 45] because they are highly applicable as a single frequency source [49]. For long delay loops, the stored energy and hence the quality factor is appreciable, leading to phase noise as low as $-140\,\mathrm{dBc/Hz}$ [49, 50, 75].[1]

Peil *et al.* [44, 45] distinguished low from high frequency oscillations and Hopf oscillations. Considering the time scales that exist in the dynamic equations, it makes sense to introduce the term "moderate frequency oscillations" for the Hopf oscillations. These oscillations the characteristics shown in table 6.1.

Type	frequency	time delay	gain
Low f. osc.	$f \sim \omega_-$	$\tau \ll 1/\omega_-$	$\gamma \sim 5$
Moderate f. osc.	$f \sim 1/\tau$	$\tau \sim 1/\omega_-$	$\gamma \sim 2$
High f. osc.	$f \sim \omega_+$	$\tau \ll 1/\omega_-$	$\gamma \sim 5$

Table 6.1: Characteristics of three periodic oscillatory states.

The three characteristic time scales in the OEO are reflected in the three oscillatory states. These time scales are $1/\omega_-$, τ and $1/\omega_+$ [45].

[1] Low phase noise OEO are built with a mode selector after the photodetector (*i.e.*, a narrow pass band) [51] and can be built with two delay lines [50].

In our experimental setup, the high frequency oscillations are not accessible. Peil *et al.* [45] reported that they can be seen in the numerics only when considering a higher order model. We do not report further on the high frequency solutions.

Peil *et al.* [45, Sec. IV] have already examined low and moderate frequency oscillations and found that the shift from moderate to low frequency oscillations (with increased γ) was "totally unexpected since it cannot be anticipated from the Hopf bifurcation analysis". We explain how both oscillation types can be explained with nullclines.

The nullcline description provides new insights into the dynamics of low and moderate frequency oscillations. For moderate frequency oscillations, we find an approximate analytic description of the waveforms and show that the oscillation pattern can be controlled with parameter m. For high frequency oscillations we calculate the wave pattern analytically from the nullclines and find a relation between period and the low frequency cutoff, ω_-. Furthermore, we apply the nullclines to breather solutions and the transition to chaos and make predictions about the time scales of these dynamics.

The Hopf bifurcation points (Sec. 4.6 on page 41) delimit the oscillatory regime for moderate and high frequency oscillations. For $\gamma < 0$ ($\gamma > 0$), upon increasing (decreasing) m, the system behavior shifts from excitable to oscillatory when the unique fixed point becomes unstable in a Hopf bifurcation (Eq. (3.7)) at $m_{\mathrm{H},1} = -(1/2)\arcsin(1/\gamma)$ and shifts from oscillatory to excitable when the fixed point becomes stable again in a second Hopf bifurcation at $m_{\mathrm{H},2} = -\operatorname{sgn}(\gamma)\pi/2 + (1/2)\arcsin(1/\gamma)$. For $\gamma < 0$, the system is in the oscillatory regime for $m \in [m_{\mathrm{H},1}, m_{\mathrm{H},2}]$ (for $\gamma > 0$: $m \in [m_{\mathrm{H},2}, m_{\mathrm{H},1}]$), intervals which lie on different sides of the $\cos^2$-transmission of the MZM (Eq. (2.3)) for different signs of γ.

6.1 Periodic moderate frequency oscillations

We discover the regime of moderate frequency oscillations with $\tau = 65\,\mathrm{ns}$ for filter #4 (*i.e.*, $1/\omega_- \approx 50\,\mathrm{ns}$). Therefore, the condition for the moderate frequency oscillations $\tau \sim 1/\omega_-$ is fulfilled.

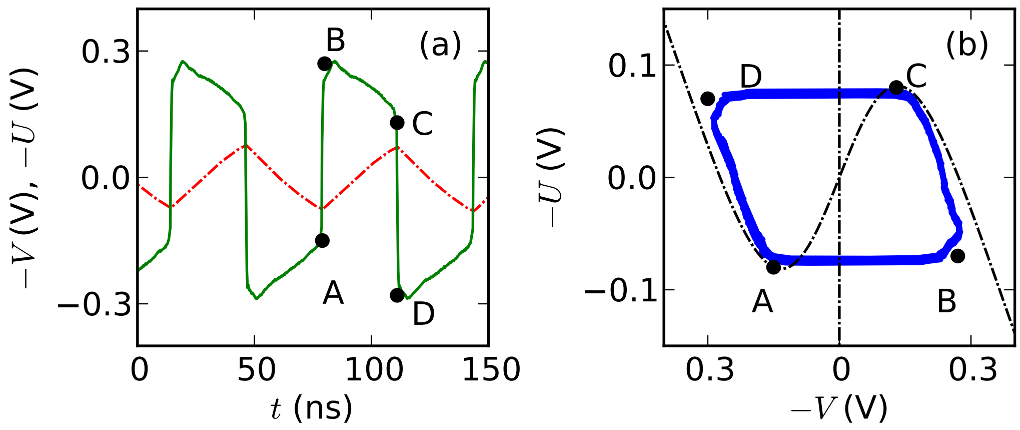


Figure 6.1: Moderate frequency oscillations for $m = \pi/4$. a) Experimental waveforms of $V(t)$ (solid green line) and $U(t)$ (dashed red line). b) Experimental phase portrait of (V, U) (blue solid line) with nullclines (Eq. (4.2)) superimposed (black dashed line). The points A to D mark the related positions on (a) and (b). The parameters for experiments and nullclines are $m = \pi/4$, $\gamma = -2.08$, $\tau = 65\,\mathrm{ns}$, filter parameters of filter #4 in table 2.1, $g = -0.28$ and $d = 1$. The variable $U(t)$ was calculated according to Eq. (4.1). The time series were averaged over 20 waveforms.

6.1.1 Experimental data

A typical experimental waveform of the moderate frequency oscillations and its corresponding phase portrait are shown in Fig. 6.1a and Fig. 6.1b, respectively. The phase shift $m = 0.25\pi$ for $\gamma = -2.08$ lies within the oscillatory regime of $[m_{\mathrm{H},1}, m_{\mathrm{H},2}] \approx [0.08\pi, 0.42\pi]$.

Let us first describe the time series, Fig. 6.1a. Starting from point A, the voltage $V(t)$ rises quickly to point B, decreases slowly until point C, and then decreases quickly to point D. Finally the voltage increases slowly to point A, where another oscillation cycle starts. The filter variable, $U(t)$, rises from point A to point C and decreases from point C to point A. τ-periodic nullclines are a valid tool to describe these dynamics because the oscillations are forced by the delayed feedback to be τ-periodic.

The phase portrait, Fig. 6.1b, reveals that the nonlinear nullcline (Eq. (4.2)) is useful to describe its trajectory. The phase space trajectory relaxes along the left branch of the nonlinear nullcline until it reaches its minimum (point A) where the velocity field is directed to the right and, therefore, the trajectory moves to the right branch of the nonlinear nullcline (point B). The trajectory moves along the right branch of the nonlinear nullcline upward until it reaches the maximum (point C), where the velocity field is directed left, so that the trajectory moves left toward the left branch (point D). Here, the trajectory moves back toward point A and starts the oscillation over again. The trajectory moves along the right and left branch of the nonlinear nullcline because they are attracted by the velocity field in the V-direction. The velocity field is directed upward (downward) along the right (left) branch. The phases from A to B and from C to D are again much faster than the phases from B to C and from D to A, because of the time scale separation, as described in Sec. 2.2.3. This explains the time series qualitatively, with fast phases from A to B and from C to D and slow phases from B to C and from D to A. The paths corresponding to the slow phases are determined by the outer branches of the nullcline, and the fast phases move horizontally in phase space.

6.1.2 Controlling the duty cycle

We describe qualitatively how a change in the parameter m changes the nullclines and hence the shape of the waveform, specifically how the times of low and high levels change. In the previous section with $m = 0.25\pi$, V spends an equal amount of time at both high and low levels (Fig. 6.1a) *i.e.*, the time from B to C (T_{BC}) equals the time from D to A (T_{DA}).

Figure 6.2 shows time series and phase portraits for $m = 0.14\pi$ and $m = 0.36\pi$, both of which remain in the oscillatory regime. In the time series (Fig. 6.2a,b), the duration of high and low levels has changed. In Fig. 6.2a (Fig. 6.2b) the time in the high level T_{BC} is short (long) and the time in the low level T_{DA} is long (short). This change can be described with the duty cycle, η, which relates the time in the active state (high level) to the whole period

$$\eta \approx \frac{T_{\mathrm{BC}}}{T_{\mathrm{BC}} + T_{\mathrm{DA}}} = \frac{1}{1 + T_{\mathrm{DA}}/T_{\mathrm{BC}}}, \tag{6.1}$$

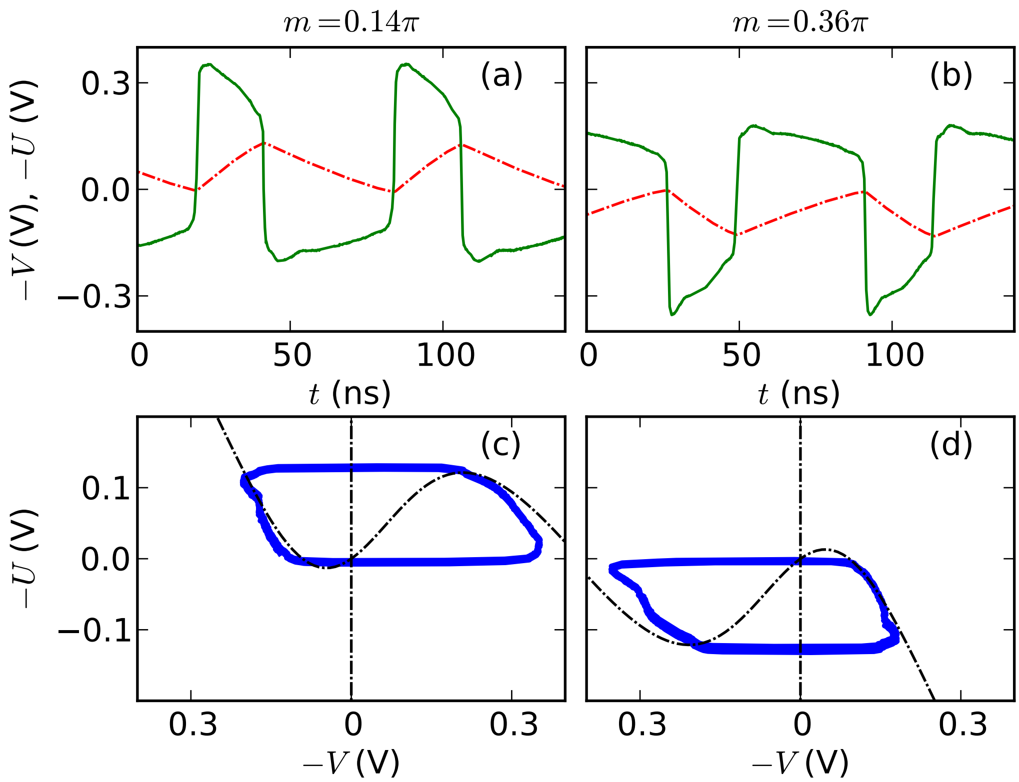


Figure 6.2: Moderate frequency oscillations for a,c $m = 0.14\pi$ and b,d $m = 0.36\pi$. The other parameters are chosen as in Fig. 6.1. a, b Waveforms of $V(t)$ (solid green line) and $U(t)$ (dashed red line) and c, d respective phase portraits of (V, U) (solid blue line) with nullclines superimposed (dashed black line). The variable $U(t)$ was calculated according to Eq. (4.1). The time series were averaged over 20 waveforms. After initializing the waveform with a pulse from the pattern generator in the oscillatory regime, m was moved to the desired value.

when neglecting the fast phases from A to B and from C to D.

In the phase portraits (Fig. 6.2c,d) the trajectories follow the nullclines. Therefore, the change of the duty cycle can be explained by the nullcline. Increasing m shifts the nullcline from the right to the left. The fixed point is located at $(U^*, V^*) = (0, 0)$ and is always located on the middle branch of the nullcline. Near the fixed point, the magnitude of the velocity field is low, because the velocity field is a (Lipschitz) continuous function and zero at the fixed point. For m close to a Hopf bifurcation, the fixed point lies close to the left or the right branch of the nullcline, resulting in a longer stay at the branch the fixed point is closer to.

In Fig. 6.2c the fixed point is close to the left branch of the nullcline, therefore T_{DA} is greater than T_{BC} and $\eta < 50\%$. In Fig. 6.2d the fixed point is close to the right branch of the nullcline, therefore T_{BC} is greater than T_{AD} and $\eta > 50\%$. For $m = \pi/4$ (Fig. 6.1b), the fixed point is located halfway between both outer branches with $T_{\mathrm{BC}} = T_{\mathrm{DA}}$, leading to $\eta = 50\%$

T_{BC} and T_{AD} cannot be calculated as in Sec. 5.10 because the approximation $V/g \gg 1$ is not valid for all m. For $m = \pi/4$, we calculate the waveform and find a relation between the period and the low frequency cutoff, ω_-, for low frequency oscillations (see Sec. 6.2).

6.1.3 Simulation data

For parameters identical to those used in the experimental waveforms depicted by Fig. 6.1 and Fig. 6.2, we simulate the dynamics with the model equations (Eq. (2.21)). The plots of the simulation data Fig. 6.3b,e and a,d and c,f correspond to Fig. 6.1a,b and Fig. 6.2a,c and b,d, respectively. In the simulations, the agreement between phase portraits and nullclines is better than in the experiments. Nevertheless, simulations and experiments agree quite well.

6.1.4 Period of moderate frequency oscillations

In the simulations, we measure the period of the moderate frequency oscillations for different ω_-. We always find that $U(t) = U(t-\tau/n)$ holds ($n \in \mathbb{N}$) because the dynamics are forced by delayed feedback. For different low frequency cutoffs, ω_-, we measure n and find that n changes linearly with ω_- but changes discretely in steps of 1 as shown in Fig. 6.4. The inverse proportionality between pulse width (here: period $T = \tau/n$) and the low frequency cutoff, ω_-, (*i.e.*, proportionality between n and ω_-) can be shown to hold for moderate frequency oscillations in a manner analogous to Sec. 5.6. One constraint is that the period can only change stepwise with $n \in \mathbb{N}$. For low frequency oscillation we derive the power law (inverse proportionality) in Sec. 6.2.1.

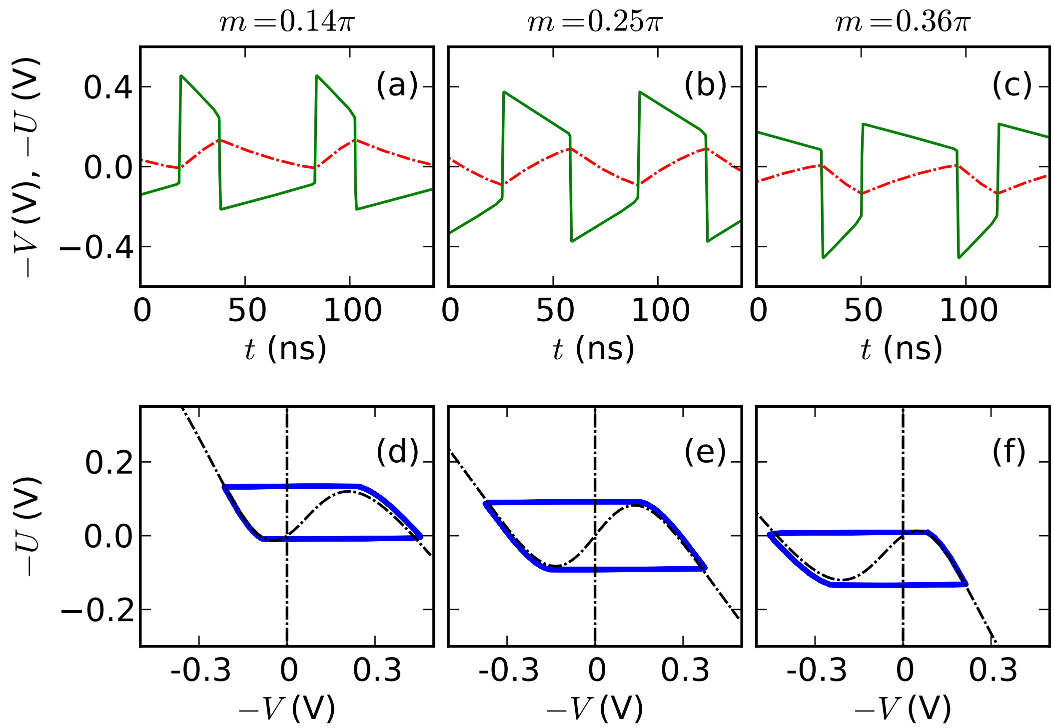


Figure 6.3: Moderate frequency oscillations for (a),(d) $m = 0.14\pi$, (b),(e) $m = 0.36\pi$, and (c),(f) $m = 0.36\pi$ from simulations of Eq. (2.21) with initial width 15 ns, initial amplitude 1 V and simulation time of 100 μs. The other parameters are chosen as in Fig. 6.1. (a), (b), (c) Waveforms of $V(t)$ (solid green line) and $U(t)$ (dashed red line) and (d), (e), (f) respective phase portraits of (V, U) (solid blue line) with nullclines (dashed black line) superimposed.

6.1.5 2τ periodic oscillations

Peil *et al.* [45, Sec. IVB] already reported on the τ and 2τ periodicity of moderate frequency oscillations. For negative γ, 2τ-periodic oscillations appear in the vicinity of $m = -\pi/4$ on the opposite side of the $\cos^2$ transmission curve as the τ-periodic oscillations that appear in the vicinity of $m = \pi/4$.

For $m = -\pi/4$ (for positive γ: $m = \pi/4$), we show that 2τ periodic oscillations can be described with nullclines and discuss the general case of other values of m. The 2τ periodic solutions ($V(t) = V(t - 2\tau)$) are in antiphase to the delayed

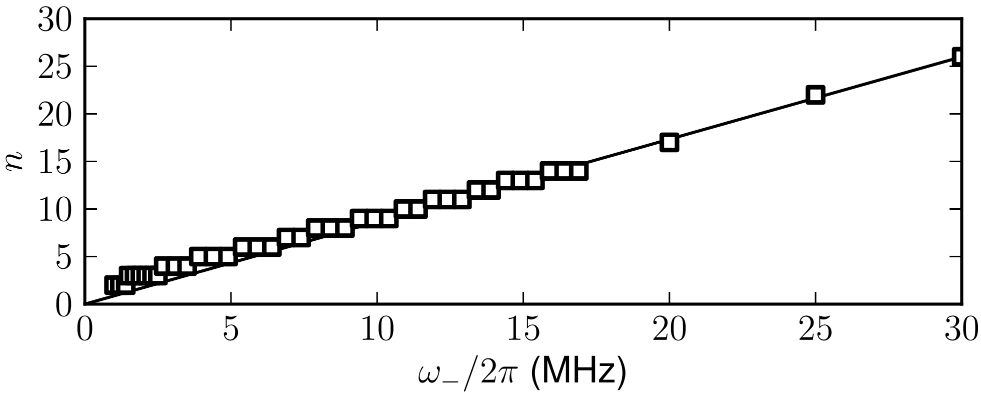


Figure 6.4: Integer n as a function of the low frequency cutoff ω_- in the simulations (squares) for moderate frequency oscillations. The straight line connects the last data point and zero. Here, n is a measure for the period: $T = \tau/n$ of moderate frequency oscillations. The parameters are $m = -0.25\pi$, $\gamma = 2$, $\omega_+/(2\pi) = 10\,\mathrm{GHz}$, $\tau = 500\,\mathrm{ns}$, $g = -0.28\,\mathrm{V}$, $d = 1.0$, initial width $10\,\mathrm{ns}$, initial amplitude $0.25\,\mathrm{V}$, simulation time $100\,\mu\mathrm{s}$.

variable, $V(t) = -V(t-\tau)$, leading to 2τ periodic nullclines:

$$U(V) = -V + \gamma G\left\{\cos^2\left[m + d\tanh\left(-V/g\right)\right] - \cos^2\left[m\right]\right\} \tag{6.2a}$$

$$V = 0. \tag{6.2b}$$

Using the fact that tanh is antisymmetric and $\cos^2$ is symmetric, we arrive at the same nullclines as those for τ-periodic oscillations but with a different sign of m

$$U(V) = -V + \gamma G\left\{\cos^2\left[-m + d\tanh\left(V/g\right)\right] - \cos^2\left[-m\right]\right\}. \tag{6.3}$$

Therefore, with 2τ-periodic oscillations, another oscillatory regime appears on the other side of the $\cos^2$ transmission curve.

For $m \neq -\pi/4$, the relation $V(t) = -V(t-\tau)$ does not hold because the oscillations are no longer symmetric. The nullclines (Eq. (6.2)) are only approximately correct, making a calculation of the Hopf bifurcation point possible that was previously found with linear stability analysis (Sec. 4.6). Figure 6.5 shows 2τ-periodic oscillations for $m = -0.14\pi$. The delayed feedback couples back the low level to the high level and vice versa. Because $V(t)$ is not symmetric ($m \neq \pi/4$),

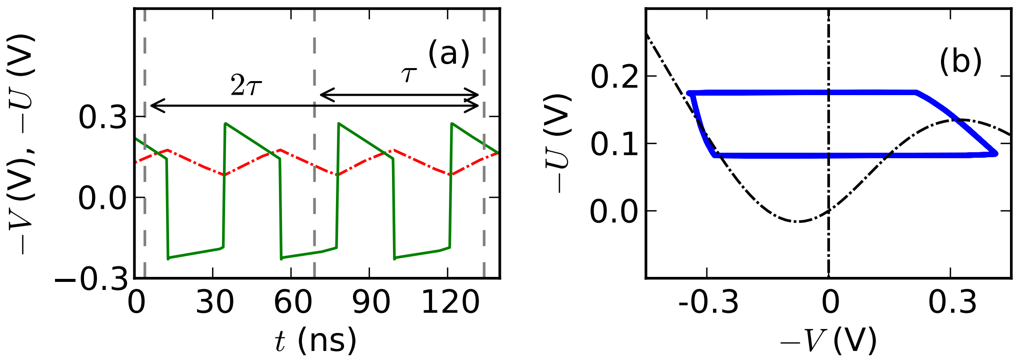


Figure 6.5: Moderate frequency oscillations for $m = -0.14\pi$ from simulations. The other parameters are chosen as in Fig. 6.1. a) Waveform of $V(t)$ (solid green line) and $U(t)$ (dashed red line) and b) respective phase portrait of (V, U) (solid blue line) with 2τ-periodic nullclines (Eq. (6.2)) superimposed (dashed black line). The waveform has a periodicity of $U(t - 2\tau/n) = U(t)$ with $n = 3$ (and therefore also the periodicity $U(t - 2\tau) = U(t)$).

$V(t) = -V(t - \tau)$ is not valid. Thus, the 2τ-periodic nullclines are no longer a good tool to describe the dynamics (for $m \neq -\pi/4$).

6.2 Periodic low frequency oscillations

Low frequency oscillations appear for small delays τ compared to the time scale of the low frequency cutoff ($\tau \ll 1/\omega_-$). We measure and simulate waveforms in the low frequency regime for filter #1 ($1/\omega_- = 18.9\,\mu$s) and $\tau = 24\,$ns, hence $\tau \ll 1/\omega_-$ is valid. Low frequency oscillations appear only on one side of the $\cos^2$ transmission curve.

In this regime, Peil *et al.* [45, Sec. IV.B.] could approximately calculate the waveforms from the dynamic equations when neglecting the time delay. We also approximate for small τ by assuming that $x(t - \tau) \approx x(t)$ for the nullclines. Consequently, the nullclines have the same form as the τ-periodic nullclines in Eq. (4.2).

A waveform of the low frequency oscillations, acquired through simulation, is shown in Fig. 6.6a. We can clearly identify fast and slow transitions due to the time scale separation and find four segments similarly to the moderate frequency

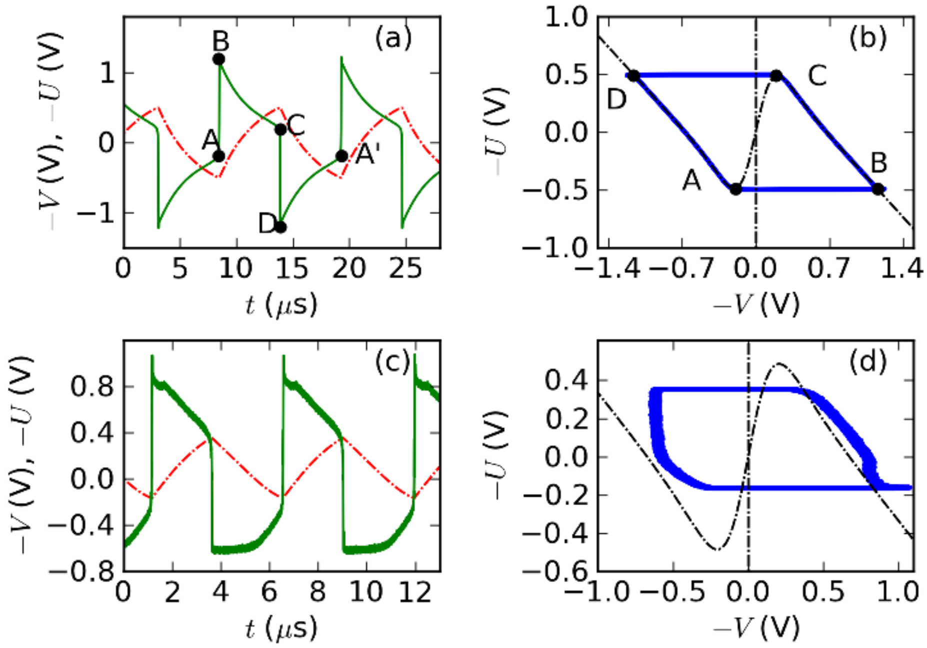


Figure 6.6: Low frequency oscillations for $m = 0.25\pi$, $\tau = 24\,\mathrm{ns}$ and filter #1 from simulations (a),(b) and from the experiments (c),(d). (a),(c) Waveform of $V(t)$ (green solid line) and $U(t)$ (red dashed line) and b,d respective phase portraits of (V, U) (blue solid line) with nullclines (Eq. (4.2)) superimposed (black dashed line). The other parameters are $\gamma = -5.2$ for simulations and nullclines, $g = -0.28$, $d = 1$. $\gamma < 0$ was not measured in the experiment. The variable $U(t)$ was calculated according to Eq. (4.1) in (c) and (d). Four respective points A-D are labeled on (a) and (b), A'=A in phase space.

oscillations in Sec. 6.1. Analogously, these can be linked to the four segments in the phase portrait Fig. 6.6b. The experimental waveform in Fig. 6.6c and its respective phase portrait in Fig. 6.6d are different from their simulated counterparts. This could be due to a faulty measurement of m in the experiments. The experimental waveform shows that we can access the low frequency oscillations in our experiments.

6.2.1 Scaling behavior of the period

The period of the low frequency oscillations is not a sub-multiple of the time delay τ as for the moderate frequency oscillations, but is instead determined by the low frequency cutoff, ω_-, as shown by Peil *et al.* [45] experimentally. In this section, we show the proportionality analytically and calculate the constant of proportionality for $m = \pi/4$.

In a manner similar to Sec. 5.1, we arrive at the scaling behavior by integrating over the path of the trajectory along the nonlinear nullcline. Again, we neglect the phases from A to B and from C to D, as they are fast compared to the phases from B to C and from D to A' in Fig. 6.6. This leads to an oscillation period of (in phase space: A'=A)

$$T = \left(\int_B^C + \int_D^A\right) \mathrm{d}t = \left(\int_{U_\mathrm{B}}^{U_\mathrm{C}} + \int_{U_\mathrm{D}}^{U_\mathrm{A}}\right) \frac{1}{\dot{U}} \mathrm{d}U. \tag{6.4}$$

Substituting $\dot{U}$ from Eq. (2.21b) into Eq. (6.4) separates the period into a frequency dependent and a parameter (m and γ) dependent function[1]

$$T = \frac{1}{\Delta\varepsilon} \left(\int_{U_\mathrm{B}}^{U_\mathrm{C}} + \int_{U_\mathrm{D}}^{U_\mathrm{A}}\right) \frac{1}{V(U)} \mathrm{d}U = f(\omega_-, \omega_+) h(m, \gamma), \tag{6.5}$$

with functions

$$f(\omega_-, \omega) := \frac{1}{\Delta\varepsilon} \approx \frac{1}{\omega_-} \tag{6.6}$$

$$h(m, \gamma) := \left(\int_{U_\mathrm{B}}^{U_\mathrm{C}} + \int_{U_\mathrm{D}}^{U_\mathrm{A}}\right) \frac{1}{V(U)} \mathrm{d}U. \tag{6.7}$$

The approximation of Eq. (6.6) holds for $\omega_- \ll \omega_+$.

Hence, for fixed m and γ, the same power law with exponent -1 (inverse proportionality) holds for the period of the low frequency oscillations as that found for the pulse width in the pulsing regime (Eq. (5.6))

$$T = h \cdot (\omega_-)^{-1}, \tag{6.8}$$

with scale factor $h(m, \gamma)$ defined by Eq. (6.7).

[1] d and g are assumed to be constant.

The function $h(m, \gamma)$, the constant of proportionality, can be approximated for $m = \pi/4$ in a manner similar to Sec. 5.1.1. Considering that $\tanh(V/g) \approx \pm 1$ ($V/g \gg 1$), the left (+) and right (−) branch of the nonlinear nullcline can be approximated by

$$U_\pm(V) = -V + F_\pm \leftrightarrow V_\pm(U) = -U + F_\pm, \tag{6.9}$$

with the constant

$$F_\pm = \frac{\gamma g}{d} \left\{ \cos^2 [m \pm d] - \cos^2 m \right\}. \tag{6.10}$$

Inserting the approximate branches into the integral for $h(m, d)$ (Eq. (6.7)), gives

$$h(m, \gamma) = \int_{U_B}^{U_C} \frac{1}{-U + F_-} \mathrm{d}U + \int_{U_D}^{U_A} \frac{1}{-U + F_+} \mathrm{d}U \tag{6.11}$$

$$= \log(F_- - U_B) - \log(F_- - U_C) + \log(F_+ - U_D) - \log(F_+ - U_A). \tag{6.12}$$

For $m = \pi/4$, $\gamma = -5.2$ and $-U_A = -U_B = -0.49$, $-U_C = -U_D = 0.49$ from the maximum and minimum of the nullcline, leading to

$$h(m, d) \approx 1.9, \tag{6.13}$$

$$\Rightarrow T \approx 11.3\,\mu\text{s}, \tag{6.14}$$

for filter #1 which agrees with Fig. 6.6a (from simulations: $T = 10.8\,\mu$s).

For moderate frequency oscillations, the derivations above are also valid, with the constraint that the period can change only stepwise ($T = \tau/n$). With $m = \pi/4$, $\gamma = -2.08$ and $-U_A = -U_B = -0.09$, $-U_C = -U_D = 0.09$ from the maximum and minimum of the nullcline, we derive

$$h(m, d) \approx 1.5, \tag{6.15}$$

$$\Rightarrow T \approx 70\,\text{ns}, \tag{6.16}$$

for filter #4, which agrees with Fig. 6.1 (from experiments and simulations: $T = 65\,\text{ns} = \tau$) for the moderate frequency oscillations.

For m not in the vicinity of $m = \pi/4$ the approximation $V/g \gg 1$ breaks down for one branch and the calculation fails

6.2.2 Analytic approximation of the waveform

We can approximate the waveform of the low frequency oscillations, by evaluating the model equation Eq. (2.21b) along the approximated nullcline Eq. (6.9), giving

$$\dot{U} = \Delta\varepsilon V = \Delta\varepsilon\left(-U + F_{\pm}\right), \tag{6.17}$$

for the left $(+)$ and right $(-)$ branch of the nullcline. This linear differential equation, with initial values $U_{\mathrm{B}\to\mathrm{C}}(t_{\mathrm{B}}) = U_{\mathrm{B}}$, $U_{\mathrm{D}\to\mathrm{A}'}(t_{\mathrm{D}}) = U_{\mathrm{D}} = U_{\mathrm{C}}$, has the solution

$$U_{\mathrm{B}\to\mathrm{C}}(t) = F_{-} + (U_{\mathrm{B}} - F_{-})\exp(-\Delta\varepsilon(t - t_{\mathrm{B}})) \text{ for } t \in [t_{\mathrm{B}}, t_{\mathrm{C}}) \tag{6.18a}$$

$$U_{\mathrm{D}\to\mathrm{A}'}(t) = F_{+} + (U_{\mathrm{C}} - F_{+})\exp(-\Delta\varepsilon(t - t_{\mathrm{D}})) \text{ for } t \in [t_{\mathrm{D}}, t_{\mathrm{A}'}) \tag{6.18b}$$

for the filter variable U along the left and right nullcline. Due to the time scale separation, we assume the transition from one branch of the nullcline to the other branch to happen instantly. Therefore, $t_B = t_A$ is the arbitrary start of the oscillation, $t_C = t_{\mathrm{B}} + T/2$, $t_{\mathrm{D}} = t_{\mathrm{C}}$ and $t_{\mathrm{A}'} = t_{\mathrm{C}} + T/2 = t_{\mathrm{B}} + T$. The period T is given by Eq. (6.5). After one round trip, the oscillation starts over again, with $t_{\mathrm{A}'} = t_{\mathrm{A}} + T = t_{\mathrm{B}} + T$. The voltage variable V is given by inserting Eq. (6.18) into the approximated nullcline

$$V(U) = -U + F_{\pm}. \tag{6.19}$$

Note that τ does not appear in the analytical waveform.

With parameters as in Fig. 6.6, the approximate analytical time series is shown in Fig. 6.7a. We use $-U_A = -0.49$ and $-U_C = 0.49$ as the extrema of the nonlinear nullcline and $T = 11.3\,\mu$s from Eq. (6.14). The analytically calculated waveform is very similar to the waveform from the numerical simulation of the model equations in Fig. 6.6a. The phase portrait Fig. 6.7b shows the nonlinear nullcline and also the approximated nullclines. Both differ near the minimum and maximum of the nonlinear nullcline. When comparing the phase trajectories of the analytical to the simulated solution, one difference can be seen. While the simulated trajectory Fig. 6.6b evolves along the nonlinear nullcline, the analytically calculated trajectory Fig. 6.7b evolves along the approximated linear nullclines. This corresponds to the approximation that we use to derive the analytic waveform.

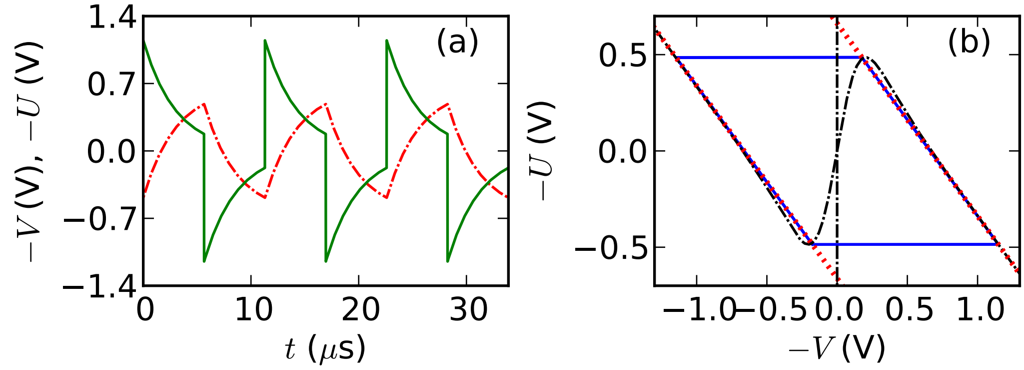


Figure 6.7: Analytical approximation of the waveform using Eq. (6.18) for $U(t)$ and Eq. (6.18) and (6.19) for $V(t)$. Parameters are the same as in Fig. 6.6a,b. (a) Waveform of $V(t)$ (green solid line) and $U(t)$ (red dashed line) and (b) respective phase portrait of (V, U) (blue solid line) with nonlinear nullcline (black dashed line) and linearized nullclines (red dotted line) superimposed.

6.3 Chaotic breather solutions

Breather solutions appear when γ is increased in the low or moderate frequency oscillatory regimes [44]. The high and low levels (phases from B to C and from D to A) of the oscillation become modulated with chaotic wiggles. Therefore, these solutions are considered chaotic. Since breathers also follow the nullcline, the nullcline description could add further information to the breather solutions. It is expected that the period of breathers follows the same inverse proportionality as the period of the oscillatory regime in Eq. (6.8).

6.4 Transition to chaos

Chaos in an OEO is an issue of current research [47, 76]. In this section we find that the nullclines help to describe the transitions to chaos. We consider the transitions to chaos for $\tau \ll \omega_-$, so that $x(t-\tau) \approx x(t)$ is valid and the nullclines Eq. (4.2) can be used. For chaotic time series $x(t-\tau) \approx x(t)$ is not true because chaos has components with frequency ω_+ [47]. The transition to chaos happens for $m = 0$ when γ is increased above a threshold γ_c that was discussed in Sec. 3.2. At this point, a train of narrow pulses, originating from a noise spike, grows and excites a long pulse that leads to chaos [47]. Figure 6.8 shows the transition to chaos as both a waveform and phase trajectory, in simulation and experiment. In the time series Fig. 6.8a,c, we see that a train of narrow pulses grows and then reaches a constant height. After 1.5 μs in the simulation and 0.5 μs in the experiment, the pulse train stops and the voltage V occupies the high level and decreases slowly for several μs. Finally, the voltage V decreases more quickly and a chaotic time series starts. In the phase portraits Fig. 6.8b,d, the initial pulse train occurs between the outer branches of the nonlinear nullcline. The slow decrease of the voltage corresponds to a movement in phase space along the right branch of the nullcline. When the maximum of the nullcline is reached, the chaotic time series starts.

The duration of the slow phase of the transition to chaos along the right branch of the nullcline is expected to scale inversely proportional to ω_{-}, a feature which can be derived analogously to Eq. (5.6).

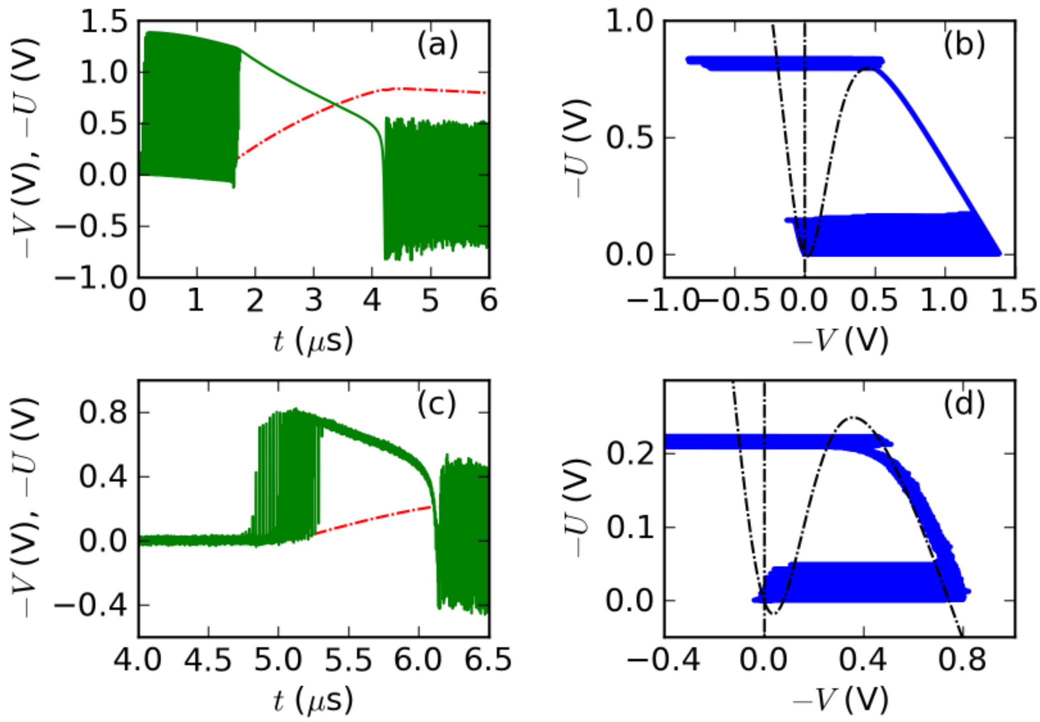


Figure 6.8: Transition to chaos in simulation (a),(b) and experiment (c),(d). The time series (a),(c) show the voltage V (green solid line) and the variable U (red dashed line). The phase portraits (b),(d) show the trajectory (V, U) (blue solid line) and the nullclines (black dashed line). The parameters are $m = 0$, $\gamma = -3.8$ for nullcline in (d) and $\gamma = -5.2$ in the simulation and respective nullcline, filter parameters of filter #1, $g = -0.28\,\mathrm{V}$, and $d = 1.0$. In the simulation, the transition to chaos was initiated with a Gaussian pulse of amplitude 0.1 V and width 0.1 ns, and in the experiments γ was increased until the transition happened. Therefore the experimental value of γ is unknown. The variable $U(t)$ was calculated according to Eq. (4.1).

7
Discussion

7.1 Conclusion

In this thesis, we study an OEO for various low frequency cutoffs of the bandpass characteristic. We find novel pulse train solutions consisting of pulses that repeat periodically with the time delay forced by time delayed feedback. The dynamics exhibits multi stability because the pulse trains coexist with the stable fixed point. Pulse trains can be excited out of the stable fixed point by means of an external pulse, giving strong evidence that the OEO is an excitable system.

This thesis deals with these pulse train solutions in experiments, simulations and analytic studies. Experimentally, these pulse trains were realized by extending the setup of Callan *et al.* [47, 64]. We add a device to excite the system with an initial pulse, and, furthermore, we gain precise control of important system parameters by automating the setup with LabVIEW. When simulating the model equations, we find that the professional integrator RADAR5 is more suitable than an integrator programmed from scratch because of two strongly separated time scales in the system. Using τ-periodic nullclines and the time scale separation, we can analytically explain the pulsing behavior.

We find an analogy between the OEO and the delayed feedback FHN model in neuroscience. Both systems produce similar pulse trains, and, similar to the FHN model, the OEO can be tuned to exhibit either excitable or oscillatory behavior. Being a generic model for excitability, the analogy to the FHN model provides further evidence that the OEO is an excitable system.

We apply τ-periodic nullclines, an analytical tool that proves useful for describing pulse trains, to other dynamic states that the scientific literature has recently explored. In that manner we use nullclines to analyze oscillatory states, a dynamic scenario that is used to build ultra-stable oscillators [75]. We derive an approximate waveform for the oscillations and show how the duty cycle can be tuned with the system parameters. Furthermore we show that nullclines are useful for analyzing breather solutions and the transition from the fixed point to chaos. The nullclines provide another method to calculate the Hopf bifurcation point that separates the oscillatory and the excitable regime. This method is superior to the current calculation method because it is strictly analytic and exact.

For the pulse trains and the oscillations we analytically show that a power law with an exponent of -1 (*i.e.*, an inverse proportionality) describes the relationship between the characteristic time scale of the system (*i.e.*, pulse width and oscillation period) and the low frequency cutoff of the bandpass filter. We approximate the scaling factor of the power law that agrees with experiments and simulations. Exploiting the power law, we can tune the pulse width and the period of the oscillations by changing bandpass filters. In the experiments, we tune the pulse width over three orders of magnitude with pulse widths as narrow as 150 ps, and, in the simulation, we find that pulse widths become even narrower with higher values of the low frequency cutoff and larger bandwidths. The quantitative relation between the characteristic time scale and the low frequency cutoff can be seen as an extension of earlier work about dynamics in bandpass filtered systems [41, 42, 47].

7.2 Applications and outlook

The OEO that we use is built with off-the-shelf telecommunications equipment and is therefore relatively cheap, increasing its applicability. The following ideas for applications of the pulse train solutions found in this thesis could be examined in future works.

The OEO that we use acts as an excitable system in some parameter regimes. We find that steady pulse trains can be excited. We hypothesize that the excitability of the OEO can be exploited to build a dynamic memory device [77, 78]. A binary code, stored as a pattern of pulses, could travel as electronic and optical pulses around the delay loop. For that application, long delays are needed to store a preferably long digital code. The advantage over present memory media is that the OEO can be written and read quickly, a property that could be exploited in radar applications.

Networks of coupled OEOs in the excitable regime are also interesting from a computational point of view, because coupled excitable systems could be used to form logic gates without a ruling clock. Motoike *et al.* [79] demonstrated this application with the excitable Belousov-Zhabotinsky chemical reaction. The advantage over regular computers is seen in having no ruling clock. A strong coupling between excitable media would represent an “and” and a weak coupling an “or” connection. This advantage could even be combined with the application of the OEO as a dynamic memory.

We find similar behavior between the OEO and the delay coupled FHN model in neuroscience. Small network motifs of FHN models have been theoretically analyzed [35]. Nevertheless, an experimental realization of these network motifs of FHN models has not been realized to date. A small number of coupled OEOs could be coupled to build small motifs of artificial neural networks that could be studied experimentally. Whether a network of OEOs would reproduce the theoretical findings remains an open question.

Steady pulse trains have a frequency spectrum that consists of pulses modulated by a $\sin(x)/x$ function, namely a frequency comb. The narrower the pulses, the further their frequency spectrum is extended. Scientists are interested in frequency combs with a broad frequency spectrum that can be technologically exploited [80, 81]. Therefore the ability to control the pulse width is a very desirable property to explore. The phase noise is a measure for the purity of a frequency source [75]. In future work the phase noise of the pulse trains, a measure of its applicability as a frequency source, could be determined.

8 Appendix: Materials & methods

8.1 Numerical Simulation

At various passages in this thesis, we use numerical simulations to test theoretical predictions. In numerical simulation methods, the solution $x(t)$ of a differential equation is approximated by a sequence $\{x_k\}$ where x_k corresponds to $x(k\mathrm{d}t)$, with $\mathrm{d}t$ being the time step. The approximation improves as the time step, $\mathrm{d}t$, decreases with the caveat that the computation time increases. A variable time step improves the computational efficiency especially for systems with a time scale separation where two largely different time scales are present.

To perform numerical simulations, we have two integrators from which to choose. We programmed one integrator ourselves with a 4th order Adams-Bashforth-Moulton algorithm that uses a constant time step $\mathrm{d}t$. We found that the professional integrator RADAR5 [82] is superior over the self-programmed integrator because it calculates smoother waveforms in less time. Its superiority is due to the use of a variable time steps. In this thesis, we use RADAR5 unless otherwise stated.

8.1.1 Adams-Bashforth-Moulton algorithm

The fourth order Adams-Bashforth-Moulton algorithm is a multi-step predictor-corrector method [83]. We consider the differential equation

$$x\prime = h(x). \tag{8.1}$$

As an initial guess, we approximate x_{n+1} as

$$x_{n+1} = x_n + \frac{\mathrm{d}t}{24}\left[55h(x_n) - 59h(x_{n-1}) + 37h(x_{n-2}) - 9h(x_{n-3})\right], \tag{8.2}$$

which is the prediction step. With our initial guess of x_{n+1}, we can improve the approximation of x_{n+1}, by using sampling points before (x_{n-1}, x_{n-2}) and after (x_{n+1}) the step n

$$x_{n+1} = x_n + \frac{\mathrm{d}t}{24}\left[9h(x_{n+1}) + 19h(x_n) - 5h(x_{n-1}) + 1h(x_{n-2})\right], \tag{8.3}$$

which is the corrector step.

The coefficients in front of $h(\cdot)$ are chosen to obtain the highest order of convergence [83].

To handle DDE with constant delays, we add another variable $x\,(t-\tau) \approx x_{n-N}$, where $N = \lfloor \tau/\mathrm{d}t \rfloor$ corresponds to the time delay.[1]

8.1.2 RADAR5

The problem we are dealing with (Eq. (2.21)) contains a time-scale separation, ε, which can be as low as 10^{-5}, leading to vastly separated time scales. In this manner, pulses in chapter 4 have slow refractory and pulsing phases and fast transition phases. When fast transitions occur the time step should be small and when the system is in a slow phase, such as resting in the fixed point and waiting to get excited again, the time step should be large to save computation time.

A so-called variable time step was realized for time delay systems by Ernst Hairer and Nicola Guglielmi [82] with the professional integrator RADAR5.

This integrator uses an implicit Runge-Kutta method with an adaptive step size. Time series for the OEO (Eq. (2.21)) calculated with the RADAR5 integrator look smoother, and require less computation time, compared to the Adams-Bashforth-Moulton algorithm.

As initial history functions we use Gaussian functions or square pulses. The square pulses were approximated by two sigmoids (Fermi functions).

[1] $\lfloor \cdot \rfloor$ denotes integer division.

8.2 Nullcline analysis

Nullclines are defined for differential equations as curves in phase space where the velocity field is zero in one direction. These curves can be calculated by setting the left hand side of the respective differential equation to zero. Where all nullclines intersect, the velocity field is zero in all directions and, therefore, a fixed point exists. The number of nullclines equals the number of system variables and, therefore, also the dimensions in phase space.

For continuous velocity fields, nullclines structure the phase space because in between a nullcline the velocity in the respective direction has the same sign and switches sign at the nullcline. Therefore, nullclines enable qualitative statements to be made about the velocity field and the trajectory in phase space. When the velocity field above and below one nullcline is directed toward the nullcline, a trajectory on this nullcline is confined to this curve until the trajectory leaves the nullcline at its borders. Together with a time scale separation, limit cycles can be calculated from nullclines as in Sec. 6.2.2.

For DDE, nullclines depend on a delayed variable, are no longer unique curves in phase space, and, therefore, possess their own dynamics which are termed dynamical nullclines. To keep the nullclines of DDE constant, it can either be assumed that the delayed variable is constant, an approach that is used for the one-dimensional map in Sec. 3.2, or are periodic in τ (*e.g.*, $x(t-\tau) = x(t)$) as used throughout this thesis.

8.3 Automation with LabVIEW

We automate the control of two parameters, m and γ, and an electric switch that allows us to propagate initial waveforms with the visual programming language LabVIEW. This software is commonly used for automating measuring equipment in laboratory setups. The LabVIEW board (model number SCB-68) allows to control several digital output voltages: ports that can only be switched from off ($0\,\mathrm{V}$) to on ($5\,\mathrm{V}$) and two analog output voltages that range from $-10\,\mathrm{V}$ to $10\,\mathrm{V}$. In addition, the board allows us to measure several analog and digital voltages. In Fig. 8.1, the connections to the LabVIEW ports of the setup are labeled.

Figure 8.1: Setup of the system. The LabView ports are labeled. The module to measure ϕ was not shown earlier, but is always present in the setup.

8.3.1 Controlling m

As described in Sec. 2.1.1, the phase shift, m, is set by applying a bias V_{DC} to the DC-port of the MZM which is supplied by an analog output port of the LabVIEW board. The phase shift, m, consists of the DC voltage, V_{DC}, and a slowly varying phase shift, ϕ. From Eq. (2.2) on page 8, we find V_{DC} for a desired phase shift m

$$V_{\mathrm{DC}} = \frac{2V_{\pi,\mathrm{DC}}}{\pi}(m - \phi), \tag{8.4}$$

with the π-voltage of the rf port of the MZM $V_{\pi,\mathrm{DC}}$. We also determine the slowly varying phase shift, ϕ, with LabVIEW. To do so, we vary V_{DC} from -10 to $10\,\mathrm{V}$ in steps of $0.2\,\mathrm{V}$ and measure the transmitted power with a DC photodetector (New Focus, model 2011) that is permanently deployed in the system and separated from the system by a 50-50 optical coupler (see Fig. 8.1). The transfer characteristic of the DC port of the MZM is fitted with a least square procedure to the $\cos^2$ characteristics of the MZM with the LabVIEW program (shown in Fig. 2.2b on page 8), a method which gives the slowly varying phase shift ϕ, as described in Sec. 2.1.1 on page 8. This procedure requires low γ to guarantee

Figure 8.2: Circuit diagram to control the stepping motor of the optical attenuator for one inductor. Switching the digital ports from (a) (on, off) to (b) (off, on), changes the sign of the current the inductor M is powered with.

that the system is in the resting state. We measure ϕ prior to each measurement in this thesis.

8.3.2 Controlling γ

We use a step-wise variable optical attenuator (JDS Uniphase VCB-Z013) after the laser diode to control the attenuation of the light incident on the MZM and therefore the gain, γ. In variable optical attenuators in fiber optics, two loose fiber ends are torn apart to adjust the attenuation. The optical attenuator in our system is electrically controlled by a stepping motor that consists of two inductors. These must be powered with opposite polarity depending on the step, a task that was implemented into the LabView program.

Therefore, we use four digital output ports of our LabView board which are connected to a circuit that we designed with the help of H. L. D. de S. Cavalcante. We use one of these circuits, shown in Fig. 8.2, for each inductor. In this circuit, the power for the inductor comes from an external voltage source and the direction is controlled by the logic ports of the LabView board. The separation of the logic ports from the transistors (Central Semiconductor Corporation 2N1308) with $100\,\mathrm{k\Omega}$ resistors protects the logic ports from supplying too much current. The arrows in Fig. 8.2 illustrate the direction of the current for two different states of the logic gates.

Filter	company	model number	description
#1	Picosecond Pulse Labs	5828-108	ultra-broadband amplifier
#2	Mini-Circuits	BLK-18-S+	DC-block
#3	Mini-Circuits	ZX60-33ln-S+	Low noise amplifier
#4	Mini-Circuits	ZX60-4016E-S+	Amplifier
#5	Mini-Circuits	ZX60-3011+	Low Noise Amplifier
#6	Mini-Circuits	SHP-100A+	Highpass filter

Table 8.1: Model numbers of the filters used.

A series of "on" and "off" combinations of the logic gates leads to a movement of the stepping motor and changes the attenuation of the laser light and therefore γ. The stepping motor also controls a potentiometer to which we apply a constant voltage from an analog output port and measure the resulting voltage with an analog input port of the LabVIEW board, thereby giving the state of the stepping motor. With the knowledge of the state of the stepping motor, we implemented a mechanism that prevents the stepping motor from exceeding certain limits.

8.3.3 Controlling the switch

The electric switch (Mini-Circuits ZYSWA-2-50DR) separates the signal generator from the experimental setup. The signal generator is used to provide one pulse as an initial condition but it actually repeats its pulse pattern with a period set to $2\,\mu$s.

We open the switch by applying a short rectangular signal with a digital off-on-off series from a digital output port of the LabVIEW board, a procedure that opens the switch for approximately $2\,\mu$s.

8.4 Experimental devices of the setup

8.4.1 Model numbers

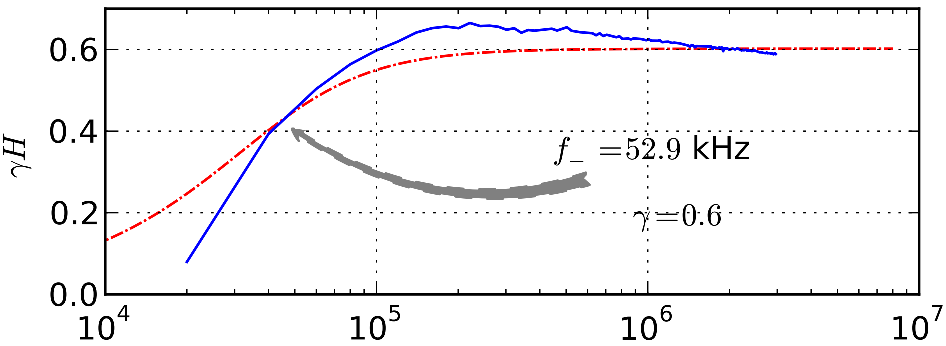
For measuring the transfer function in our system, we use a function generator (HP Agilent E8267D) coupled with a spectrum analyzer (HP Agilent E4440A). To measure the gain, we input sine waves with another function generator (Tektronix afg3251) and monitor the input and output signals with the oscilloscope as described in Sec. 2.1.4. To input initial pulses, we use a 12 Gbit/s error performance analyzer coupled with the measurement system HP Agilent 70004A as a pattern generator. The output of our system is measured with a high speed oscilloscope (Infiniium DSO 80804b) with an analog bandwidth of 8 GHz and a sampling rate of 40 GSa/s.

The MD driver has the model number JDS Uniphase H301-2310, the MZM the model number OTI 791003301, the AC photodiode the model number MITEQ DR-125G-A, the laser has the model number Sumitomo SEI SLT5411.

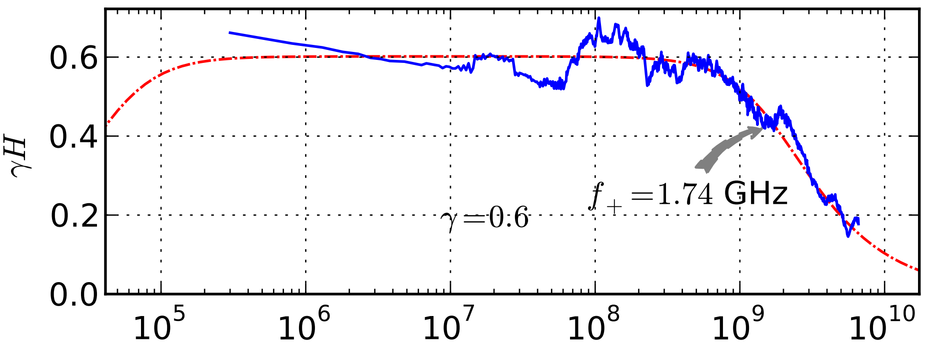
The model numbers of the bandpass filters are shown in table 8.1. All amplifiers were preceded by a 20 dB attenuator.

8.4.2 Transfer functions

In this section, we show the transfer functions of the filters of table 8.1.

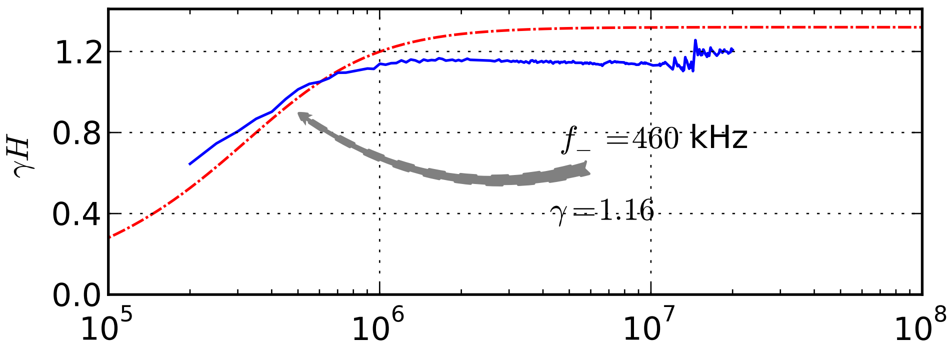


(a) Filter #1, low frequency cutoff

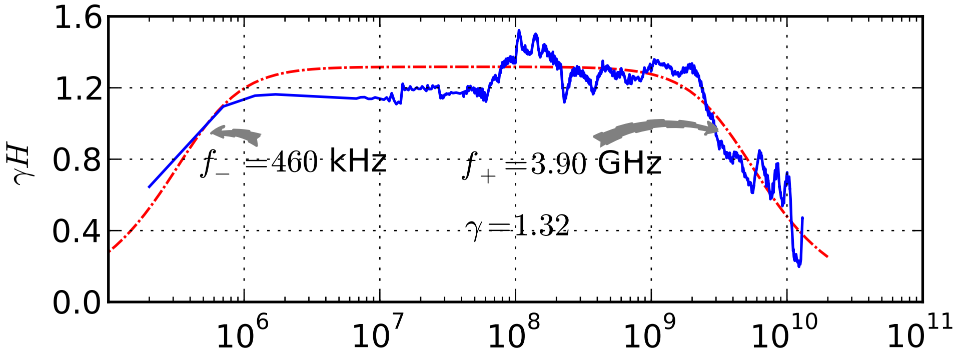


(b) Filter #1

Figure 8.3: Picosecond labs amplifier (Filter #1). Measured transferfunction (blue solid line) and least square fit of two pole bandpass filter with parameters $\omega_-/(2\pi) = 52.9\,\mathrm{kHz}$, $\omega_+/(2\pi) = 1.74\,\mathrm{GHz}$, $\gamma = 0.6$ (red dashed line).

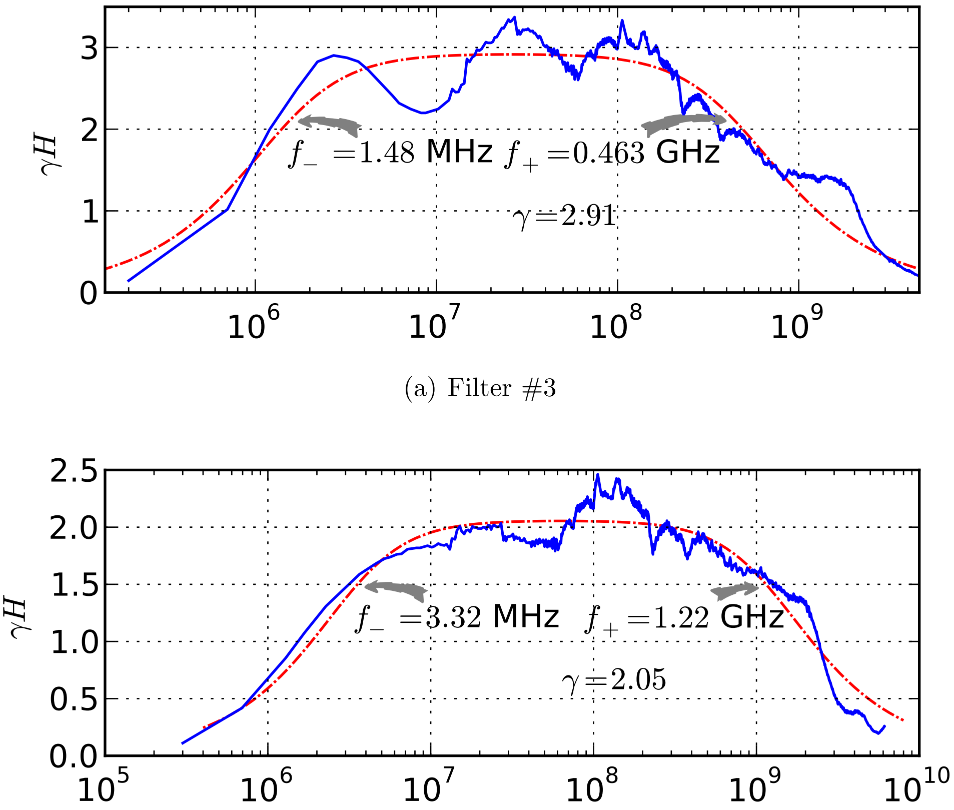


(a) Filter #2, low frequency cutoff



(b) Filter #2

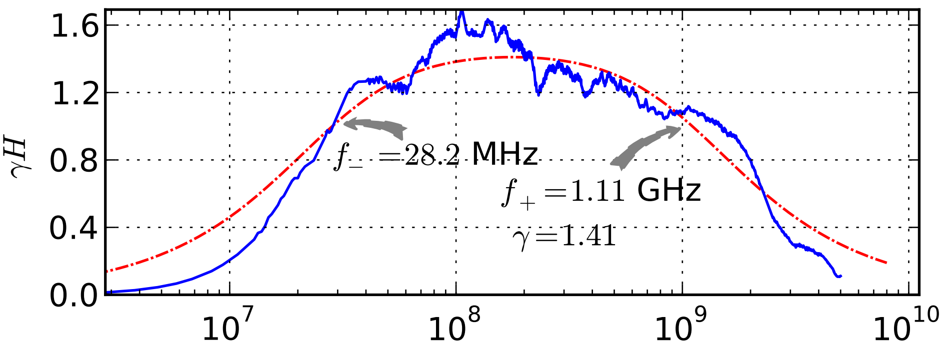
Figure 8.4: BLK18 (Filter #2). Measured transferfunction (blue solid line) and least square fit of two pole bandpass filter with parameters $\omega_-/(2\pi) = 460\,\mathrm{kHz}$, $\omega_+/(2\pi) = 3.90\,\mathrm{GHz}$, $\gamma = 1.32$ (red dashed line).



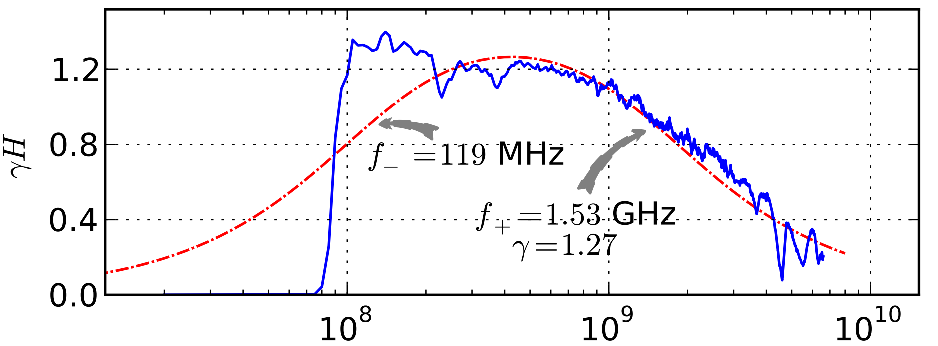
(a) Filter #3

(b) Filter #4

Figure 8.5: (a) amplifier 33ln (Filter #3) and (b) amplifier 4016 (Filter #4). Measured transferfunction (blue solid line) and least square fit of two pole band-pass filter (red dashed line) with parameters (a) $\omega_-/(2\pi) = 1.48\,\text{MHz}$, $\omega_+/(2\pi) = 463\,\text{MHz}$, $\gamma = 2.91$ and (b) $\omega_-/(2\pi) = 3.32\,\text{MHz}$, $\omega_+/(2\pi) = 1.22\,\text{GHz}$, $\gamma = 2.05$.



(a) Filter #5



(b) Filter #6

Figure 8.6: (a) amplifier 3011 (Filter #5) and (b) highpass shp100a (Filter #6). Measured transferfunction (blue solid line) and least square fit of two pole bandpass filter (red dashed line) with parameter (a) $\omega_-/(2\pi) = 28.2\,\mathrm{MHz}$, $\omega_+/(2\pi) = 1.11\,\mathrm{GHz}$, $\gamma = 1.41$ and (b) $\omega_-/(2\pi) = 119\,\mathrm{MHz}$, $\omega_+/(2\pi) = 1.53\,\mathrm{GHz}$, $\gamma = 1.27$.

Bibliography

[1] E. Izhikevich: *Dynamical systems in neuroscience: The geometry of excitability and bursting* (The MIT press, Cambridge, MA, 2007).

[2] J. Keener and J. Sneyd: *Mathematical Physiology* (Springer, New York, NY, 1998).

[3] F. Plaxa, M. G. Velarde, F. T. Arecchi, S. Boccaletti, and M. Ciofini: *Excitability following an avalanche-collapse process*, Europhys. Lett. **38**, 85 (1997).

[4] D. Gomila, M. Matias, and P. Colet: *Excitability mediated by localized structures in a dissipative nonlinear optical cavity*, Phys. Rev. Lett. **94**, 063905 (2005).

[5] M. Bache, F. Prati, R. K. G Tissoni, L. Lugiato, I. Protsenko, and M. Brambilla: *Cavity soliton laser based on vcsel with saturable absorber*, Appl. Phys. B **81**, 913 (2005).

[6] P. Genevet, S. Barland, M. Giudici, and J. Tredicce: *Cavity soliton laser based on mutually coupled semiconductor microresonators.*, Phys. Rev. Lett. **101**, 123905 (2008).

[7] A. Hodgkin: *The local electric changes associated with repetitive action in a non-medullated axon*, The Journal of Physiology **107**, 165 (1948).

[8] A. L. Hodgkin and A. F. Huxley: *A quantitative description of membrane current and its application to conduction and excitation in nerve*, J. Physiol. **117**, 500 (1952).

[9] P. Nelson: *Biological Physics* (W. H. Freeman, New York, NY, 2007).

[10] R. FitzHugh: *Mathematical models of threshold phenomena in the nerve membrane*, Bull. Math. Biol. **17**, 257 (1955).

[11] J. Nagumo, S. Arimoto, and S. Yoshizawa.: *An active pulse transmission line simulating nerve axon.*, Proc. IRE **50**, 2061 (1962).

[12] K. Pyragas: *Continuous control of chaos by self-controlling feedback*, Phys. Lett. A **170**, 421 (1992).

[13] E. Schöll and H. G. Schuster (Editors): *Handbook of Chaos Control* (Wiley-VCH, Weinheim, 2008), second completely revised and enlarged edition.

[14] W. Just, A. Pelster, M. Schanz, and E. Schöll: *Delayed complex systems*, Theme Issue of Phil. Trans. R. Soc. A **368** (2010), pp.301-513.

[15] C. U. Choe, T. Dahms, P. Hövel, and E. Schöll: *Controlling synchrony by delay coupling in networks: from in-phase to splay and cluster states*, Phys. Rev. E **81**, 025205(R) (2010).

[16] F. M. Atay (Editor): *Complex Time-Delay Systems*, Understanding Complex Systems (Springer, Berlin Heidelberg, 2010).

[17] V. Flunkert, S. Yanchuk, T. Dahms, and E. Schöll: *Synchronizing distant nodes: a universal classification of networks*, Phys. Rev. Lett. **105**, 254101 (2010).

[18] M. A. Dahlem, G. Hiller, A. Panchuk, and E. Schöll: *Dynamics of delay-coupled excitable neural systems*, Int. J. Bifur. Chaos **19**, 745 (2009).

[19] E. Schöll, P. Hövel, V. Flunkert, and M. A. Dahlem: *Time-delayed feedback control: from simple models to lasers and neural systems*, in *Complex Time-Delay Systems*, edited by F. M. Atay (Springer, Berlin, 2009).

[20] N. B. Janson, A. G. Balanov, and E. Schöll: *Delayed feedback as a means of control of noise-induced motion*, Phys. Rev. Lett. **93**, 010601 (2004).

[21] A. G. Balanov, N. B. Janson, and E. Schöll: *Control of noise-induced oscillations by delayed feedback*, Physica D **199**, 1 (2004).

[22] B. Hauschildt, N. B. Janson, A. G. Balanov, and E. Schöll: *Noise-induced cooperative dynamics and its control in coupled neuron models*, Phys. Rev. E **74**, 051906 (2006).

[23] T. Prager, H. P. Lerch, L. Schimansky-Geier, and E. Schöll: *Increase of coherence in excitable systems by delayed feedback*, J. Phys. A **40**, 11045 (2007).

[24] P. Hövel, M. A. Dahlem, and E. Schöll: *Synchronization of noise-induced oscillations by time-delayed feedback*, in *Proc. 19th Internat. Conf. on Noise and Fluctuations (ICNF-2007)* (American Institute of Physics, College Park, Maryland 20740-3843, 2007), vol. 922, pp. 595–598, ISBN 0-7354-0432-8.

[25] P. Hövel, M. A. Dahlem, T. Dahms, G. Hiller, and E. Schöll: *Time-delayed feedback control of delay-coupled neurosystems and lasers*, in *Preprints of the Second IFAC meeting related to analysis and control of chaotic systems (CHAOS09)* (World Scientific, 2009), (arXiv:0912.3395).

[26] P. Hövel, S. A. Shah, M. A. Dahlem, and E. Schöll: *Feedback-dependent control of stochastic synchronization in coupled neural systems*, Proc. 4th International Scientific Conference on Physics and Control (PhysCon 09), ed. L. Fortuna and M. Frasca, IPACS Open Access Library http://lib.physcon.ru (e-Library of the International Physics and Control Society) (2009).

[27] P. Hövel, M. A. Dahlem, and E. Schöll: *Control of synchronization in coupled neural systems by time-delayed feedback*, Int. J. Bifur. Chaos **20**, 813 (2010).

[28] J. Lehnert: *Dynamics of Neural Networks with Delay*, Master's thesis, Technische Universität Berlin (2010).

[29] N. Kouvaris, L. Schimansky-Geier, and E. Schöll: *Control of coherence in excitable systems by the interplay of noise and time-delay*, Eur. Phys. J. ST **191**, 29 (2010).

[30] M. A. Dahlem, F. M. Schneider, and E. Schöll: *Failure of feedback as a putative common mechanism of spreading depolarizations in migraine and stroke*, Chaos **18**, 026110 (2008).

[31] F. M. Schneider, E. Schöll, and M. A. Dahlem: *Controlling the onset of traveling pulses in excitable media by nonlocal spatial coupling and time delayed feedback*, Chaos **19**, 015110 (2009).

[32] M. A. Dahlem, R. Graf, A. J. Strong, J. P. Dreier, Y. A. Dahlem, M. Sieber, W. Hanke, K. Podoll, and E. Schöll: *Two-dimensional wave patterns of spreading depolarization: retracting, re-entrant, and stationary waves*, Physica D **239**, 889 (2010).

[33] J. Hizanidis, R. Aust, and E. Schöll: *Delay-induced multistability near a global bifurcation*, Int. J. Bifur. Chaos **18**, 1759 (2008).

[34] R. Aust, P. Hövel, J. Hizanidis, and E. Schöll: *Delay control of coherence resonance in type-I excitable dynamics*, Eur. Phys. J. ST **187**, 77 (2010).

[35] E. Schöll, G. Hiller, P. Hövel, and M. A. Dahlem: *Time-delayed feedback in neurosystems*, Phil. Trans. R. Soc. A **367**, 1079 (2009).

[36] S. A. Brandstetter, M. A. Dahlem, and E. Schöll: *Interplay of time-delayed feedback control and temporally correlated noise in excitable systems*, Phil. Trans. R. Soc. A **368**, 391 (2010).

[37] D. Rosin: *Dynamic scenarios in two delay-coupled excitable systems*, Bachelor's thesis (2009).

[38] K. Ikeda: *Multiple-valued stationary state and its instability of the transmitted light by a ring cavity system*, Optics Communications **30**, 257 (1979).

[39] H. Gibbs, F. Hopf, D. Kaplan, and R. Shoemaker: *Observation of chaos in optical bistability*, Phys. Rev. Lett. **46**, 474 (1981).

[40] A. Neyer and E. Voges: *Dynamics of electrooptic bistable devices with delayed feedback*, Quantum Electronics, IEEE Journal of **18**, 2009 (1982).

[41] L. Illing and D. Gauthier: *Hopf bifurcations in time-delay systems with band-limited feedback*, Physica D: Nonlinear Phenomena **210**, 180 (2005).

[42] L. Illing and D. J. Gauthier: *Ultra-high-frequency chaos in a time-delay electronic device with band-limited feedback*, Chaos **16**, 033119 (2006).

[43] A. Argyris, D. Syvridis, L. Larger, V. Annovazzi-Lodi, P. Colet, I. Fischer, J. García-Ojalvo, C. Mirasso, L. Pesquera, and K. Shore: *Chaos-based communications at high bit rates using commercial fibre-optic links*, Nature **438**, 343 (2005).

[44] Y. C. Kouomou, P. Colet, L. Larger, and N. Gastaud: *Chaotic breathers in delayed electro-optical systems*, Phys. Rev. Lett. **95**, 203903 (2005).

[45] M. Peil, M. Jacquot, Y. K. Chembo, L. Larger, and T. Erneux: *Routes to chaos and multiple time scale dynamics in broadband bandpass nonlinear delay electro-optic oscillators*, Phys. Rev. E **79**, 026208 (2009).

[46] Y. Kouomou Chembo, L. Larger, H. Tavernier, R. Bendoula, E. Rubiola, and P. Colet: *Dynamic instabilities of microwaves generated with optoelectronic oscillators*, Optics letters **32**, 2571 (2007).

[47] K. E. Callan, L. Illing, Z. Gao, D. J. Gauthier, and E. Schöll: *Broadband chaos generated by an opto-electronic oscillator*, Phys. Rev. Lett. **104**, 113901 (2010).

[48] L. Larger and J. Dudley: *Nonlinear dynamics: Optoelectronic chaos*, Nature **465**, 41 (2010).

[49] X. Yao and L. Maleki: *Optoelectronic microwave oscillator*, JOSA B **13**, 1725 (1996).

[50] X. Yao and L. Maleki: *Multiloop optoelectronic oscillator*, Quantum Electronics, IEEE Journal of **36**, 79 (2000).

[51] S. Romisch, J. Kitching, E. Ferrè-Pikal, L. Hollberg, and F. Walls: *Performance evaluation of an optoelectronic oscillator*, Ultrasonics, Ferroelectrics and Frequency Control, IEEE Transactions on **47**, 1159 (2000).

[52] L. Illing, G. Hoth, L. Shareshian, and C. May: *Scaling behavior of oscillations arising in delay-coupled optoelectronic oscillators*, Phys. Rev. E **83**, 026107 (2011).

[53] T. Erneux: *Applied delay differential equations* (Springer, New York, NY, 2009).

[54] J. P. Richard: *Time-delay systems: an overview of some recent advances and open problems*, Automatica **39**, 1667 (2003).

[55] E. Schöll: *Neural control: Chaos control sets the pace*, Nature Physics **6**, 161 (2010).

[56] P. Hövel and E. Schöll: *Control of unstable steady states by time-delayed feedback methods*, Phys. Rev. E **72**, 046203 (2005).

[57] A. Ahlborn and U. Parlitz: *Controlling dynamical systems using multiple delay feedback control*, Phys. Rev. E **72**, 016206 (2005).

[58] K. B. Blyuss, Y. N. Kyrychko, P. Hövel, and E. Schöll: *Control of unstable steady states in neutral time-delayed systems*, Eur. Phys. J. B **65**, 571 (2008).

[59] A. Gjurchinovski and V. Urumov: *Stabilization of unstable steady states by variable delay feedback control*, Europhys. Lett. **84**, 40013 (2008).

[60] A. Gjurchinovski and V. Urumov: *Variable-delay feedback control of unstable steady states in retarded time-delayed systems*, Phys. Rev. E **81**, 16209 (2010).

[61] Y. Kuang: *Delay differential equations* (Academic Press, San Diego, CA, 1993).

[62] J. K. Hale and S. M. Verduyn Lunel: *Introduction to Functional Differential Equations* (Springer, New York, 1993).

[63] A. Bellen and M. Zennaro: *Numerical methods for delay differential equations* (Oxford University Press, USA, New York, NY, 2003).

[64] K. E. Callan: *Broadband chaos generated in a time-delayed opto-electronic device*, Master's thesis, Department of Physics, Graduate School of Duke University (2008).

[65] G. Ghione: *Semiconductor Devices for High-Speed Optoelectronics* (Cambridge University Press, Cambridge, UK, 2009).

[66] D. J. Gauthier: *Private communication* (2011).

[67] V. Udaltsov, L. Larger, J. Goedgebuer, M. Lee, E. Genin, and W. Rhodes: *Bandpass chaotic dynamics of electronic oscillator operating with delayed nonlinear feedback*, IEEE transactions on circuits and systems I-fundamental theory and applications **49**, 1006 (2002).

[68] Y. Kouomou Chembo, L. Larger, H. Tavernier, R. Bendoula, E. Rubiola, and P. Colet: *Dynamic instabilities of microwaves generated with optoelectronic oscillators*, Optics letters **32**, 2571 (2007).

[69] S. H. Strogatz: *Nonlinear Dynamics and Chaos* (Westview Press, Cambridge, MA, 1994).

[70] M. Desroches, B. Krauskopf, and H. Osinga: *Mixed-mode oscillations and slow manifolds in the self-coupled fitzhugh nagumo system*, Chaos **18**, 015107 (2008).

[71] E. M. Izhikevich and R. FitzHugh: *Fitzhugh-nagumo model*, Scholarpedia **1**, 1349 (2006).

[72] M. Wechselberger: *Canards*, Scholarpedia **2**, 1356 (2007).

[73] K. E. Callan: *Private communication* (2011).

[74] P. Emplit, J. P. Hamaide, F. Reynaud, C. Froehly, and A. Barthelemy: *Picosecond steps and dark pulses through nonlinear single mode fibers*, Optics Communications **62**, 374 (1987).

[75] E. Rubiola: *Phase Noise and Frequency Stability in Oscillators* (Cambridge University Press, Cambridge, MA, 2010).

[76] M. C. Soriano, L. Zunino, L. Larger, I. Fischer, and C. R. Mirasso: *Distinguishing fingerprints of hyperchaotic and stochastic dynamics in optical chaos from a delayed opto-electronic oscillator*, Optics Letters **36**, 2212 (2011).

[77] T. Aida and P. Davis: *Oscillation modes of laser diode pumped hybrid bistable system with large delay and application to dynamical memory*, Quantum Electronics, IEEE Journal of **28**, 686 (1992).

[78] J. Foss, A. Longtin, B. Mensour, and J. Milton: *Multistability and delayed recurrent loops*, Phys. Rev. Lett. **76**, 708 (1996).

[79] I. Motoike and K. Yoshikawa: *Information operations with an excitable field*, Phys. Rev. E **59**, 5354 (1999).

[80] T. Udem, R. Holzwarth, and T. Hänsch: *Optical frequency metrology*, Nature **416**, 233 (2002).

[81] Y.-S. Juan and F.-Y. Lin: *Ultra broadband microwave frequency combs generated by an optical pulse-injected semiconductor laser*, Opt. Express **17**, 18596 (2009).

[82] N. Guglielmi and E. Hairer: *Users guide for the code radar5-version 2.1* (2005).

[83] W. Press, B. Flannery, S. Teukolsky, W. Vetterling, *et al.*: *Numerical recipes*, vol. 3 (Cambridge Univ Press, New York, NY, 2007).

Eidesstattliche Erklärung

Die selbstständige und eigenhändige Anfertigung der vorliegenden Arbeit versichere ich an Eides statt.

Berlin, den

David Rosin